

Document
Identification >>>

HOST PROTOCOL

USYS32

Version 054

Handbook
USYS.012.0001.E

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1 HOST / USYS Communication

A Host Computer may upload or download data to the USYS system. The communication can be done either by a serial link (RS232) or an Ethernet LAN connection.

In the Ethernet link, the TCP/IP protocol is used to encapsulate the ZUMBACH ASCII protocol. The USYS will act as Server and will permit only one client connected, in this case the ASCII protocol in Multi-drop mode is not allowed.

1.1 CHARACTER SET

ASCII characters are used for data transmission except for the Wide-String fields. The Wide-String items are UNICODE strings encoded in UTF-8. The UTF-8 encoding has the advantage that the characters between 0x00 and 0x7F are equivalent with the ASCII ones so that, if we do not need to use localized strings, the variables using the Wide String format can be directly encoded in plain ASCII.

Non-printable characters are shown by placing the name of the character between "<" and ">".

The following non-printable characters are used in this manual:

Character	Function	Decimal	Hex
<ACK>	Acknowledge	6	06
<HT>	Horizontal Tab	9	09
<LF>	Line Feed	10	0A
<CR>	Carriage return	13	0D
<NAK>	Negative Acknowledge	21	15
<ESC>	Escape	27	1B
<SP>	Space	32	20
<EOT>	End of Transmission	4	04
<ENQ>	End of Enquiry	5	05
<STX>	Start of Text	2	02
<ETX>	End of Text	3	03

1.2 COMMAND AND DATA FORMAT - ZUMBACH ASCII PROTOCOL

1.2.1 Single messages

Data, device commands, and control signals transmitted between the Host and the USYS have the following format:

Group - Number - Message - Termination Character.

Example:

C2100<SP>1.175<CR>

The Group and Number fields are separated from the rest of the message with a space <SP>.

Group

Transmitted commands and data are divided into groups. Each group is designated with a letter. An upper case (capital) letter is used for a **download** sequence (Data transmitted from the Host to the USYS). A lower case letter is used for an **upload** sequence (Data transmitted from the USYS to the Host)

Number

Each member of a group is designated with a number. The numbers range from 0000 to 9999.

Suffix Codes

Some orders can extend the Number by a suffix. The suffix is a one or two digits number added at the end of the order number and separated by a ".".

Message

The message may be a value, a string, a selection or a Multiple Selection.

Value

A value consists only of numbers (0...9), prefixes (+ -), and a decimal point.

String

A string is a series of characters. Any printable ASCII character is allowed. The maximum length of the string is the same as if the string were entered manually.

Wide String

A UNICODE string coded in UTF-8

Selection

A selection from a list can be made by sending a number that corresponds to the position of the desired selection in the list. The first selection is 0.

Multiple

A multiple selection is used when more than one option may be selected at one time. The selections are made by adding the values of the choices desired.

Termination Character

Data transmitted from the USYS to the Host is always terminated with a <CR> except for the <ACK> and <NAK> messages. Data transmitted from the Host to the USYS may be terminated with a <CR><LF> or with simply a <CR>.

1.2.2 Suffix Codes

Some orders can extend the Number by a suffix. The suffix is one or two digits number added at the end of the order number and separated by the character ".".

The suffix codes have been introduced to facilitate the access to a group of related messages and extend the number of possible host codes. This makes much easier the communication with the USYS32 host protocol, and permits the extension of the codes range.

The previous codes are kept for compatibility purposes.

For example, the messages to upload the characteristic's current values where:

d2000 for the first characteristic, *d2500* for the second, *d3000* for the third ... and so on.

This restricted to 10 the maximum number of codes allowed.

With the Suffix codes we can use:

d2000.1 to upload the first characteristic, *d2000.2* the second, *d2000.27* the characteristic number 27 ... etc.

When commands using Suffix do not include it, 1 is assumed. Suffix always starts at number 1.

Suffix 0 is allowed, referring to all the allowed suffixes for this command. For example, on a product with 4 characteristics a query of `c2000.0` will receive the answer of four commands: `c2000.1`, `c2000.2`, `c2000.3` and `c2000.4`.

Range suffixes are also allowed, adding a second suffix to the command as the end of the range. On the example before, a query of `c2000.1.3` will receive the answer of three commands: `c2000.1`, `c2000.2` and `c2000.3`.

Changing one value using all or range suffixes will change the same value on all the implicated values. For example, `c2100.1.2 5.30` will set the same nominal value for characteristics 1 and 2.

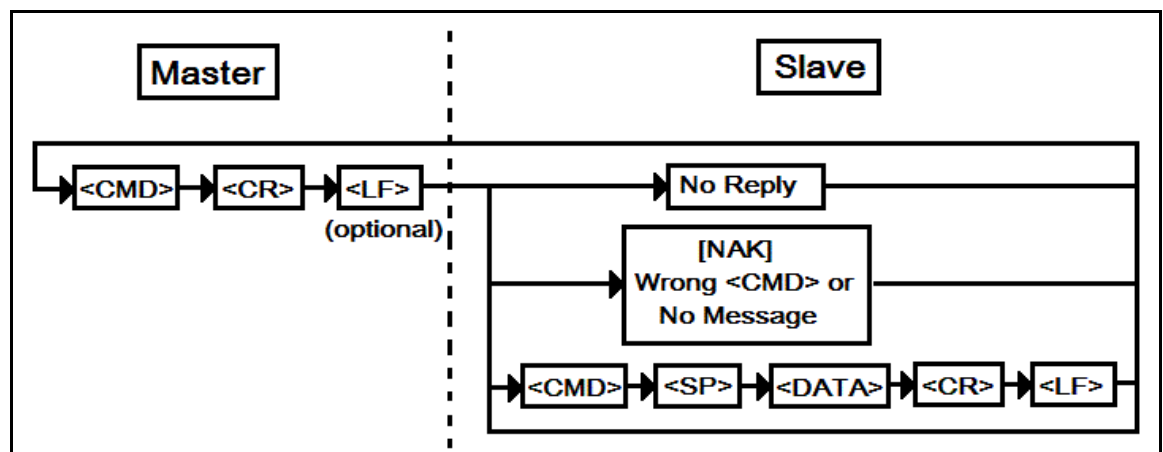
1.2.3 Grouped messages

The grouped messages contain several values in the same telegram. The format is the same than the Single messages but the message part is composed of several values separated by the character `<HT>` (Horizontal TAB).

Example:

```
m1005<SP>5.123<HT>10.432<HT>7.5431<HT>430<HT>-0.005<CR>
```

1.2.4 Upload



Used when the Host requires some information from the USYS

Masters request

<CMD> Five character sequence, C1, C2, C3, C4, C5 representing the parameter code. For example, the Slave software version number is requested with the `<CMD>` of 'a', '9', '9', '0', '0' (lower case letter!).

<Suffix> Optional. The suffix is a one or two digits number added at the end of the order number and separated by a ".".

<CR> ASCII carriage return terminates the request.

<LF> ASCII Line Feed. (Optional) Ignored by the Slave.

Response from Slave

<CMD> Echo of the four character sequence representing the parameter code.

<SP> ASCII space character.

<DATA> A variable number of characters (D1, D2, D3, ... Dn), representing the value requested.

<CR> ASCII carriage return. Indicates the end of the data.
<LF> ASCII Line Feed. The output of this character may be disabled in some devices, to reduce the telegram length.

Example 1:

Request for software version:

a9900<CR><LF>

Slave response to request for software version:

a9900<SP>8K1E3105<CR><LF>

Example 2:

Request for measured value of Characteristic 2:

d2000.2<CR><LF>

Slave response to request for measured value:

d2000.2<SP>2.345<CR><LF>

Example 3:

Request for all characteristic measured value:

d2000.0<CR><LF>

Slave response to request for measured value:

d2000.1<SP>2.34<CR><LF>

d2000.2<SP>1.23<CR><LF>

d2000.3<SP>2.323<CR><LF>

d2000.4<SP>0.123<CR><LF>

d2000.5<SP>23<CR><LF>

Example 4:

Request for measured value form characteristic 1 to 3:

d2000.1.3<CR><LF>

Slave response to request for measured value:

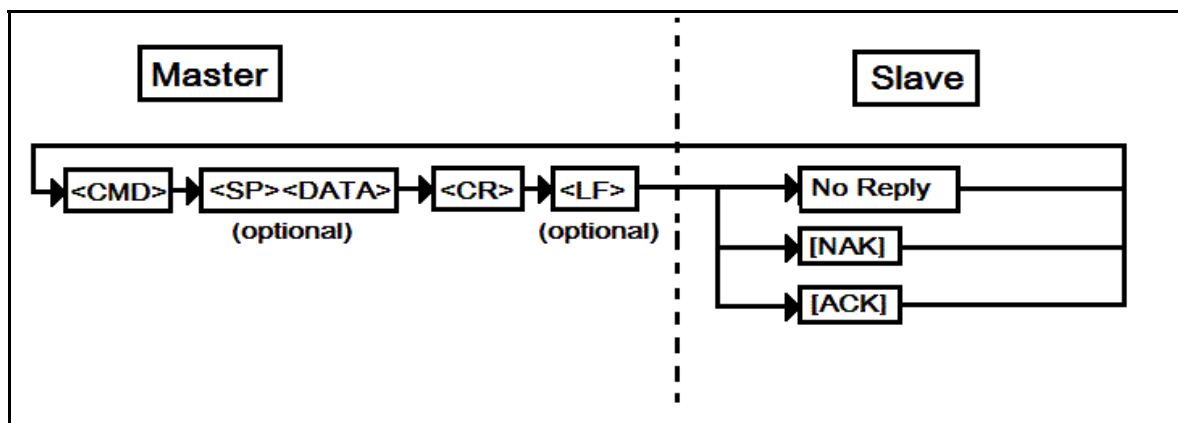
d2000.1<SP>2.34<CR><LF>

d2000.2<SP>21.234<CR><LF>

d2000.3<SP>23<CR><LF>

1.2.5 Download

The Host uses the download commands to program some value on the USYS.



Masters request

- <CMD> Five-character sequence, C1, C2, C3, C4, C5, representing the parameter code. For example, the Printer Default Language is downloaded with the <CMD> 'A' '2' '2' '0' '0' (upper case letter!).
- <Suffix> Optional. The suffix is a one or two digits number added at the end of the order number and separated by a ".".
- <SP> ASCII space character.
- <DATA> A variable number of characters (D1, D2, D3, ... Dn), representing the value downloaded.
- <CR> ASCII carriage return terminates the request.
- <LF> ASCII Line Feed. (Optional) Ignored by the Slave.

Response from Slave

- <ACK> ASCII Acknowledge. Sent if data is accepted.

or

- <NAK> ASCII Negative Acknowledge. Sent if data is not accepted due to an error.

Example 1:

Set the printer default language to French:

A2200<SP>02<CR><LF>

Slave response to downloaded Printer Default Language:

<ACK>

Example 2:

Set the product nominal value of characteristic 2 to 2.1:

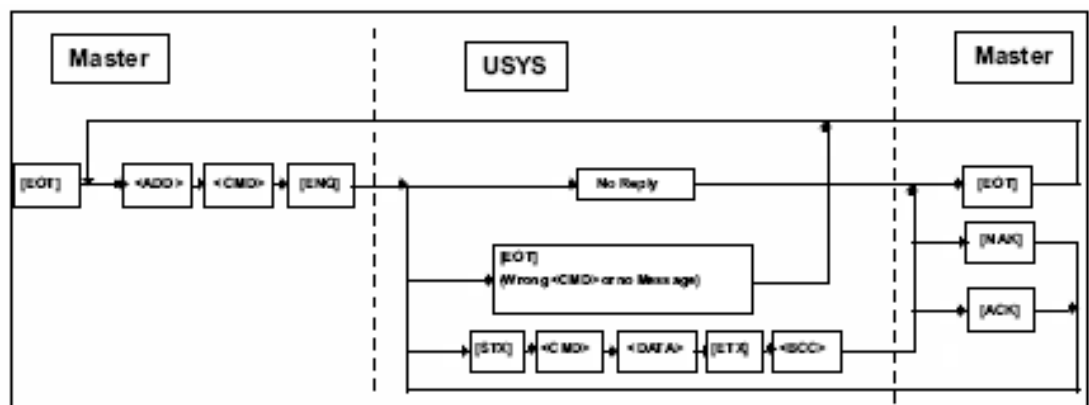
C2000.2<SP>2.1<CR><LF>

Slave response to this downloaded:

<ACK>

1.3 COMMAND AND DATA FORMAT - ANSI STANDARD X3.28-2.5-A4

1.3.1 Upload



Masters request:

- <EOT> ASCII control character. Resets the communication link passing control back to the control station (your PCB). Alerts all devices on the link to examine the next four characters to see if they correspond with their group / unit address identifier.
- [ADDR] Five character sequence representing the address of the instrument. The address is broken down into a most significant digit (MSD) and a least significant digit (LSD). Each digit is repeated for security. The format of the address, is MSD MSD LSD LSD. For example, the "[ADDR]" for device number 10 is '1' '1' '0' '0'.
- [CMD] Five character sequence, C1, C2, C3, C4, C5 representing the parameter code. For example, the USYS software version number is requested with the [CMD] of 'a' '9' '9' '0' '0'.
- <ENQ> ASCII control character. Indicates the end of an inquiry message.

Response from USYS:

- <EOT> ASCII control character. If the <CMD> is not recognized or if the USYS has no value to send, the USYS will send an <EOT> to return control to the control station (your PCB).

Or

- <STX> ASCII control character indicating the Start of Text.
- [CMD] Echo of the five character sequence representing the parameter code.
- [DATA] A variable number of characters (D1, D2, D3, ... Dn), representing the value requested.
- <ETX> ASCII control character. Indicates the end of the message.
- [BCC] Block Check Character. Generated by taking the Exclusive OR of the values of the ASCII characters following and excluding <STX> up to and including <ETX>.
- i.e.: [BCC] = C1 XOR C2 XOR C3 XOR C4 XOR D1 XOR D2 XOR D3 XOR D4...XOR Dn XOR ETX.

Response from master:

- <EOT> ASCII control character. Returns control to the station. This may be in response to a valid response from the USYS or to a "Time Out" if no reply was received.

Or

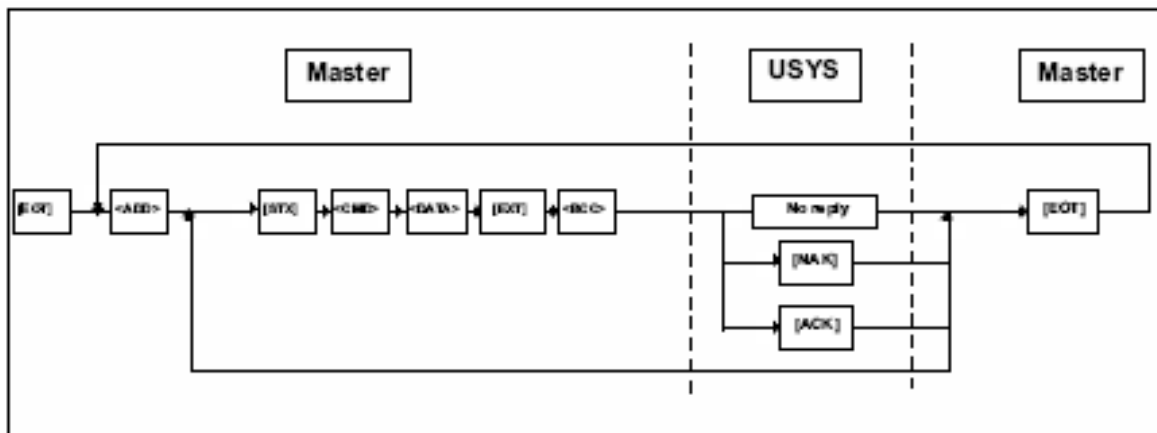
- <NAK> ASCII control character. Used to indicate an unsuccessful receipt of a message. This will cause the USYS to repeat the transmission.

Or

- <ACK> ASCII control character. Used to indicate a successful receipt of a message. This will cause the USYS to repeat the transmission.

1.3.2 Download

Master Message:



<EOT> ASCII control character. Resets the communication link passing control back to the control station (your PCB). Alerts all devices on the link to examine the next four characters to see if they correspond with their group / unit address identifier.

[ADDR] Five character sequence representing the address of the instrument. The address is broken down into a most significant digit (MSD) and the least significant digit (LSD). Each digit is repeated for security. The format of the address, is MSD MSD LSD LSD. For example, the "[ADDR]" for device number 10 is '1' '1' '0' '0'.

<STX> ASCII control character indicating the Start of Text.

[CMD] Five character sequence, C1, C2, C3, C4, C5 representing the parameter code. For example, the USYS software version number is requested with the [CMD] of 'a' '9' '9' '0' '0'.

[DATA] A variable number of characters (D1, D2, ... Dn), representing the value requested.

<ETX> ASCII control character. Indicates the end of the message.

[BCC] Block Check Character. Generated by taking the Exclusive OR of the values of the ASCII characters following and excluding <STX> up to and including <ETX>.

Response from USYS

<ACK> ASCII control character. Used to indicate a successful receipt of a message. This will cause the USYS to repeat the transmission.

Or

<NAK> ASCII control character. Used to indicate an unsuccessful receipt of a message. This will cause the USYS to repeat the transmission.

Response from master

<EOT> ASCII control character. Resets the communication link passing control back to the control station (itself).

2 Upload/Download Codes

The upload download codes for the USYS configuration are as follows:

2.1 GROUP A - SYSTEM

2.1.1 Localization

a1000	Date format 00 MM/DD/YYYY (month/day/year) 01 DD/MM/YYYY (day/month/year) 02 YYYY/MM/DD (year/month/day)	Selection	R/W
a1010	Date mm/dd/yyyy (month/day/year if date format is 00) dd/mm/yyyy (day/month/year if date format is 01) yyyy/mm/dd (year/month/day if date format is 02)	String	R/W
a1020	Time format 00 HH:MM (Hours:Minutes 24h.) 01 HH:MM AM (Hours:Minutes am/pm) 02 HH:MM:SS (Hours:Minutes:Seconds 24h.) 03 HH:MM:SS AM (Hours:Minutes:Seconds am/pm)	Selection	R/W
a1030	Time hh:mm (if time format is 00) hh:mm AM/PM (if time format is 01) hh:mm:ss (if time format is 02) hh:mm:ss AM/PM (if time format is 03)	String	R/W
a1031	System Time Zone -1 Undefined 0.. 350 Time Zone Code (see Appendix C)	Value	R/W
a1032	Standard Time Zone Abbreviation xxxxx (5 characters)	Wide-String	R/W

To define the DST it is needed to define the start and the end of the daylight saving time:

a1033	Daylight Saving Time (DST) 00 Off 01 On	Selection	R/W
a1034	DST Time Zone Abbreviation xxxxx (5 characters)	Wide-String	R/W
a1035	Start DST Week of the month 00 First 01 Second 02 Third 03 Forth 04 Last	Selection	R/W
a1036	Start DST Day 00 Sunday 01 Monday 02 Tuesday 03 Wednesday	Selection	R/W

04 Thursday
05 Friday
06 Saturday

a1037	Start DST Month	Selection	R/W
	00 January		
	01 February		
	02 March		
	03 April		
	04 May		
	05 June		
	06 July		
	07 August		
	08 September		
	09 October		
	10 November		
	11 December		
a1038	Start DST Time	String	R/W
	hh:mm (Fixed format)		

The start and the end of the daylight saving time must be defined using suffix:

Start DST	End DST
a1035.1	a1035.2
a1036.1	a1036.2
a1037.1	a1037.2
a1038.1	a1038.2

a1040	System Language	Selection	R/W
	00 English		
	01 French		
	02 German		
	03 Spanish		
	04 Dutch		
	05 Italian		
	06 Portuguese		
	08 Turkish		
	09 Swedish		
	10 Slovak		
	11 Polish		
	14 Japanese		
	15 Hangul		
	20 Simplified Chinese		
	21 Traditional Chinese		
a1050	System Language Operator Configurable	Selection	R/W
	00 Not allowed		
	01 Allowed		
a1060	International Keyboard Language	Selection	R/W
	00 Belgium		
	01 Brazil		
	02 Canada (F)		
	03 Denmark		
	04 Finland		
	05 France		
	06 Germany		
	07 Italy		

08 Latin America
 09 Netherlands
 10 Norway
 11 Portugal
 12 Spain
 13 Sweden
 14 Switzerland (F)
 15 Switzerland (G)
 16 United Kingdom
 17 United States

a1070	Personalized name for "Session"	Wide-String	R/W
a1080	Personalized name for "Coil"	Wide-String	R/W

The codes for other special texts are as the next list:

a3000	Last coil text - "Last Reel"
a3001	Units for analog gauge 1- "Units 1"
a3002	Units for analog gauge 2- "Units 2"
a3003	Units for analog gauge 3- "Units 3"
a3004	Units for analog gauge 4- "Units 4"
a3005	Units for analog gauge 1 with parenthesis - "(Units 1)"
a3006	Units for analog gauge 2 with parenthesis - "(Units 2)"
a3007	Units for analog gauge 3 with parenthesis - "(Units 3)"
a3008	Units for analog gauge 4 with parenthesis - "(Units 4)"
a3009	Data screen, page actual coil - "Present Reel"
a3010	Data screen, page session log - "Lot Log"
a3011	Data screen, coil number - "Reel Nr."
a3012	Data screen, coil length - "Reel Length"
a3013	Data screen, session length - "Lot Length"
a3014	Data screen, actual coil - "Piece"
a3015	Data screen, actual session - "Lot"
a3016	Start/Stop menu, coil - "Reel"
a3017	Start/Stop menu, session - "Lot"
a3018	Incidence coil begin - "Reel Begin"
a3019	Incidence coil end - "Reel End"
a3020	Incidence new coil number - "New Reel Number"
a3021	Incidence session begin - "Lot Begin"
a3022	Incidence lot end - "Lot End"
a3023	IO signal coil change - "Piece Change Signal"
a3024	IO signal session start/stop - "Lot Start/Stop Signal"
a3025	IO input for analog gauge 1 - "Analog Gauge 1"
a3026	IO input for analog gauge 2 - "Analog Gauge 2"
a3027	IO input for analog gauge 3 - "Analog Gauge 3"
a3028	IO input for analog gauge 4 - "Analog Gauge 4"
a3029	Printer, current session log. - "Current Lot Log"
a3030	Printer, coil report - "Reel Report"
a3031	Printer, session data log - "Lot Data Log"
a3032	Printer, session report - "Lot Report"
a3033	coil change - "Reel Change"

2.1.2 Global

a1100	Measuring Units	Selection	R/W
	00 mm		
	01 inch		

a1110	Global Resolution 00 0.1 mm or 0.005 inch 01 0.05 mm or 0.002 inch 02 0.02 mm or 0.001 inch 03 0.01 mm or 0.0005 inch 04 0.005 mm or 0.0002 inch 05 0.002 mm or 0.0001 inch 06 0.001 mm or 0.00005 inch 07 0.0005 mm or 0.00002 inch 08 0.0002 mm or 0.00001 inch 09 0.0001 mm or 0.000005 inch 10 0.00001 mm or 0.000002 inch	Selection	R/W
a1111	Area Units 00 mm ² 01 in ²	Selection	R/W
a1112	Resolution for Area measurement 00 1 mm ² or 0.05 in ² 01 0.5 mm ² or 0.02 in ² 02 0.2 mm ² or 0.01 in ² 03 0.1 mm ² or 0.005 in ² 04 0.05 mm ² or 0.002 in ² 05 0.02 mm ² or 0.001 in ² 06 0.01 mm ² or 0.0005 in ² 07 0.005 mm ² or 0.0002 in ² 08 0.002 mm ² or 0.0001 in ² 09 0.001 mm ² or 0.00005 in ² 10 0.0001 mm ² or 0.00002 in ²	Selection	R/W
a1115	Temperature Units 00 °C 01 °F	Selection	R/W
a1117	Resolution for Temperature measurement 00 1 °C or °F 01 0.5 °C or °F 02 0.2 °C or °F 03 0.1 °C or °F	Selection	R/W
a1120	Length Units 00 meters 01 feet	Selection	R/W
a1130	Resolution for Length measurement 00 1 m or feet 01 0.5 m or feet 02 0.2 m or feet 03 0.1 m or feet 04 0.05 m or feet	Selection	R/W
a1135	Capacitance Units 00 pF/m 01 pF/ft	Selection	R/W

a1137	Resolution for Capacitance measurement 00 0.2 pF/m or pF/ft 01 0.1 pF/m or pF/ft 02 0.05 pF/m or pF/ft 03 0.02 pF/m or pF/ft	Selection	R/W
a1140	Tolerance Type 00 Relative (to nominal) 01 Absolute	Selection	R/W
a1150	Prealarm Type 00 None 01 % 02 Relative 03 Absolute	Selection	R/W
a1160	Prealarm Setting 00 Product 01 Global	Selection	R/W
a1170	Prealarm Value x.xx (% , fixed resolution)	Value	R/W
a1171	FFT Frequency Units 00 Hertz 01 KHertz	Selection	R/W
a1172	Resolution for FFT Freq. measurement 00 1 Hertz or KHertz 01 0.5 Hertz or KHertz 02 0.2 Hertz or KHertz 03 0.1 Hertz or KHertz 04 0.05 Hertz or KHertz 05 0.02 Hertz or KHertz 06 0.01 Hertz or KHertz 07 0.005 Hertz or KHertz 08 0.002 Hertz or KHertz 09 0.001 Hertz or KHertz 10 0.0001 Hertz or KHertz	Selection	R/W
a1173	SRL Frequency Units 00 MHertz 01 GHertz	Selection	R/W
a1174	Resolution for SRL Freq. measurement 00 1 MHertz or GHertz 01 0.5 MHertz or GHertz 02 0.2 MHertz or GHertz 03 0.1 MHertz or GHertz 04 0.05 MHertz or GHertz 05 0.02 MHertz or GHertz 06 0.01 MHertz or GHertz 07 0.005 MHertz or GHertz 08 0.002 MHertz or GHertz 09 0.001 MHertz or GHertz 10 0.0001 MHertz or GHertz	Selection	R/W

a1175	WSWR Frequency Units 00 MHertz 01 GHertz	Selection	R/W
a1176	Resolution for VSWR Freq. measurement 00 1 MHertz or GHertz 01 0.5 MHertz or GHertz 02 0.2 MHertz or GHertz 03 0.1 MHertz or GHertz 04 0.05 MHertz or GHertz 05 0.02 MHertz or GHertz 06 0.01 MHertz or GHertz 07 0.005 MHertz or GHertz 08 0.002 MHertz or GHertz 09 0.001 MHertz or GHertz 10 0.0001 MHertz or Ghertz	Selection	R/W
a1180	Offset Type 00 Absolute 01 Factor 02 %	Selection	R/W
a1190	Resolution for Expansion measurement 00 1 % 01 0.5 % 02 0.2 % 03 0.1 % 04 0.05 % 05 0.02 % 06 0.01 %	Selection	R/W
a1195	Weight Units 00 g/m 01 g/3ft	Selection	R/W
a1197	Resolution for Weight measurement 00 1 g/m or g/3ft 01 0.5 g/m or g/3ft 02 0.2 g/m or g/3ft 03 0.1 g/m or g/3ft 04 0.05 g/m or g/3ft 05 0.02 g/m or g/3ft 06 0.01 g/m or g/3ft 07 0.005 g/m or g/3ft 08 0.002 g/m or g/3ft 09 0.001 g/m or g/3ft	Selection	R/W
a6500	Throughput Units 00 Kg/h 01 g/min 02 lb/h	Selection	R/W
a6505	Resolution for Throughput measurement 00 1 Kg/h, 0.1 g/min or lb/h 01 0.1 Kg/h, 0.01 g/min or lb/h 02 0.01 Kg/h, 0.001 g/min or lb/h 03 0.001 Kg/h, 0.0001 g/min or lb/h	Selection	R/W

a6510	Screw Yield Units 00 g/turn 01 Kg/h/rpm	Selection	R/W
a6515	Resolution for Screw Yield measurement 00 0.1 g/turn or 1 Kg/h/rpm 01 0.01 g/turn or 0.1 Kg/h/rpm 02 0.001 g/turn or 0.01 Kg/h/rpm 03 0.0001 g/turn or 0.001 Kg/h/rpm	Selection	R/W
a6520	Density Units 00 kg/m ³ 01 lb/ft ³	Selection	R/W
a6525	Resolution for Density measurement 00 1 Kg/m ³ or lb/ft ³ 01 0.1 Kg/m ³ or lb/ft ³ 02 0.01 Kg/m ³ or lb/ft ³ 03 0.001 Kg/m ³ or lb/ft ³ 04 0.0005 Kg/m ³ or lb/ft ³ 05 0.0002 Kg/m ³ or lb/ft ³ 06 0.0001 Kg/m ³ or lb/ft ³ 07 0.00005 Kg/m ³ or lb/ft ³ 08 0.00002 Kg/m ³ or lb/ft ³ 09 0.00001 Kg/m ³ or lb/ft ³	Selection	R/W
a1200	Display Values Averaging 00 Averaged 01 Fast 02 Moving Average	Selection	R/W
a1210	Statistics Averaging 00 Averaged 01 Fast 02 Moving Average	Selection	R/W
a1220	Alarms Averaging 00 Averaged 01 Fast 02 Moving Average	Selection	R/W
a1230	SPC Averaging 00 Averaged 01 Fast 02 Moving Average	Selection	R/W
a1240	Analog Output Averaging 00 Averaged 01 Fast 02 Moving Average	Selection	R/W
a1250	Averaging based On 00 Length 01 Time	Selection	R/W
a1251	Display Values Averaging based On 00 Length 01 Time	Selection	R/W

a1252	Statistic Values Averaging based On 00 Length 01 Time	Selection	R/W
a1253	Alarm Values Averaging based On 00 Length 01 Time	Selection	R/W
a1254	SPC Values Averaging based On 00 Length 01 Time	Selection	R/W
a1255	Analog Output Values Averaging based On 00 Length 01 Time	Selection	R/W
a1260	Averaging Range x.xxxx (length resolution, Length units if averaging unit is 00) x.x (fixed resolution, seconds if averaging unit is 01)	Value	R/W
a1261	Averaging Range for Display Values x.xxxx (length resolution, Length units if averaging unit is 00) x.x (fixed resolution, seconds if averaging unit is 01)	Value	R/W
a1262	Averaging Range for Statistic Values x.xxxx (length resolution, Length units if averaging unit is 00) x.x (fixed resolution, seconds if averaging unit is 01)	Value	R/W
a1263	Averaging Range for Alarm Values x.xxxx (length resolution, Length units if averaging unit is 00) x.x (fixed resolution, seconds if averaging unit is 01)	Value	R/W
a1264	Averaging Range for SPC Values x.xxxx (length resolution, Length units if averaging unit is 00) x.x (fixed resolution, seconds if averaging unit is 01)	Value	R/W
a1265	Averaging Range for Analog Output Values x.xxxx (length resolution, Length units if averaging unit is 00) x.x (fixed resolution, seconds if averaging unit is 01)	Value	R/W
a1270	High Speed Gauges: Average for Fast Values x.xx (fixed resolution, seconds)	Value	R/W
a1271	Moving Average Number for Display Values xxx(fixed resolution)	Value	R/W
a1272	Moving Average Number for Statistic Values xxx(fixed resolution)	Value	R/W
a1273	Moving Average Number for Alarm Values xxx(fixed resolution)	Value	R/W
a1274	Moving Average Number for SPC Values xxx(fixed resolution)	Value	R/W
a1275	Moving Average Number for Analog Output Value xxx(fixed resolution)		R/W

2.1.3 Screens

a1300	Strip Charts: X-Scale Unit 00 Length 01 Time	Selection	R/W
a1310	Strip Charts: X-Scale Division x.xxxx (length resolution, Length units if X-Scale unit is 00) x.x (fixed resolution, seconds if X-Scale unit is 01)	Value	R/W
a1320	Strip Charts Visualization 00 No Lines 01 Maximum Line 02 Average Line 04 Minimum Line 08 Coil Change	Multiple	R/W
a1372	UMAC Tolerance Variation % xx.xx (% , fixed resolution)	Value	R/W
a1374	UMAC Overall Wall Color 000..255 000 Black 001 Dark blue 002 Green 003 Light blue 004 Red 005 Pink 006 Brown 007 Light grey ...	Selection	R/W
a1375	UMAC resolution for Wallthickness Displays 00 0.1 mm or 0.005 inch 01 0.05 mm or 0.002 inch 02 0.02 mm or 0.001 inch 03 0.01 mm or 0.0005 inch 04 0.005 mm or 0.0002 inch	Selection	R/W
a1376	UMAC resolution for Eccentricity Displays 00 0.1 mm or 0.005 inch 01 0.05 mm or 0.002 inch 02 0.02 mm or 0.001 inch 03 0.01 mm or 0.0005 inch 04 0.005 mm or 0.0002 inch	Selection	R/W
a1377	UMAC resolution for Position Displays 00 0.1 mm or 0.005 inch 01 0.05 mm or 0.002 inch 02 0.02 mm or 0.001 inch 03 0.01 mm or 0.0005 inch 04 0.005 mm or 0.0002 inch	Selection	R/W
a1378	UMAC Displays Zoom 00 x4 01 x3 02 x2 03 x1	Selection	R/W

a1380	RAYEX Overall Color Same as a1374	Selection	R/W
a1381	RAYEX Mark Position Active 00 Off 01 On	Selection	R/W
a1382	RAYEX Mark Position Value x.xxxx (global units, global resolution)	Value	R/W
a1383	RAYEX resolution for Wallthickness Displays 00 0.1 mm or 0.005 inch 01 0.05 mm or 0.002 inch 02 0.02 mm or 0.001 inch 03 0.01 mm or 0.0005 inch 04 0.005 mm or 0.0002 inch	Selection	R/W
a1384	RAYEX resolution for Eccentricity Displays 00 0.1 mm or 0.005 inch 01 0.05 mm or 0.002 inch 02 0.02 mm or 0.001 inch 03 0.01 mm or 0.0005 inch 04 0.005 mm or 0.0002 inch	Selection	R/W
a1385	RAYEX resolution for Position Displays 00 0.1 mm or 0.005 inch 01 0.05 mm or 0.002 inch 02 0.02 mm or 0.001 inch 03 0.01 mm or 0.0005 inch 04 0.005 mm or 0.0002 inch	Selection	R/W
a1386	RAYEX Displays Zoom 00 x4 01 x3 02 x2 03 x1	Selection	R/W
a1395	Initial Screen 00 Main 01 Displays Characteristics 02 Displays UMAC #1 03 Displays UMAC #2 04 Displays ODEX #1 05 Displays ODEX #2 06 Displays ODEX #3 07 Displays ODEX #4 08 Displays ODEX #5 09 Displays ODEX #6 10 Displays RAYEX #1 12 Displays Event Log 13 Data Summary 14 Data Current Reel 15 Data Lot 16 Data Current SPC 17 Data Lot Log 18 Charts Strip Charts 19 Charts SPC 20 Product UMAC Layers #1 21 Product UMAC Layers #2	Selection	R/W

22 Product RAYEX Layers
 23 Product Characteristics
 24 Product SetPoints
 25 Product Devices
 26 Product Controllers
 27 Product Reports
 28 Product Texts
 29 Controllers
 30 DLPs

2.1.4 Statistics

a1400	SPC Collection interval based On 00 Length 01 Time	Selection	R/W
a1405	SPC Samples per piece xxx (fixed resolution)	Value	R/W
a1410	SPC Sample interval x.xxxx (length resolution, Length units if interval unit is 00) x.x (fixed resolution, seconds if interval unit is 01)	Value	R/W
a1415	SPC Pieces between subgroups xxx (fixed resolution)	Value	R/W
a1420	SPC Subgroup size xx (2 to 25, fixed resolution)	Value	R/W
a1430	SPC Control Limits Mode 00 Global 01 Product	Selection	R/W
a1440	Minimum SPC Subgroups xxx (fixed resolution)	Value	R/W
a1450	SPC Automatic Start/Stop - External Signal 00 Off 01 On	Selection	R/W
a1451	SPC Automatic Start/Stop – Shift Change 00 Off 01 On	Selection	R/W
a1452	SPC Automatic Start/Stop – Max. Subgroups 00 Off 01 On	Selection	R/W
a1455	SPC Shift Change 1 hh:mm (hours:minutes in 24h. format)	String	R/W
a1456	SPC Shift Change 2 hh:mm (hours:minutes in 24h. format)	String	R/W
a1457	SPC Shift Change 3 hh:mm (hours:minutes in 24h. format)	String	R/W
a1460	SPC Charts Scale 00 Automatic 01 Control Limits	Selection	R/W

a1470	Max. Subgroup Stored xxx (10 to 500, fixed resolution)	Value	R/W
a1480	SPC Standard Deviation 00 Estimated 01 Calculated	Selection	R/W
a1491	SPC Type 00 Standard 01 Moving Range	Selection	R/W
a1492	SPC Sampling Type 00 Continuous 01 Reels	Selection	R/W
a1500	Pause Criteria 00 None 01 External Signal 02 Minimum Line Speed	Multiple	R/W
a1505	Minimum Line Speed x.xx (% , fixed resolution)	Value	R/W
a1510	Session Automatic Start/Stop 00 Off 01 External Signal 02 Shift Change	Multiple	R/W
a1512	Reel Automatic Start/Stop 00 Off 01 External Signal 02 Reel Length 04 Shift Change 08 Measurement	Multiple	R/W
a1514	SPC Automatic Start/Stop 00 Off 01 External Signal 02 Max SPC Subgroups 04 Shift Change	Multiple	R/W
a1520	Shift Change 1 hh:mm (hours:minutes in 24h. format)	String	R/W
a1522	Reel Start / Stop Dependencies 00 Disabled 01 Lot 02 SPC	Multiple	R/W
a1524	Lot Start / Stop Dependencies 00 Disabled 01 Reel 02 SPC	Multiple	R/W
a1526	SPC Start / Stop Dependencies 00 Disabled 01 Reel 02 Lot	Multiple	R/W

a1530	Shift Change 2 hh:mm (hours:minutes in 24h. format)	String	R/W
a1540	Shift Change 3 hh:mm (hours:minutes in 24h. format)	String	R/W
a1544	Trigger Level x.xx (% , fixed resolution)	Value	R/W
a1546	Trigger Level Begin Mode 00 All Characteristics Triggered On 01 Some Characteristics Triggered On	Selection	R/W
a1548	Trigger Level End Mode 00 All Characteristics Triggered Off 01 Some Characteristics Triggered Off	Selection	R/W
a1550	Periodic Data 00 Disabled 01 Enabled	Selection	R/W
a1560	Periodic Data Interval x.xxxx (length resolution, Length units if output interval unit is 00) x.x (fixed resolution, seconds if output interval unit is 01)	Value	R/W
a1570	Periodic Data Output Interval Based On 00 Length 01 Time	Selection	R/W
a1580	Periodic Data Values 00 None 01 Minimum values 02 Average values 04 Maximum values	Multiple	R/W
a1590	Minimum Coil Length x.xxxx (length resolution)	Value	R/W
a1592	Tolerance Filter x.xx (fixed resolution)	Value	R/W
a1594	Exclusion Alarms/PreAlarms Values 00 None Exclusion 01 Alarms Values 02 PreAlarms Values	Selection	R/W
a1595	Pause Inhibits Length 00 Off 01 On	Selection	R/W
a1596	Pause Inhibits Digital Outputs 00 Off 01 On	Selection	R/W
a1598	Coil Automatic Start/Stop - External Signal 00 Off 01 On	Selection	R/W

a1600	Ends Filter 00 Disabled 01 Enabled	Selection	R/W
2.1.5 Security			
a1700	Product Access Permission 00 Partial Access 01 Full Access 02 No Access	Selection	R/W
a1710	Product Partial Access 00 None 01 Offsets 02 Nominals and Limits 04 Device Settings 08 Controller Settings 16 Reports 32 Texts 64 Save Operation 128 UMAC Layers 256 UMAC UPS 512 RAYEX Layers	Multiple	R/W
a1750	Password #1 xxxxxxxxxxxxxxx (15 characters)	Wide-String	R/W
a1755	Use of Password #1 00 None 01 System 02 Elements 04 Calibration 08 Peripherals 16 Configuration 32 Product 64 Service 128 Change Mode 256 Exit USYS	Multiple	R/W
a1760	Password #2 xxxxxxxxxxxxxxx (15 characters)	Wide-String	R/W
a1765	Use of Password #2 00 None 01 System 02 Elements 04 Calibration 08 Peripherals 16 Configuration 32 Product 64 Service 128 Change Mode 256 Exit USYS	Multiple	R/W
a1770	Product View 00 None 01 Characteristics 02 Set Points 04 Devices 08 Controllers	Multiple	R/W

16 Reports
 32 Texts
 64 Product Table
 128 FFT
 256 UMAC Layers
 512 UMAC UPS
 1024 RAYEX Layers

2.1.6 Printers

a1800	Printer 01 Text Mode 02 Graphic Mode	Selection	R/W
a1805	Printer #1 Setup String xxxxx...xxxxx (40 characters)	String	R/W
a1810	Printer Model 00 Text Printer(Only Text Mode) 01 Epson LQ (Only Graphic Mode) 02 Epson LX (Only Graphic Mode) 03 Epson Stylus (Only Graphic Mode) 04 HP DeskJet (Only Graphic Mode) 05 HP LaserJet 4 (Only Graphic Mode) 06 IBM Proprinter (Only Graphic Mode) 07 Postscript (Only Graphic Mode)	Selection	R/W
a1812	Printer Resolution (Only Graphic Mode) xx<SP>xx (vertical and horizontal resolution, DPI units)	String	R/W
a1813	Printer Character Set (Only Text Mode) 00 ANSI 7bits 01 PC-437 02 PC-737 03 PC-850 04 PC-852 05 PC-855 06 PC-857 07 PC-860 08 PC-861 09 PC-862 10 PC-863 11 PC-865 12 PC-866 13 PC-869	Selection	R/W
a1815	Printer Paper 00 Letter 01 Legal 02 A4 03 Continuous (Only Text Mode) 04 Special (Only Text Mode)	Selection	R/W
a1817	Printer Paper Orientation (Only Graphic Mode) 00 Landscape 01Portrait	Selection	R/W
a1822	Printer Text Columns (Only Text Mode) xx (fixed resolution)	Value	R/W

a1824	Printer Text Lines (Only Text Mode) xx (fixed resolution)	Value	R/W
a1826	Printer Line Feeds Number (Only Text Mode) xx (fixed resolution)	Value	R/W
a1828	Printer New Page (Only Text Mode) 00 Form Feed 01 Line Feeds	Selection	R/W
a1890	Printer Reports 00 None 01 Session [1] 02 Session [2] 04 Coil [1] 08 Coil [2] 16 SPC Summary 32 SPC Charts 64 Session Log 128 Event Log 256 System 512 Print Screen 1024 FFT 2048 FFT Summary	Multiple	R/W

The codes for Printers are as the next table:

Printer #1	Printer #2	Printer #3
a1800	a1900	a2000
a1805	a1905	a2005
a1810	a1910	a2010
a1812	a1912	a2012
a1813	a1913	a2013
a1815	a1915	a2015
a1817	a1917	a2017
a1822	a1922	a2022
a1824	a1924	a2024
a1826	a1926	a2026
a1828	a1928	a2028
a1890	a1990	a2090

a2200	Printer Default Language 00 English 01 French 02 German 03 Spanish 04 Dutch 05 Italian 06 Portuguese 08 Turkish 09 Swedish 20 Simplified Chinese 21 Traditional Chinese	Selection	R/W
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a2210	Printer Default Date format 00 MM/DD/YYYY (month/day/year) 01 DD/MM/YYYY (day/month/year) 02 YYYY/MM/DD (year/month/day)	Selection	R/W
a2220	Printer Default Time format 00 HH:MM (Hours:Minutes 24h.) 01 HH:MM AM (Hours:Minutes am/pm) 02 HH:MM:SS (Hours:Minutes:Seconds 24h.) 03 HH:MM:SS AM (Hours:Minutes:Seconds am/pm)	Selection	R/W

Suffix Codes are available for Printers (n=1..3):

Printer n
a2200.n
a2210.n
a2220.n

2.1.7 Remote Displays

a2300	Remote Displays 00 Disabled 01 Enabled	Selection	R/W
a2310	Remote Displays Type 00 NP 6155 01 VORNE 77 02 VORNE GY2200 03 SIEBERT	Selection	R/W
a2320	Refresh time x.x (seconds, fixed resolution) (only VORNE GY2200 type)	Value	R/W
a2340	Remote Display 1 selection 0 None 01 .. 99 Allowed Characteristic (old Usys2000 codes) 150 Controlled characteristic 1 151 Line speed 152 Coil length 153 Session length 1000.. 9999 Allowed Characteristic (see Appendix A) 10001 Controlled characteristic 1 10002 Controlled characteristic 2 10003 Controlled characteristic 3 10004 Controlled characteristic 4	Value	R/W
a2342	Remote Display 1 selection type 00 Current value 01 Maximum 02 Minimum 03 Average 04 Standard Deviation 05 CpK 06 Previous Maximum 07 Previous Minimum 08 Previous Average 09 Previous Standard Deviation 10 Previous CpK 11 Gauge	Selection	R/W

a2344	Remote Display 1 identification selector xx (2 characters) (only VORNE GY2200 type)	String	R/W
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The codes for other remote displays are as the next table:

R.D. 1	R.D. 2	R.D. 3	R.D. 4	R.D. 5	R.D. 6	R.D. 7	R.D. 8	R.D. 9	R.D.10
a2340	a2350	a2360	a2370	a2380	a2390	a2400	a2410	a2420	a2430
a2342	a2352	a2362	a2372	a2382	a2392	a2402	a2412	a2422	a2432
a2344	a2354	a2364	a2374	a2384	a2394	a2404	a2414	a2424	a2434

2.1.8 Marker

a2505	Marker Pulse Type 00 Length 01 Time	Selection	R/W
a2510	Marker Minimum Pulse Length x.xxxx (length resolution, Length units if Pulse Type is 00) x.x (fixed resolution, seconds if Pulse Type is 01)	Value	R/W
a2515	Marker Pulse Extension Type 00 Length 01 Time	Selection	R/W
a2516	Marker Pulse Extension Length x.xxxx (length resolution, Length units if Pulse Type is 00) x.x (fixed resolution, seconds if Pulse Type is 01)	Value	R/W
a2520	Marker Trigger 1 - Source 00 None 01 .. 99 Allowed Characteristic (see Usys2000 codes on Appendix A) 150 Controlled characteristic 1 151..249 Allowed Event (Usys2000 event codes + 150) 251 Any Characteristic 252 Any Event 1000.. 9999 Allowed Characteristic (see Appendix A) 10000 Controlled characteristic 1 10001 Controlled characteristic 2 10002 Controlled characteristic 3 10003 Controlled characteristic 4 10100..19999 Allowed Event (Appendix B codes + 10000)	Selection	R/W
a2521	Marker Trigger 1 – Characteristic Alarms High 00 Off 01 On	Selection	R/W
a2522	Marker Trigger 1 – Characteristic Alarms Low 00 Off 01 On	Selection	R/W
a2523	Marker Trigger 1 – Characteristic PreAlarms High 00 Off 01 On	Selection	R/W
a2524	Marker Trigger 1 – Characteristic PreAlarms Low 00 Off 01 On	Selection	R/W

a2525	Marker Trigger 1 – Characteristic No Measure 00 Off 01 On	Selection	R/W
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The codes for other Triggers of the Marker are as the next table:

Trigger 1	Trigger 2	Trigger 3	Trigger 4	Trigger 5	Trigger 6	Trigger 7	Trigger 8	Trigger 9	Trigger 10
a2520	a2530	a2540	a2550	a2560	a2570	a2580	a2590	a2600	a2610
a2521	a2531	a2541	a2551	a2561	a2571	a2581	a2591	a2601	a2611
a2522	a2532	a2542	a2552	a2562	a2572	a2582	a2592	a2602	a2612
a2523	a2533	a2543	a2553	a2563	a2573	a2583	a2593	a2603	a2613
a2524	a2534	a2544	a2554	a2564	a2574	a2584	a2594	a2604	a2614
a2525	a2535	a2545	a2555	a2565	a2575	a2585	a2595	a2605	a2615

2.1.9 UMAC Manual Compensation.

a2710	Acquisition Interval 00 Length 01 Time	Selection	R/W
a2720	Interval Size x.xxxx (length resolution, Length units if acquisition interval type is 00) x.x (fixed resolution, seconds if acquisition interval type is 01)	Value	R/W
a2730	Statistics Type 00 Average 01 Minimum 02 Maximum	Selection	R/W
a2740	Max. Diff. from Nominal x.xx (% , fixed resolution)	Value	R/W
a2750	Min. Diff. from Nominal x.xx (% , fixed resolution)	Value	R/W
a2760	Start Mode 00 On request 01 At product activation	Selection	R/W
a2765	Individual wall compensation by transducer 00 Off 01 On	Selection	R/W
a2770	Individual OD compensation by transducer 00 Off 01 On	Selection	R/W

2.1.10 UMAC Automatic Compensations.

a2800	WallThickness Characteristic to apply process 0 None 1.. 99 Allowed Characteristic (see Usys2000 codes on Appendix A) 100 .. 9999 Allowed Characteristic (see Appendix A)	Value	R/W
a2805	Self Tuning 00 Off 01 On	Selection	R/W
a2810	Control limits (on Offset) x.xx (% of nominal, fixed resolution)	Value	R/W

a2815	Correction Factor x.xx (% , fixed resolution)	Value	R/W
a2820	Start Mode 00 Manually 01 Automatically	Selection	R/W
a2825	Gain Up modifier x.xxxx (global resolution)	Value	R
a2830	Gain Down modifier x.xxxx (global resolution)	Value	R
a2835	Averaging modifier x.xxxx (global resolution)	Value	R
a2840	Computed Dead band x.xxxx (global resolution)	Value	R
a2845	Calculated Offset x.xxxx (global resolution)	Value	R
a2850	Status 00 Off 01 Ready 02 Dead Time 03 Start Acquisition 04 Acquisition 05 No Action 06 Increase 07 Decrease 08 Waiting 09 Error	Selection	R

Suffix Codes are available for Automatic Compensations (n=1..4):

Compensation n
a2800.n
a2805.n
a2810.n
a2815.n
a2820.n
a2825.n
a2830.n
a2835.n
a2840.n
a2845.n
a2850.n

2.1.11 Host Computer

a2900	Host Protocol Selection 00 Zumbach ASCII 01 ANSI Standard	Selection	R/W
a2905	Unicode on Wide-String Commands 00 UTF-8 01 Zumbach Unicode	Selection	R/W

a2907	Return error on char. actual when not allowed 00 Off 01 On	Selection	R/W
a2910	Address of USYS for Multidrop xx (fixed resolution)	Value	R/W
a2920	Termination Character (Zumbach ASCII) 00 <CR> 01 <CR><LF>	Selection	R/W
a2930	PC-Link Used 00 Off 01 On	Selection	R/W
a2940	Save Automatic & Grouped Configuration (Save Current Automatic & Grouped Configuration to Disk)	--	W
a2950	Reset Automatic & Grouped Configuration (Load Default Configuration)	--	W

2.1.12 RAYEX Shrinkage

a3100	Diameter Characteristic to apply process 0 None 1.. 99 Allowed Characteristic (see Usys2000 codes on Appendix A) 100 .. 9999 Allowed Characteristic (see Appendix A)	Value	R/W
a3105	Self Tuning 00 Off 01 On	Selection	R/W
a3110	Control limits (on Offset) x.xx (% of nominal, fixed resolution)	Value	R/W
a3115	Correction Factor x.xx (% , fixed resolution)	Value	R/W
a3120	Start Mode 00 Manually 01 Automatically	Selection	R/W

2.1.13 Controllers

a3125	Gain Up modifier x.xxxx (global resolution)	Value	R
a3130	Gain Down modifier x.xxxx (global resolution)	Value	R
a3135	Averaging modifier x.xxxx (global resolution)	Value	R
a3140	Computed Dead band x.xxxx (global resolution)	Value	R
a3145	Calculated Offset x.xxxx (global resolution)	Value	R
a3150	Status 00 Off 01 Ready	Selection	R

- 02 Dead Time
- 03 Start Acquisition
- 04 Acquisition
- 05 No Action
- 06 Increase
- 07 Decrease
- 08 Waiting
- 09 Error

2.1.14 Analog Peripherals

a4000	Analog Peripheral 1 00 Disabled 01 Enabled	Selection	R/W
a4010	Analog Peripheral 1 Characteristic 0 None 1 .. 99 Allowed Characteristic (see Usys2000 codes on Appendix A) 100.. 9999 Allowed Characteristic (see Appendix A)	Value	R/W
a4020	Analog Peripheral 1 FSR Minimum x.xxxx (global units, global resolution)	Value	R/W
a4030	Analog Peripheral 1 FSR Maximum x.xxxx (global units, global resolution)	Value	R/W
a4040	Analog Peripheral 1 Mode 00 Deviation 01 Absolute	Selection	R/W
a4050	Analog Peripheral 1 Reset Voltage x.xxx (voltage, fixed resolution)	Value	R/W
a4060	Analog Peripheral 1 Pause Value 00 Last Value 01 Reset Voltage 02 Current Value	Selection	R/W

The codes for other analog peripherals are as the next table:

Analog Periph. 1	Analog Periph. 2	Analog Periph. 3	Analog Periph. 4
a4000	a4100	a4200	a4300
a4010	a4110	a4210	a4310
a4020	a4120	a4220	a4320
a4030	a4130	a4230	a4330
a4040	a4140	a4240	a4340
a4050	a4150	a4250	a4350
a4060	a4160	a4260	a4360

Suffix Codes are available for Analog Peripherals (n=1..12):

Analog Peripheral n
a4000.n
a4010.n
a4020.n
a4030.n
a4040.n
a4050.n
a4060.n

2.1.15 Reports

a5000	Report Session 1Type 00 List 01 Table	Selection	R/W
a5002	Report Session 1Automatic 00 Off 01 On	Selection	R/W
a5004	Report Session 1 New Page Begin 00 Off 01 On	Selection	R/W
a5005	Report Session 1 New Page End 00 Off 01 On	Selection	R/W
a5010	Report Session 1 Header Contents 00 None 01 User Text #1 02 Product Name 04 Date 08 User Text #2 16 Line Name 32 Time 64 User Text #3 128 Length 256 Coil Number (only Session 1 and Session 2 reports)	Multiple	R/W
a5020	Report Session 1 Values Contents 00 None 01 Nominal 02 Limits Setup 04 Average 08 Alarms 16 Offset 32 PreAlarms Setup 64 Maximum 128 Std. Deviation 256 Minimum 512 CpK 1024 StripChart Maximum (only for Coils) 2048 StripChart Mean (only for Coils) 4096 StripChart Minimum (only for Coils)	Multiple	R/W
a5030	Report Session 1 Events Contents 00 None 01 Event Counters 02 Fault Detectors	Multiple	R/W

The codes for Session and Coil reports are as the next table:

Session 1	Session 2	Coil 1	Coil 2
a5000	a5050	a5100	a5150
a5002	a5052	a5102	a5152
a5004	a5054	a5104	a5154
a5005	a5055	a5105	a5155
a5010	a5060	a5110	a5160
a5020	a5070	a5120	a5170
a5030	a5080	a5130	a5180

a5200	Report SPC Summary Type 00 List 01 Table	Selection	R/W
a5202	Report SPC Summary Automatic 00 Off 01 On	Selection	R/W
a5204	Report SPC Summary New Page Begin 00 Off 01 On	Selection	R/W
a5205	Report SPC Summary New Page End 00 Off 01 On	Selection	R/W
a5210	Report SPC Summary Header Contents 00 None 01 User Text #1 02 Product Name 04 Date 08 User Text #2 16 Line Name 32 Time 64 User Text #3	Multiple	R/W
a5220	Report SPC Summary Values Contents 00 None 01 Cp 02 CpK 04 X bar 08 Range 16 Std. deviation 32 SPC UCL X bar 64 SPC LCL X bar 128 SPC UCL Range 256 Samples 512 Subgroups 1024 X double bar + 2 * standard deviation 2048 X double bar - 2 * standard deviation 4096 X double bar + 3 * standard deviation 8192 X double bar - 3 * standard deviation	Multiple	R/W
a5230	Report SPC Summary Alarms Contents 00 None 01 Upper UCL number 02 Lower LCL number 04 Upper UCL percentage 08 Lower LCL percentage 16 Total out of limits number 32 Total alarms number 64 Total alarms percentage	Multiple	R/W
a5252	Report SPC Charts Automatic 00 Off 01 On	Selection	R/W

a5260	Report SPC Charts Header Contents 00 None 01 User Text #1 02 Product Name 04 Date 08 User Text #2 16 Line Name 32 Time 64 User Text #3	Multiple	R/W
a5270	Report SPC Charts Values Contents 00 None 01 Cp 02 CpK 04 X bar 08 Range 16 Std. deviation 32 SPC UCL X bar 64 SPC LCL X bar 128 SPC UCL Range 256 Samples 512 Subgroups	Multiple	R/W
a5280	Report SPC Charts Selection 00 None 01 Histogram 02 X bar Range	Multiple	R/W
a5300	Report Session Log Type 00 Normal 01 Inverted	Selection	R/W
a5302	Report Session Log Automatic 00 Off 01 On	Selection	R/W
a5310	Report Session Log Header Contents 00 None 01 User Text #1 02 Product Name 04 Date 08 User Text #2 16 Line Name 32 Time 64 User Text #3 128 Length	Multiple	R/W
a5320	Report Session Log Values Contents 00 None 01 Alarms and Events 02 SPC Alarms 04 Periodic Data 08 Line Status 16 Pause	Multiple	R/W

2.1.16 FFT

a5510	Test Length x.xxx (voltage, fixed resolution)	Value	R/W
a5520	Window Type 00 None 01 Hanning 02 Hamming 03 Triangle 04 Blackman 05 Exact Blackman 06 Blackman-Harris 07 Flat Top 08 Super Gaussian	Selection	R/W
a5530	Frame Size xxx (fixed resolution. 4 to 8192)	Value	R/W

2.1.17 Software version

a9900	Software Version	String	R
a9910	Software Version Date	String	R

2.1.18 Examples

Define (GMT+01:00) Brussels, Copenhagen, Madrid, Paris as Time Zone:

```
A1031<SP>125<CR>
```

All default settings of this zone will be loaded automatically by USYS:

```
Time Zone          (GMT+01:00) Brussels, Copenhagen, Madrid, Paris
Standard Abbrev:  CET
DST:              On
DST Abbrev:       CEST
DST Start Week:   Last
DST Start Day:    Sunday
DST Start Month:  March
DST Start Hour:   02:00
DST End Week:     Last
DST End Day:      Sunday
DST End Month:    October
DST End Hour:     03:00
```

if you want to change them, you can use the next commands:

```
A1032<SP>+01<CR>    //Standard abbreviation
A1033<SP>1<CR>       //Activate Daylight Saving Time
A1034<SP>+02<CR>     //Daylight Saving Time abbreviation
A1035.1<SP>04<CR>    //Daylight Saving Time Start: forth
A1036.1<SP>06<CR>    //Daylight Saving Time Start: Saturday
A1037.1<SP>03<CR>    //Daylight Saving Time Start: April
A1038.1<SP>00:00<CR> //Daylight Saving Time Start: 00:00
A1034.2<SP>03<CR>    //Daylight Saving Time End: forth
A1035.2<SP>06<CR>    //Daylight Saving Time End: Saturday
A1036.2<SP>10<CR>    //Daylight Saving Time End: November
A1037.2<SP>02:00<CR> //Daylight Saving Time End: 02:00
```

Change SPC sampling

```
A1400<SP>0<CR>      // Sampling by Length
A1410<SP>100<CR>     // Sample interval of 100m
A1420<SP>5<CR>       // Samples per subgroup
F1110                // SPC Stop/Start
                    //      (see Group F - Control Functions)
```

2.2 GROUP B - LINE AND CONTROL

2.2.1 Controllers

b1020	Upper Limit x.xxxx (Trim Regulating Device Mode: no units, global resolution) x.xx (Absolute Regulating Device Mode: %, fixed resolution) x.xxx (Voltage Regulating Device Mode: voltage, fixed resolution)	Value	R/W
b1030	Lower Limit x.xxxx (Trim Regulating Device Mode: no units, global resolution) x.xx (Absolute Regulating Device Mode: %, fixed resolution) x.xxx (Voltage Regulating Device Mode: voltage, fixed resolution)	Value	R/W
b1050	Self Tuning 00 Off 01 On	Selection	R/W
b1070	Control limits x.xx (% , fixed resolution)	Value	R/W
b1080	Minimum averaging time x.x (seconds, fixed resolution)	Value	R/W
b1090	Minimum Dead Time x.x (seconds, fixed resolution)	Value	R/W
b1100	Controller correction factor x.xx (% , fixed resolution)	Value	R/W
b1110	Through Travel Time (Through Regulated Device) x.x (seconds, fixed resolution) Acceleration (SRD Regulating Device Type) x.x (seconds, fixed resolution) Relative Rate (Pulse, Extruder or Capstan Regulated Device) x.xx (%/second, fixed resolution) Absolute Rate (Pulse, other Regulated Devices) x.xxxx (global units/second, global resolution)	Value	R/W
b1120	Controller minimum line speed x.xx (% of nominal line speed, fixed resolution)	Value	R/W
b1150	Nominal SRD out (Heat or Gas controllers) x.xxx (voltage, fixed resolution)	Value	R/W
b1160	Full Travel (Through controller) x.xxxx (global units, global resolution)	Value	R/W
b1170	Manual Dead Band 00 Off 01 On	Selection	R/W

b1175	I Term 00 Disabled 01 Enabled	Selection	R/W
b1300	Gain Up modifier x.xxxx (global resolution)	Value	R
b1310	Gain Down modifier x.xxxx (global resolution)	Value	R
b1320	Averaging modifier x.xxxx (global resolution)	Value	R
b1330	Computed Dead band x.xxxx (global resolution)	Value	R
b1340	In/Out Ratio x.xxxx (global resolution)	Value	R
b1390	Controller Status 00 Off 01 Ready 02 Dead Time 03 Waiting For Object 04 Acquisition 05 No Action 06 Increase 07 Decrease 08 Waiting 09 Error	Selection	R

The codes for other controllers are as the next table:

Controller 1	Controller 2	Controller 3	Controller 4
b1020	b2020	b3020	b4020
b1030	b2030	b3030	b4030
b1050	b2050	b3050	b4050
b1070	b2070	b3070	b4070
b1080	b2080	b3080	b4080
b1090	b2090	b3090	b4090
b1100	b2100	b3100	b4100
b1110	b2110	b3110	b4110
b1120	b2120	b3120	b4120
b1150	b2150	b3150	b4150
b1160	b2160	b3160	b4160
b1170	b2170	b3170	b4170
b1175	b2175	b3175	b4175
b1300	b2300	b3300	b4300
b1310	b2310	b3310	b4310
b1320	b2320	b3320	b4320
b1330	b2330	b3330	b4330
b1340	b2340	b3340	b4340
b1390	b2390	b3390	b4390

2.2.2 Dual Loop

b5050	Self Tuning 00 Off 01 On	Selection	R/W
-------	--------------------------------	-----------	-----

b5070	Control limits (on Offset) x.xx (% of nominal, fixed resolution)	Value	R/W
b5100	Correction Factor x.xx (% , fixed resolution)	Value	R/W
b5300	DLP Gain Up modifier x.xxxx (global resolution)	Value	R
b5310	DLP Gain Down modifier x.xxxx (global resolution)	Value	R
b5320	DLP Averaging modifier x.xxxx (global resolution)	Value	R
b5330	DLP Computed Dead band x.xxxx (global resolution)	Value	R
b5350	Calculated Offset x.xxxx (global resolution)	Value	R
b5390	Dual Loop Status 00 Off 01 Ready 02 Dead Time 03 Start Acquisition 04 Acquisition 05 No Action 06 Increase 07 Decrease 08 Waiting 09 Error	Selection	R

The codes for other Dual Loops are as the next table:

Dual Loop 1	Dual Loop 2	Dual Loop 3	Dual Loop 4
b5050	b6050	b6550	b7050
b5070	b6070	b6570	b7070
b5100	b6100	b6600	b7100
b5300	b6300	b6800	b7300
b5310	b6310	b6810	b7310
b5320	b6320	b6820	b7320
b5330	b6330	b6830	b7330
b5350	b6350	b6850	b7350
b5390	b6390	b6890	b7390

2.2.3 Controller & Dual Loop Assignments

b8000	Activate and Save Assignments configuration	--	W
-------	---	----	---

Following group B orders modify only local configuration when Download. One order B8000 must be sent to activate and save changes.

b8100	Regulated Device 00 None 01 Extruder 02 Capstan 03 Trough 04 Heat 05 Offset	Selection	R/W
-------	---	-----------	-----

06 Pressure
07 Vacuum
08 Mechanical
09 Preheater
10 Gas
11 Conveyor
12 Other

b8110	Regulating Device Type 00 None 01 Pulse 02 SRD	Selection	R/W
b8120	Regulating Device Mode 00 Off 01 Trim 02 Absolute 03 Voltage	Selection	R/W

The codes for other controllers are as the next table:

Controller 1	Controller 2	Controller 3	Controller 4
b8100	b8200	b8300	b8400
b8110	b8210	b8310	b8410
b8120	b8220	b8320	b8420

b8500	DLP Characteristic 0 None xxxx Dual Loop combined characteristic code (see Appendix A)	Value	R
-------	--	-------	---

The codes for other Dual Loops are as the next table:

Dual Loop 1	Dual Loop 2	Dual Loop 3	Dual Loop 4
b8500	b8600	b8700	b8800

2.2.4 Line Layout

b9000	Activate Line (Activate line changes previously sent)	--	W
b9010	Line Name xxxxxxxxxxxxxxxx (15 characters)	Wide-String	R/W
b9020	Line Direction 00 Left 01 Right	Selection	R/W
b9100	Device #1 0001 Void 0010 Preheater 0100 Extruder 0105 Extruder CV 0107 Gravimetric Extruder 0110 Mechanical 0112 Grinding 0114 Peeling 0116 Rolling 0200 ODAC Single 0300 ODAC XY 0310 2 ODAC Singles Synchronized -> XY	Selection	R/W

0600 2 ODAC Singles -> Dual XY
 0650 2 ODAC XY -> Dual 2 XY (4 axes)
 0750 2 ODAC Singles Synchronized in Penetration Mode
 0770 ODAC CV J
 0780 ODAC TRIO
 0800 ODEX
 0900 MSD XY J
 0910 MSD XY RS
 0920 MSD XY/4 RS
 1000 Trough
 1150 CAPAC J
 1160 CAPAC RS
 1170 Analog CAPAC
 1200 UMAC
 1210 UMAC Basic
 1250 RAYEX D
 1255 RAYEX CV
 1260 RAYEX S
 1265 RAYEX D XT
 1301 SPARK TESTER IO
 1302 SPARK TESTER RS
 1401 KW IO
 1402 KW RS
 1500 Event Gauge
 1600 Analog Gauge
 1900 MARKER
 2000 Capstan
 2005 Conveyer
 2500 Puller
 2510 Accumulator
 3000 LINE END: Reel

b9102	Device #1 with Statistics 00 Off 01 On	Selection	R/W
b9115	Distance from Device #2 to previous one x.xxxx (length resolution)	Value	R/W

The codes for other devices are as the next table:

Dev. 1	Dev. 2	Dev. 3	Dev. 4	Dev. 5	Dev. 6	Dev. 7	Dev. 8	Dev. 9	Dev 10
b9100	b9110	b9120	b9130	b9140	b9150	b9160	b9170	b9180	b9190
b9102	b9112	b9122	b9132	b9142	b9152	b9162	b9172	b9182	b9192
	b9115	b9125	b9135	b9145	b9155	b9165	b9175	b9185	b9195
Dev 11	Dev 12	Dev 13	Dev 14	Dev 15	Dev 16	Dev 17	Dev 18	Dev 19	Dev 20
b9200	b9210	b9220	b9230	b9240	b9250	b9260	b9270	b9280	b9290
b9202	b9212	b9222	b9232	b9242	b9252	b9262	b9272	b9282	b9292
b9205	b9215	b9225	b9235	b9245	b9255	b9265	b9275	b9285	b9295

2.2.5 Length Detector

b9500	Length Detector 00 Off 01 On	Selection	R/W
b9510	Length Pulses x.xxxxxx (Length Counter resolution)	Value	R/W

b9520	Line Speed x.xxxx (length resolution, m/s or ft/s)	Value	R/W
-------	---	-------	-----

2.2.6 Accumulator

b9530	Length Pulses x.xxxxxx (Length Counter resolution)	Value	R/W
-------	---	-------	-----

2.2.7 Example

Modify and activate a new line configuration

This example modifies the current line configuration in order to increase/decrease the length of a trough.

First of all, we need to check the number of the device that follows the trough on the System- Line Layout screen. Let's say it is #6.

```
B9155<SP>110.0<CR>    // Distance from Device #6 to previous one
B9155<CR>              // Activate line
```

An important fact to take in account when new line is activated is that it forces the close of all currently opened statistical periods.

2.3 GROUP C - PRODUCT PARAMETERS

Using the group C messages you can change the set points and other measuring conditions of the USYS system.

There are two working modes with this group, one for reading current activated product and other for modifying this product and any other from products table. Next command is used for switching between both modes.

c0100	Product Working Mode 00 Work with Product Active 01 Work with Working Area	Selection	R/W
-------	--	-----------	-----

On active product mode, the default option, all parameters have only reading access, meaning downloads are not possible and write-only parameters are not allowed. Upload request will return the corresponding value to active product.

The Product modification mode is used when a product is going to be modified or activated. This mode is explained on next item.

2.3.1 Product modification

The HOST product modification parameters work on a special working area, used as a temporary buffer. Before making any product operations, the Host must gain access to a product recipe placing it on the working area, either loading it from USYS local table, or making a copy of the active product.

After making the desired changes, the HOST can save the product again on USYS local table and/or activate it. Be aware that most of the changes will reset the current statistics when activated. When the modified product is saved into the local product database the active product will remain unchanged.

In the case we do not want to introduce changes in an existing product, there is also a parameter to activate a product directly from the USYS product table.

At the reception of a download parameter USYS will make a preliminary check of the received value concerning only to the message format, and answers with <ACK> or <NAK>. At the reception of the Activation message it will validate in deep all the

messages received (for example checking the Tolerances coherence) and will send the answer to the HOST (<ACK> or <NAK>).

At the reception of any <NAK> HOST computer can ask for the type of the error by the means of the group E parameters. When activation parameters returns an error on download, and the error code on e1000 parameter is 203, it means that there are some validation problem on the product to activate. It is possible to know the exact problem using e1030 and e1035 parameters.

c1000	Activate product (Activate the product previously open)	--	W
c1010	Open Active Product (Make the product parameters visible from host)	--	W
c1020	Open Product from Table xxx Number of the table product to make visible from host	Value	W
c1025	Open Product from Table (by Name) xxxxx...xxxxx (20 characters)	Wide-String	W
c1030	Save Product to Table (Save the product previously open)	--	W
c1040	Activate Product from Table xxx Number of the product to activate	Value	W
c1050	Delete Product from Table xxx Number of the table product to make visible from host	Value	W
c1055	Delete Product from Table (by Name) xxxxx...xxxxx (20 characters)	Wide-String	W

2.3.2 Product Parameters

The meaning and access of group C parameters depends on working mode (see previous items). On active product mode, they contain current USYS active product. On modification mode, they contain the working area product. When selecting modification mode, into the working area will be loaded as default the current active product.

Remember that on active product mode only read access is allowed (descriptions show access for modification mode).

2.3.3 Not Characteristic related Product Parameters

c1100	Product number xxx (fixed resolution, 0 to 999)	Value	R/W
c1110	Product Name xxxxx...xxxxx (20 characters)	Wide-String	R/W
c1115	Product Description xxxxx...xxxxx (40 characters)	Wide-String	R/W
c1120	Comment 1 xxxxx...xxxxx (70 characters)	Wide-String	R/W
c1130	Comment 2 xxxxx...xxxxx (70 characters)	Wide-String	R/W

Suffix Codes are available for Comments (n=1..8):

Comment n
c1120.n

c1140	Nominal Line Speed x.xxxx (length resolution, m/s or ft/s)	Value	R/W
c1145	Nominal Reel Length x.xxxx (length resolution)	Value	R/W
c1150	SPC Control Limits 00 Entered 01 Calculated	Selection	R/W
c1160	Nominal Diameter Zero x.xxxx (global resolution)	Value	R/W
c1170	Global Prealarm value x.xx (% , fixed resolution)	Value	R/W
c1175	OD Meter Weight Density x.xxxx (Density resolution)	Value	R/W
c1176	OD Meter Weight Correction Factor x.xx (% , fixed resolution)	Value	R/W
c1180	Piece Start Inhibit Value x.xxxx (length resolution)	Value	R/W
c1190	Piece End Inhibit Value x.xxxx (length resolution)	Value	R/W
c1192	Custom Alarm Tolerance x.xxxx (Global resolution)	Value	R/W

Suffix Codes are available for Custom Alarms (n=1..4):

Custom Alarm n
c1192.n

c1195	Number of active Characteristics xx (integer, 0 to 10) Note: This is a read-only value. To add or delete characteristics use the order c2010.	Value	R
c1200	Wire Type (CM) 00 Solid 01 Foam 02 Foam / Skin 03 Skin / Foam / Skin	Selection	R/W
c1210	Process Control Characteristic (CM) 0 None 5000 .. 5099 Allowed Expansion Characteristic (see Appendix A)	Selection	R/W
c1230	Conductor Type (CM) 00 Solid 01 Stranded	Selection	R/W

c1240	Foam Mode (CM) 00 Chemical (Heat control) 01 Physical (Gas control)	Selection	R/W
c1350	Foam Gravity (CM) x.xx (grams, fixed resolution)	Value	R/W
c1351	Foam Dielectric (CM) x.xx (% , fixed resolution)	Value	R/W
c1360	Inner Skin Gravity (CM) x.xx (grams, fixed resolution)	Value	R/W
c1361	Inner Skin Dielectric (CM) x.xx (% , fixed resolution)	Value	R/W
c1362	Inner Skin Thickness (CM) x.xxxx (global resolution)	Value	R/W
c1370	Outer Skin Gravity (CM) x.xx (grams, fixed resolution)	Value	R/W
c1371	Outer Skin Dielectric (CM) x.xx (% , fixed resolution)	Value	R/W
c1372	Outer Skin Thickness (CM) x.xxxx (global resolution)	Value	R/W
c1380	Strand Coefficient (CM) x.xx (% , fixed resolution)	Value	R/W

2.3.4 Event related Product Parameters

c1500	Event 1 Name xxxxxxxxxxxxxxx (15 characters)	Wide-String	R/W
c1510	Event 1 Type code 0 None 1.. 99 Allowed Event (see Usys2000 codes on Appendix B) 100 .. 9999 Allowed Event (see Appendix B)	Value	R/W
c1521	Event 1 Reports Selection 00 None 01 Session #1 02 Session #2 04 Coil #1 08 Coil #2	Multiple	R/W

The codes for other events are as the next table:

Event 1	Event 2	Event 3	Event 4	Event 5
c1500	c1550	c1600	c1650	c1700
c1510	c1560	c1610	c1660	c1710
c1521	c1571	c1621	c1671	c1721

Suffix Codes are available for events (n=1..6).

Event n
c1500.n
c1510.n
c1521.n

2.3.5 Characteristic related Product Parameters

c2000	Characteristic 1 Name xxxxxxxxxxxxxxx (15 characters)	Wide-String	R/W
c2010	Characteristic 1 Type code 0 None 1.. 99 Allowed Characteristic (see Usys2000 codes on Appendix A) 100 .. 9999 Allowed Characteristic (see Appendix A)	Value	R/W
c2020	Characteristic 1 Strip Charts 00 Off 01 On	Selection	R/W
c2025	Characteristic 1 Statistics 00 Off 01 On	Selection	R/W
c2030	Characteristic 1 S.P.C. 00 Off 01 On	Selection	R/W
c2035	Characteristic 1 Alarms 00 Off 01 On	Selection	R/W
c2036	Characteristic 1 FFT 00 Off 01 On	Selection	R/W
c2037	Characteristic 1 SRL 00 Off 01 On	Selection	R/W
c2039	Characteristic 1 VSWR 00 Off 01 On	Selection	R/W
c2040	Characteristic 1 resolution List depending on the Characteristic type: Capacity characteristics: 00 0.5 pF/m or 0.2 pF/ft 01 0.2 pF/m or 0.1 pF/ft 02 0.1 pF/m or 0.05 pF/ft 03 0.05 pF/m or 0.02 pF/ft Voltage characteristics: 00 1 V 01 0.5 V 02 0.2 V 03 0.1 V	Selection	R/W

04 0.05 V
 05 0.02 V
 06 0.01 V
 07 0.005 V
 08 0.002 V
 09 0.001 V

Percentage characteristics:

00 1 %
 01 0.5 %
 02 0.2 %
 03 0.1 %
 04 0.05 %
 05 0.02 %
 06 0.01 %

Analog Gauge characteristics:

00 1
 01 0.5
 02 0.2
 03 0.1
 04 0.05
 05 0.02
 06 0.01
 07 0.005
 08 0.002
 09 0.001
 10 0.0005
 11 0.0002
 12 0.0001
 13 0.00005
 14 0.00002
 15 0.00001

Other characteristics:

00 0.1 mm or 0.005 inch
 01 0.05 mm or 0.002 inch
 02 0.02 mm or 0.001 inch
 03 0.01 mm or 0.0005 inch
 04 0.005 mm or 0.0002 inch
 05 0.002 mm or 0.0001 inch
 06 0.001 mm or 0.00005 inch
 07 0.0005 mm or 0.00002 inch
 08 0.0002 mm or 0.00001 inch
 09 0.0001 mm or 0.000005 inch

c2051	Characteristic 1 Reports Selection	Multiple	R/W
	00 None		
	01 Session #1		
	02 Session #2		
	04 Coil #1		
	08 Coil #2		
	16 S.P.C. Summary		
	32 S.P.C. Charts		
	64 Session Log		
c2100	Characteristic 1 Nominal x.xxxx (characteristic units and resolution)	Value	R/W

c2110	Characteristic 1 Offset x.xxxx (characteristic units and resolution)	Value	R/W
c2120	Characteristic 1 Upper Tolerance x.xxxx (characteristic units and resolution)	Value	R/W
c2130	Characteristic 1 Upper Pre-Alarm x.xxxx (characteristic units and resolution)	Value	R/W
c2140	Characteristic 1 Lower Tolerance x.xxxx (characteristic units and resolution)	Value	R/W
c2150	Characteristic 1 Lower Pre-Alarm x.xxxx (characteristic units and resolution)	Value	R/W
c2160	Characteristic 1 UCLx x.xxxx (characteristic units and resolution)	Value	R/W
c2170	Characteristic 1 LCLx x.xxxx (characteristic units and resolution)	Value	R/W
c2180	Characteristic 1 UCLr x.xxxx (characteristic units and resolution)	Value	R/W
c2190	Characteristic 1 Offset DLP x.xxxx (characteristic units and resolution)	Value	R/W

The codes for other characteristics are as the next table:

Char. 1	Char. 2	Char. 3	Char. 4	Char. 5	Char. 6	Char. 7	Char. 8	Char. 9	Ch. 10
c2000	c2500	c3000	c3500	c4000	c4500	c5000	c5500	c6000	c6500
c2010	c2510	c3010	c3510	c4010	c4510	c5010	c5510	c6010	c6510
c2020	c2520	c3020	c3520	c4020	c4520	c5020	c5520	c6020	c6520
c2025	c2525	c3025	c3525	c4025	c4525	c5025	c5525	c6025	c6525
c2030	c2530	c3030	c3530	c4030	c4530	c5030	c5530	c6030	c6530
c2035	c2535	c3035	c3535	c4035	c4535	c5035	c5535	c6035	c6535
c2036	c2536	c3036	c3536	c4036	c4536	c5036	c5536	c6036	c6536
c2037	c2537	c3037	c3537	c4037	c4537	c5037	c5537	c6037	c6537
c2039	c2539	c3039	c3539	c4039	c4539	c5039	c5539	c6039	c6539
c2040	c2540	c3040	c3540	c4040	c4540	c5040	c5540	c6040	c6540
c2050	c2550	c3050	c3550	c4050	c4550	c5050	c5550	c6050	c6550
c2100	c2600	c3100	c3600	c4100	c4600	c5100	c5600	c6100	c6600
c2110	c2610	c3110	c3610	c4110	c4610	c5110	c5610	c6110	c6610
c2120	c2620	c3120	c3620	c4120	c4620	c5120	c5620	c6120	c6620
c2130	c2630	c3130	c3630	c4130	c4630	c5130	c5630	c6130	c6630
c2140	c2640	c3140	c3640	c4140	c4640	c5140	c5640	c6140	c6640
c2151	c2651	c3151	c3651	c4151	c4651	c5151	c5651	c6151	c6651
c2160	c2660	c3160	c3660	c4160	c4660	c5160	c5660	c6160	c6660
c2170	c2670	c3170	c3670	c4170	c4670	c5170	c5670	c6170	c6670
c2180	c2680	c3180	c3680	c4180	c4680	c5180	c5680	c6180	c6680
c2190	c2690	c3190	c3690	c4190	c4690	c5190	c5690	c6190	c6690

Suffix Codes are available for characteristics (n=1..20).

Char. n
c2000.n
c2010.n
c2020.n
c2025.n
c2030.n
c2035.n
c2036.n
c2037.n
c2039.n
c2040.n
c2050.n
c2100.n
c2110.n
c2120.n
c2130.n
c2140.n
c2151.n
c2160.n
c2170.n
c2180.n
c2190.n

2.3.6 Controllers related Product Parameters

Foam Controllers (CM)

c7000	Control Priority 00 Diameter 01 Capacitance	Selection	R/W
c7010	Trough Control 00 Disabled 01 Enabled	Selection	R/W
c7015	Gas Control 00 MFC 01 Expansion	Selection	R/W
c7020	Trough Range (Length) x.xxxx (global units, global resolution)	Value	R/W
c7030	Trough Range (Capacitance) x.xx (% , fixed resolution)	Value	R/W
c7040	Heat Range (Capacitance) x.xx (% , fixed resolution)	Value	R/W
c7045	Gas Range (Capacitance) x.xx (% , fixed resolution)	Value	R/W
c7050	Control Nominal Inner Diameter x.xxxx (global units, global resolution)	Value	R/W
c7060	Control Weight Absolute Mode Ratio x.xx (% , fixed resolution)	Value	R/W
c7080	Foam Ctrl 1 Manual Upper Dead Band Value x.xxxx (capacitance units, capacitance resolution)	Value	R/W

c7081	Foam Ctrl 1 Manual Lower Dead Band Value x.xxxx (capacitance units, capacitance resolution)	Value	R/W
-------	--	-------	-----

Suffix Codes are available for foam Ctrl Dead Band Values (n=1..3):

Value n
c7080.n
c7081.n

Solid Controllers

c7105	Controller 1 Controlled Characteristic 00 None 01..08 Characteristic number	Selection	R/W
c7120	Controller 1 Control Based On 00 Average 01 Minimum 02 Maximum	Selection	R/W
c7130	Controller 1 CPK Pilot 00 "Off" 01 "3.0 : 1.00 CpK" 02 "3.5 : 1.16 CpK" 03 "4.0 : 1.33 CpK" 04 "4.5 : 1.50 CpK" 05 "5.0 : 1.67 CpK" 06 "5.5 : 1.83 CpK" 07 "6.0 : 2.00 CpK"	Selection	R/W
c7140	Controller 1 Absolute Mode Ratio x.xx (% , fixed resolution)	Value	R/W
c7150	Controller 1 Nominal Inner Diameter x.xxxx (global units, global resolution)	Value	R/W
c7160	Controller 1 Manual Upper Dead Band Value x.xxxx (global units, global resolution)	Value	R/W
c7170	Controller 1 Manual Lower Dead Band Value x.xxxx (global units, global resolution)	Value	R/W
c7180	Controller 1 SRD Range x.xxxx (global units, global resolution)	Value	R/W
c7190	Controller 1 Absolute Rate x.xxxx (characteristic units and resolution * 0.01)	Value	R/W

The codes for other controllers are as the next table:

Controller 1	Controller 2	Controller 3	Controller 4
c7105	c7205	c7305	c7405
c7120	c7220	c7320	c7420
c7140	c7240	c7340	c7440
c7150	c7250	c7350	c7450
c7160	c7260	c7360	c7460
c7170	c7270	c7370	c7470
c7180	c7280	c7380	c7480
c7190	c7290	c7390	c7490

Suffix Codes are available for Controllers (n=1..4):

Controller 1
c7105.n
c7120.n
c7140.n
c7150.n
c7160.n
c7170.n
c7180.n
c7190.n

2.3.7 Localization related Product Parameters

c8000	Printer User Text #1 xxxxx...xxxxx (65 characters)	Wide-String	R/W
c8010	Printer User Text #2 xxxxx...xxxxx (65 characters)	Wide-String	R/W
c8020	Printer User Text #3 xxxxx...xxxxx (65 characters)	Wide-String	R/W

2.3.8 Devices related Product Parameters

c8100	Spark Tester Voltage kVAC or kVDC x.xxx (kV, fixed resolution)	Value	R/W
c8110	Spark Tester CI-SP/RS(Generic & Series A) Tolerance x.xxx (kV, fixed resolution)	Value	R/W
c8120	Spark Tester CI-SP/RS(Series A) Sensitivity x.xxx (kV, fixed resolution)	Value	R/W
c8130	Spark Tester CI-SP/RS(Series A) Min Bare Patch Span x.xxx (kV, fixed resolution)	Value	R/W
c8140	Spark Tester CI-SP/RS(Series A) Test Frequency x (fixed resolution)	Value	R/W

Suffix Codes are available for Spark Tester Gauges (n=1..6):

Spark Tester n
c8100.n
c8110.n
c8120.n
c8130.n
c8140.n

c8150	Preheater WST Nominal Temperature x.xxxx (global units, global resolution)	Value	R/W
c8152	Preheater WST Temperature Offset x.xxxx (global units, global resolution)	Value	R/W
c8154	Preheater WST Wire cross section x.xxxx (global units, global resolution)	Value	R/W

c8156	Preheater WST Wire Diameter x.xxxx (global units, global resolution)	Value	R/W
c8158	Preheater WST Material Material Identifier	Value	R/W
c8160	Preheater WST Material Electrical Resistivity x.x (Fixed resolution)	Value	R/W
c8162	Preheater WST Material Temperature Coefficient x.xxxx (Fixed resolution)	Value	R/W
c8164	Preheater WST Material Specific Heat Capacity x.x (Fixed resolution)	Value	R/W
c8166	Preheater WST Material Density x.xxxx (global units, global resolution)	Value	R/W
c8200	Lump/Neck Level (Lump Level if KW32) x.xxxx (global units, global resolution)	Value	R/W
c8210	Neck Level (if KW32) x.xxxx (global units, global resolution)	Value	R/W

Suffix Codes are available for KW Gauges (n=1..6):

KW n
c8200.n
c8210.n

c8250	Accumulator Length (if Accumulator) x.xxxx (global units, global resolution)	Value	R/W
c8401	OD Gauge 1 Acquisition Mode X 00 Average 01 Minimum 02 Maximum	Value	R/W
c8402	OD Gauge 1 Acquisition Mode Y 00 Average 01 Minimum 02 Maximum	Value	R/W
c8403	OD Gauge 1 Acquisition Mode Z 00 Average 01 Minimum 02 Maximum	Value	R/W
c8404	OD Gauge 1 Acquisition Mode T 00 Average 01 Minimum 02 Maximum	Value	R/W
c8409	OD Gauge 1 Stranded 00 Off 01 On	Value	R/W
c8410	OD Gauge 1 Lump Level x.xxxx (global units, global resolution)	Value	R/W

c8411	OD Gauge 1 Abs Lump Level x.xxxx (global units, global resolution)	Value	R/W
c8415	OD Gauge 1 Neck Level x.xxxx (global units, global resolution)	Value	R/W
c8416	OD Gauge 1 Abs Neck Level x.xxxx (global units, global resolution)	Value	R/W
c8417	OD Gauge Measured Objects XX (1 to MaxWires)	Value	R/W
c8418	OD Gauge 1 Statistical Period Based On 00 Length 01 Time	Selection	R/W
c8419	OD Gauge 1 Statistical Period Size x.xxxx (length resolution, Length units if interval unit is 00) x.x (fixed resolution, seconds if interval unit is 01)	Value	R/W

The codes for other OD gauges are as the next table:

OD 1	OD 2	OD 3	OD 4	OD 5	OD 6
c8401	c8421	c8441	c8461	c8481	c8501
c8402	c8422	c8442	c8462	c8482	c8502
c8403	c8423	c8443	c8463	c8483	c8503
c8404	c8424	c8444	c8464	c8484	c8504
c8409	c8429	c8449	c8469	c8489	c8509
c8410	c8430	c8450	c8470	c8490	c8510
c8411	c8431	c8451	c8471	c8491	c8511
c8415	c8435	c8455	c8475	c8495	c8515
c8416	c8436	c8456	c8476	c8496	c8516
c8417	c8437	c8457	c8477	c8497	c8517
c8418	c8438	c8458	c8478	c8498	c8518
c8419	c8439	c8459	c8479	c8499	c8519

Suffix Codes are available for OD Gauges (n=1..8).

OD n
c8409.n
c8410.n
c8411.n
c8415.n
c8416.n
c8417.n
c8418.n
c8419.n
c8420.n
c8421.n
c8422.n
c8423.n

c8585	OD Gauge 1 Stranded - Number of Strands xx (2 to 10, fixed resolution.)	Value	R/W
c8590	OD Gauge 1 Stranded - Lay Length x.xxxx (length resolution, m/s or ft/s)	Value	R/W

c8592	OD Gauge 1 Stranded - Acquisition Mode 00 Minimum 01 Maximum	Selection	R/W
c8595	OD Gauge 1 Stranded - Averaging Time xx (equal or bigger than 1, fixed resolution)	Value	R/W
UMAC			
c8650	Select Current Transducer (1 .. 8)	Value	R/W
c8651	Select Current Layer (1.. Max Physical Layer) 1 is the outer physical Layer (water)	Value	R/W
c8652	Extended Layer 00 Simple 01 Extended	Selection	R
c8653	Extended Transducer 00 Simple 01 Extended	Selection	R
c8654	Number of Active Transducers XX (Global Resolution)	Value	R
c8660	Number of Measured Layer XX (Global Resolution)	Value	R/W

Select current layer (C8651) before to execute a command.

Select current transducer (C8650) before to execute a command in extended mode.

c8661	Nominal WallThickness x.xxxx (Global Units, Global Resolution)	Value	R/W
c8662	Layer Material XX (Index of the material)	Selection	R/W
c8663	UMAC Twist Angle Rotation Sense 00 Counter-clockwise 01 Clockwise	Selection	R/W
c8664	Twist Angle x.xxxx (global units, global resolution)	Value	R/W

The codes for other UMAC gauges are as the next table:

UMAC 1	UMAC 2
c8652	c8752
c8653	c8753
c8654	c8754
c8660	c8760
c8661	c8761
c8662	c8762
c8663	c8763
c8664	c8764

UMAC Compensation

c8665 Wallthickness Offset Correction Value R/W
x.xxxx (global units, global resolution)

Suffix Codes are available for Other layers (n=1..Max Physical Layer):

UMAC 1 Compensation n	UMAC 2 Compensation n
c8665.n	c8765.n

Note: Select the desired transducer (c8650) before using this command if the compensation's method is Manual and the Manual's configuration is individual by transducer (a2765's value is 01).

c8667 Outer diameter Offset Correction Value R/W
x.xxxx (global units, global resolution)

UMAC 1	UMAC 2
c8667	c8767

Note: Select the first transducer number of the desired pair of transducers using c8650 before using this command if the compensation's method is Manual and the Manual's configuration is individual by transducer (a2765's value is 01).

UMAC Basic

c8666 Sound Velocity (first layer) Value R/W
x.xxxx (global units, global resolution)

c8670 Measure deadtime Value R/W
xx (ns)

UMAC Basic 1	UMAC Basic 2
c8666	C8766
c8670	c8770

Gravimetric Gauge

c8800 Gravimetric Extruder #1 Throughput Value R/W
xx (Throughput resolution, Kg/h, Gr/min or Pounds/hour)

c8801 Gravimetric Extruder #1 Weight Value R/W
xx (Weight resolution, g/m, g/3ft or Lb/3ft)

c8802 Gravimetric Extruder #1 Density Value R/W
xx (Density resolution, Kg/m³ or Kg/3ft³)

The codes for other Gravimetric gauges are as the next table:

Gravimetric #1	Gravimetric #2	Gravimetric #3	Gravimetric #4
c8800	c8810	c8820	c8830
c8801	c8811	c8821	c8831
c8802	c8812	c8822	c8832

ODEX

c8840 ODEX 1 Lump Value Value R/W
xx (global units, global resolution)

c8841	ODEX 1 Neck Value xx (global units, global resolution)	Value	R/W
c8842	ODEX 1 Inner Diameter xx (global units, global resolution)	Value	R/W
c8843	ODEX 1 Twist Angle Rot. Sense 00 Counter-clockwise 01 Clockwise	Selection	R/W
c8844	ODEX 1 Twist Angle Value x.xxxx (global units, global resolution)	Value	R/W
c8845	ODEX 1 Statistical Period Based On 00 Length 01 Time	Selection	R/W
c8846	ODEX 1 Statistical Period Size x.xxxx (length resolution, Length units if interval unit is 00) x.x (fixed resolution, seconds if interval unit is 01)	Value	R/W
c8847	ODEX 1 Abs Lump Value xx (global units, global resolution)	Value	R/W
c8848	ODEX 1 Abs Neck Value xx (global units, global resolution)	Value	R/W
c8849	ODEX 1 Auto-ref Min Speed x.xx (% of line speed, fixed resolution)	Value	R/W

Suffix Codes are available for ODEX Gauges (n=1..6):

ODEX n
c8840.n
c8841.n
c8842.n
c8843.n
c8844.n
c8845.n
c8846.n
c8847.n
c8848.n
c8849.n

RAYEX D

c8900	RAYEX OSL Layer Material Material Identifier	String	R/W
c8901	RAYEX OSL Absorption Coefficient xx (%)	Value	R/W
c8902	RAYEX OSL Shrinkage Factor xx (%)	Value	R/W
c8905	RAYEX ILL Layer Material Material Identifier	String	R/W
c8906	RAYEX ILL Absorption Coefficient xx (%)	Value	R/W

c8907	RAYEX ILL Shrinkage Factor xx (%)	Value	R/W
c8910	RAYEX ISL Layer Material Material Identifier	String	R/W
c8911	RAYEX ISL Absorption Coefficient xx (%)	Value	R/W
c8912	RAYEX ISL Shrinkage Factor xx (%)	Value	R/W
c8915	RAYEX CoL Layer Material Material Identifier	String	R/W
c8916	RAYEX CoL Absorption Coefficient xx (%)	Value	R/W
c8950	Shrinkage Characteristic Type code 0 None 1.. 99 Allowed Characteristic (see Usys2000 codes on Appendix A) 100 .. 9999 Allowed Characteristic (see Appendix A)	Value	R/W
c8955	RAYEX Shrinkage Offset xx (global units, global resolution, Current Offset Type)	Value	R/W
c8956	RAYEX Twist Angle Rotation Sense 00 Counter-clockwise 01 Clockwise	Selection	R/W
c8957	RAYEX Twist Angle x.xxxx (global units, global resolution)	Value	R/W
RAYEX S/D XT			
c8960	RAYEX S/D XT Layer 1 Material Material Identifier	String	R/W
c8961	RAYEX S/D XT Layer 1 Absorption x.xxxx (fixed resolution)	Value	R/W
c8962	RAYEX S/D XT Layer 1 Shrinkage Factor x.xx (% , fixed resolution)	Value	R/W
c8963	RAYEX D XT Layer 1 Fixed Absorption 00 Yes 01 No (Not available for RAYEX S)	Selection	R/W

Suffix Codes are available for other Layers (n=1..5):

Layer n
c8960.n
c8961.n
c8962.n
c8963.n

c8965	RAYEX S/D XT Shrinkage Charact. Type code 0 None 1.. 99 Allowed Characteristic (see Usys2000 codes on Appendix A) 100 .. 9999 Allowed Characteristic (see Appendix A)	Value	R/W
c8966	RAYEX S/D XT Shrinkage Offset xx (global units, global resolution, Current Offset Type)	Value	R/W
c8967	Number of Measured Layer xx (Global Resolution)	Value	R/W
c8968	RAYEX S/D XT Twist Angle Rotation Sense 00 Counter-clockwise 01 Clockwise	Selection	R/W
c8969	RAYEX S/D XT Twist Angle x.xxxx (global units, global resolution)	Value	R/W
Weight Cutter			
c8970	Weight Cutter Desired Weight x.xxxx (kg or Lb units, Density resolution)	Value	R/W
c8971	Weight cutter Specific Weight x.xxxx (Density units, Density resolution)	Value	R/W
c8972	Weight Cutter Recoil Length x.xxxx (Length units, fixed resolution 0.0001)	Value	R/W
Host Gauge			
c8980	Host Gauge #1 Density Value xx (Density resolution, Kg/m ³ or Kg/3ft ³)	Value	R/W

Suffix Codes are available for other Host Gauges (n=1..4):

Host Gauge n
c8980.n

2.3.9 Examples

Activate product from table

```
C1040<SP>6<CR>          // Activate product #6 directly from table
```

Modify a table product

```
C0100<SP>1<CR>          // Open work area with current active
                          // product
C1020<SP>6<CR>          // Load product #6 into the work area
C2100.1<SP>6.2<CR>      // Set characteristic #1 Nominal.
C1030<CR>              // Save product to table
C0100<SP>0<CR>          // Close work area to avoid modifications on
                          // product
```

Modify and activate UMAC product (3 Measured layers)

```
C0100<SP>1<CR>          // Open work area with current active
                          // product
C8651<SP>2<CR>          // Select layer to be modified (1 is the
                          // outer physical Layer (water), 2 is the
                          // first outer measured layer... )
```

```

C8661<SP>6.45<CR>    // Set Nominal WallThickness for selected
                        // layer
C8651<SP>3<CR>        // Select layer to be modified
C8661<SP>0.35<CR>    // Set Nominal WallThickness for selected
                        // layer
C8651<SP>4<CR>        // Select layer to be modified
C8661<SP>1.8<CR>    // Set Nominal WallThickness for selected
                        // layer
C1000<CR>            // Activate this product
C0100<SP>0<CR>        // Close work area to avoid modifications on
                        // product

```

UMAC Manual Compensation

```

C0100<SP>1<CR>        // Open work area with current active
                        // product
C8665.1<SP>0.12<CR>    // Set correction for first measured layer
C1000<SP>1<CR>        // Activate this product
C0100<SP>0<CR>        // Close work area to avoid modifications on
                        // product

```

UMAC Manual Compensation (Individual by transducer, 8 transducers)

```

C0100<SP>1<CR>        // Open work area with current active
                        // product
C8650<SP>3<CR>        // Select transducer to be modified
C8665.1<SP>0.12<CR>    // Set correction for first measured layer
C1000<SP>1<CR>        // Activate this product
C0100<SP>0<CR>        // Close work area to avoid modifications on
                        // product

```

UMAC Manual Outer Diameter Compensation

```

C0100<SP>1<CR>        // Open work area with current active
                        // product
C8667<SP>0.02<CR>    // Set outer diameter correction
C1000<SP>1<CR>        // Activate this product
C0100<SP>0<CR>        // Close work area to avoid modifications on
                        // product

```

UMAC Manual Outer Diameter Compensation (Individual by transducer, 8 transducers)

```

C0100<SP>1<CR>        // Open work area with current active
                        // product
C8650<SP>2<CR>        // Select the first transducer number of the
                        // desired pair of transducers [2,6]
C8667<SP>0.02<CR>    // Set outer diameter correction
C1000<SP>1<CR>        // Activate this product
C0100<SP>0<CR>        // Close work area to avoid modifications on
                        // product

```

Create a new RAYEX S product

This example opens the product modification mode, sets the main commands of new product (the rest of product values remain unchanged from last active one) and finally saves the product into table and activates it.

```

C0100<SP>1<CR>        // Open work area with current
                        // active product
C1110<SP>Product Example<CR> // Set product name, used as file
                        // name in table
C1115<SP>This is an example with 7 characteristics<CR> // Set
                        //product description
C1140<SP>10.5<CR>    // Set nominal line speed

```

```

C2010.0<SP>0<CR>           // Clear all characteristics
C2010.1<SP>18400<CR>        // Characteristic 1: OD Diameter
                               // Average Overall
C2010.2<SP>18411<CR>        // Characteristic 2: WallThickness
                               // Average
C2010.3<SP>18413<CR>        // Characteristic 3: WallThickness
                               // Minimum
C2010.4<SP>18418<CR>        // Characteristic 4: Eccentricity
                               // Shift of Center Overall
C2010.5<SP>18450<CR>        // Characteristic 5: WallThickness
                               // Average L1
C2010.6<SP>18500<CR>        // Characteristic 6: WallThickness
                               // Average L2
C2010.7<SP>18550<CR>        // Characteristic 7: WallThickness
                               // Average L3
C2040.1<SP>3<CR>            // Characteristic 1: Resolution 0.01
C2040.2<SP>3<CR>            // Characteristic 2: Resolution 0.01
C2040.3<SP>3<CR>            // Characteristic 3: Resolution 0.01
C2040.4<SP>6<CR>            // Characteristic 4: Resolution 0.001
C2040.5<SP>3<CR>            // Characteristic 5: Resolution 0.01
C2040.6<SP>3<CR>            // Characteristic 6: Resolution 0.01
C2040.7<SP>3<CR>            // Characteristic 7: Resolution 0.01
C2100.1<SP>35.00<CR>        // Characteristics 1 Nominal value
C2120.1<SP>35.20<CR>        // Characteristics 1: Upper tolerance
C2140.1<SP>34.80<CR>        // Characteristics 1: Lower tolerance
C2120.2.3<SP>12.5<CR>       // Characteristics 2&3: Nominal value
C2120.2.3<SP>35.20<CR>     // Characteristics 2&3: Upper
                               // tolerance
C2140.2.3<SP>34.80<CR>     // Characteristics 2&3: Lower
                               // tolerance
C2100.3<SP>0.1<CR>          // Characteristic 4: Nominal value
C2120.3<SP>0.15<CR>         // Characteristic 4: Upper tolerance
C2100.5<SP>1.2<CR>          // Characteristics 5 Nominal value
C2120.5<SP>1.3<CR>          // Characteristics 5: Upper tolerance
C2120.5<SP>1.1<CR>          // Characteristics 5: Lower tolerance
C2100.6<SP>10.0<CR>         // Characteristics 6 Nominal value
C2120.6<SP>10.1<CR>         // Characteristics 6: Upper tolerance
C2120.6<SP>9.9<CR>          // Characteristics 6: Lower tolerance
C2100.7<SP>1.3<CR>          // Characteristics 7 Nominal value
C2120.7<SP>1.4<CR>          // Characteristics 7: Upper tolerance
C2120.7<SP>1.2<CR>          // Characteristics 7: Lower tolerance
C8960.1<SP>57<CR>           // Layer 1 Material: EPR
C8960.2<SP>39<CR>           // Layer 2 Material: XLPE
C8960.3<SP>57<CR>           // Layer 2 Material: EPR
C8960.4<SP>57<CR>           // Layer 2 Material: Copper
C1030<CR>                   // Save product on table as "Product
                               // Example.PRD"
C1000<CR>                   // Activate this product
C0100<SP>0<CR>              // Close work area to avoid
                               // modifications on product

```

2.4 GROUP P - FFT PRODUCT PARAMETERS

FFT settings

p1000	Characteristic 1 FFT Cut Off x.xxxx (Vp resolution)	Value	R/W
p1010	Characteristic 1 FFT Max. Amplitude x.xxxx (Unit and resolution in dB)	Value	R/W

p1012	Characteristic 1 FFT Max. Amplitude x.xxxx (Unit and resolution in characteristic units)	Value	R/W
p1020	Characteristic 1 FFT Min. Amplitude x.xxxx (Unit and resolution in dB)	Value	R/W
p1022	Characteristic 1 FFT Min. Amplitude x.xxxx (Unit and resolution in characteristic units)	Value	R/W
p1030	Characteristic 1 FFT Num. Zones xx (Fixed Resolution, 1 to 10)	Value	R/W
p1040	Characteristic 1 FFT Zone 1 Min. Frequency x.xxxx (Hz units, Frequency resolution)	Value	R/W
p1050	Characteristic 1 FFT Zone 1 Max. Frequency x.xxxx (Hz units, Frequency resolution)	Value	R/W
p1060	Characteristic 1 FFT Zone 1 Alarm x.xxxx (Unit and resolution in dB)	Value	R/W
p1070	Characteristic 1 FFT Zone 1 PreAlarm x.xxxx (Unit and resolution in dB)	Value	R/W
p1080	Characteristic 1 FFT Zone 1 Alarm x.xxxx (Unit and resolution in characteristic units)	Value	R/W
p1090	Characteristic 1 FFT Zone 1 PreAlarm x.xxxx (Unit and resolution in characteristic units)	Value	R/W

Suffix Codes are available for characteristics (n=1..20).

Zone 1 Char. n	Zone 2 Char. n	Zone 3 Char. n	Zone 4 Char. n	Zone 5 Char. n	Zone 6 Char. n	Zone 7 Char. n	Zone 8 Char. n	Zone 9 Char. n	Zone 10 Char. n
p1040.n	p1041.n	p1042.n	p1043.n	p1044.n	p1045.n	p1046.n	p1047.n	p1048.n	p1049.n
p1050.n	p1051.n	p1052.n	p1053.n	p1054.n	p1055.n	p1056.n	p1057.n	p1058.n	p1059.n
p1060.n	p1061.n	p1062.n	p1063.n	p1064.n	p1065.n	p1066.n	p1067.n	p1068.n	p1069.n
p1070.n	p1071.n	p1072.n	p1073.n	p1074.n	p1075.n	p1076.n	p1077.n	p1078.n	p1079.n
p1080.n	p1081.n	p1082.n	p1083.n	p1084.n	p1085.n	p1086.n	p1087.n	p1088.n	p1089.n
p1090.n	p1091.n	p1092.n	p1093.n	p1094.n	p1095.n	p1096.n	p1097.n	p1098.n	p1099.n

p.e: p1000.5 Characteristic 5 FFT Cut Off.
p1040.2 Characteristic 2 FFT Zone 1 Min. Frequency.
p1053.4 Characteristic 4 FFT Zone 4 Max. Frequency.

SRL settings

p1200	Characteristic 1 SRL Cut Off x.xxxx (Vp resolution)	Value	R/W
p1210	Characteristic 1 SRL Max. Amplitude x.xxxx (Unit and resolution is dB)	Value	R/W
p1220	Characteristic 1 SRL Min. Amplitude x.xxxx (Unit and resolution is dB)	Value	R/W

p1222	Characteristic 1 SRL Velocity of propagation x.xxxx (global units, Vp resolution)	Value	R/W
p1224	Characteristic 1 SRL Inner Diameter x.xxxx (global units, global resolution)	Value	R/W
p1226	Characteristic SRL Offset (KSrl) x.xxxx (Vp resolution)	Value	R/W
p1230	Characteristic 1 SRL Num. Zones xx (Fixed Resolution, 1 to 10)	Value	R/W
p1240	Characteristic 1 SRL Zone 1 Min. Frequency x.xxxx (MHz units, Frequency resolution)	Value	R/W
p1250	Characteristic 1 SRL Zone 1 Max. Frequency x.xxxx (MHz units, Frequency resolution)	Value	R/W
p1260	Characteristic 1 SRL Zone 1 Alarm x.xxxx (Unit and resolution is dB)	Value	R/W
p1270	Characteristic 1 SRL Zone 1 PreAlarm x.xxxx (Unit and resolution is dB)	Value	R/W

Suffix Codes are available for characteristics (n=1..20).

Zone 1 Char. n	Zone 2 Char. n	Zone 3 Char. n	Zone 4 Char. n	Zone 5 Char. n	Zone 6 Char. n	Zone 7 Char. n	Zone 8 Char. n	Zone 9 Char. n	Zone 10 Char. n
p1240.n	p1241.n	p1242.n	p1243.n	p1244.n	p1245.n	p1246.n	p1247.n	p1248.n	p1249.n
p1250.n	p1251.n	p1252.n	p1253.n	p1254.n	p1255.n	p1256.n	p1257.n	p1258.n	p1259.n
p1260.n	p1261.n	p1262.n	p1263.n	p1264.n	p1265.n	p1266.n	p1267.n	p1268.n	p1269.n
p1270.n	p1271.n	p1272.n	p1273.n	p1274.n	p1275.n	p1276.n	p1277.n	p1278.n	p1279.n

p.e: p1200.5 Characteristic 5 SRL Cut Off.
p1240.2 Characteristic 2 SRL Zone 1 Min. Frequency.
p1253.4 Characteristic 4 SRL Zone 4 Max. Frequency.

VSWR settings

p1400	Characteristic 1 VSWR Cut Off x.xxxx (Vp resolution)	Value	R/W
p1410	Characteristic 1 VSWR Max. Amplitude x.xxxx (Unit and resolution is dB)	Value	R/W
p1420	Characteristic 1 VSWR Min. Amplitude x.xxxx (Unit and resolution is dB)	Value	R/W
p1422	Characteristic 1 VSWR Velocity of propagation x.xxxx (global units, Vp resolution)	Value	R/W
p1424	Characteristic 1 VSWR Inner Diameter x.xxxx (global units, global resolution)	Value	R/W
p1426	Characteristic VSWR Offset (KSWR) x.xxxx (Vp resolution)	Value	R/W
p1430	Characteristic 1 VSWR Num. Zones xx (Fixed Resolution, 1 to 10)	Value	R/W

p1440	Characteristic 1 VSWR Zone 1 Min. Frequency x.xxxx (MHz units, Frequency resolution)	Value	R/W
p1450	Characteristic 1 VSWR Zone 1 Max. Frequency x.xxxx (MHz units, Frequency resolution)	Value	R/W
p1460	Characteristic 1 VSWR Zone 1 Alarm x.xxxx (Unit and resolution is dB)	Value	R/W
p1470	Characteristic 1 VSWR Zone 1 PreAlarm x.xxxx (Unit and resolution is dB)	Value	R/W

Suffix Codes are available for characteristics (n=1..20).

Zone 1 Char. n	Zone 2 Char. n	Zone 3 Char. n	Zone 4 Char. n	Zone 5 Char. n	Zone 6 Char. n	Zone 7 Char. n	Zone 8 Char. n	Zone 9 Char. n	Zone 10 Char. n
p1440.n	p1441.n	p1442.n	p1443.n	p1444.n	p1445.n	p1446.n	p1447.n	p1448.n	p1449.n
p1450.n	p1451.n	p1452.n	p1453.n	p1454.n	p1455.n	p1456.n	p1457.n	p1458.n	p1459.n
p1460.n	p1461.n	p1462.n	p1463.n	p1464.n	p1465.n	p1466.n	p1467.n	p1468.n	p1469.n
p1470.n	p1471.n	p1472.n	p1473.n	p1474.n	p1475.n	p1476.n	p1477.n	p1478.n	p1479.n

p.e: p1400.5 Characteristic 5 VSWR Cut Off.
p1440.2 Characteristic 2 VSWR Zone 1 Min. Frequency.
p1453.4 Characteristic 4 VSWR Zone 4 Max. Frequency.

2.5 GROUP D - CURRENT MEASUREMENTS AND STATISTICS

2.5.1 Controlled Characteristics related Current Values

d1100	Controller 1 Characteristic Current Value x.xxxx (characteristic units and resolution)	Value	R
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The codes for other controllers are as the next table:

Controller 1	Controller 2	Controller 3	Controller 4
d1100	d1200	d1300	d1400

Suffix Codes are available for controllers (n=1..4).

Controller n
d1100.n

d1500	DLP 1Characteristic Cold Value x.xxxx (characteristic units and resolution)	Value	R
d1520	DLP 1Characteristic Calculated offset x.xxxx (characteristic units and resolution)	Value	R
d1530	DLP 1Characteristic Hot Value x.xxxx (characteristic units and resolution)	Value	R

The codes for other Dual Loops are as the next table:

Dual Loop 1	Dual Loop 2	Dual Loop 3	Dual Loop 4
d1500	d1600	d1700	d1800
d1520	d1620	d1720	d1820

d1530	d1630	d1730	d1830
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Suffix Codes are available for Dual Loops (n=1..4).

Dual Loop n
d1100.n
d1520.n
d1530.n

2.5.2 Characteristic related Current Values

Characteristic 1 Current Values

d2000	Current Value x.xxxx (characteristic units and resolution)	Value	R
d2005	Last Current Value x.xxxx (characteristic units and resolution)	Value	R
d2010	Current Deviation x.xxxx (characteristic units and resolution)	Value	R
d2105	Current Value average latch x.xxxx (characteristic units and resolution)	Value	R
d2115	Current Value maximum latch x.xxxx (characteristic units and resolution)	Value	R
d2125	Current Value minimum latch x.xxxx (characteristic units and resolution)	Value	R
d2135	Current Value standard deviation latch x.xxxx (characteristic units and resolution)	Value	R
d2181	Current Value Cpk latch x.xxxx (characteristic resolution)	Value	R
d2184	Current Value Cp latch x.xxxx (characteristic resolution)	Value	R

Current value: Is the last characteristic calculated value. When any error makes characteristic not to be calculated, it returns <NAK>.

Last value: Is the last characteristic value inserted into statistics at line end. While characteristic is on error, it returns the last valid value introduced into statistics. It only returns <NAK> after start of coil and before first value into statistics.

Latch value: Statistic value calculated with all characteristic values accumulated since the last time this value was uploaded. The statistic is reset at each coil start. For example, the d2115 command will return the maximum value of the first characteristic calculated since the last time d2115 was uploaded. These commands are not available for OPC servers.

Characteristic 1 Current Coil Values

d2100	Current coil average x.xxxx (characteristic units and resolution)	Value	R
d2110	Current coil maximum x.xxxx (characteristic units and resolution)	Value	R

d2120	Current coil minimum x.xxxx (characteristic units and resolution)	Value	R
d2130	Current coil standard deviation x.xxxx (characteristic units and resolution)	Value	R
d2140	Current coil high alarms xx (fixed resolution)	Value	R
d2150	Current coil low alarms xx (fixed resolution)	Value	R
d2160	Current coil high pre-alarms xx (fixed resolution)	Value	R
d2170	Current coil low pre-alarms xx (fixed resolution)	Value	R
d2179	Current coil Cp x.xxxx (characteristic resolution)	Value	R
d2180	Current coil Cpk x.xxxx (characteristic resolution)	Value	R
Characteristic 1 Current Session Values			
d2200	Current session average x.xxxx (characteristic units and resolution)	Value	R
d2210	Current session maximum x.xxxx (characteristic units and resolution)	Value	R
d2220	Current session minimum x.xxxx (characteristic units and resolution)	Value	R
d2230	Current session standard deviation x.xxxx (characteristic units and resolution)	Value	R
d2240	Current session high alarms xx (fixed resolution)	Value	R
d2250	Current session low alarms xx (fixed resolution)	Value	R
d2260	Current session high pre-alarms xx (fixed resolution)	Value	R
d2270	Current session low pre-alarms xx (fixed resolution)	Value	R
d2279	Current session Cp x.xxxx (characteristic resolution)	Value	R
d2280	Current session Cpk x.xxxx (characteristic resolution)	Value	R
Characteristic 1 Current SPC Values			
d2300	Current SPC - X double bar x.xxxx (characteristic units and resolution)	Value	R

d2310	Current SPC - R bar x.xxxx (characteristic units and resolution)	Value	R
d2320	Current SPC - Standard deviation x.xxxx (characteristic units and resolution)	Value	R
d2330	Current SPC - Cpk x.xxxx (characteristic resolution)	Value	R
d2340	Current SPC - Cp x.xxxx (characteristic resolution)	Value	R
d2350	Current SPC - # of subgroups xx (fixed resolution)	Value	R
d2360	Current SPC - # of samples xx (fixed resolution)	Value	R
d2370	Current SPC - UCLx x.xxxx (characteristic units and resolution)	Value	R
d2380	Current SPC - LCLx x.xxxx (characteristic units and resolution)	Value	R
d2390	Current SPC - UCLr x.xxxx (characteristic units and resolution)	Value	R
d2400	Current SPC – X bar alarms xx (fixed resolution)	Value	R
d2410	Current SPC – Range alarms xx (fixed resolution)	Value	R

The codes for the current values of other characteristics are as the next table:

Char. 1	Char. 2	Char. 3	Char. 4	Char. 5	Char. 6	Char. 7	Char. 8	Char. 9	Ch.10
d2000	d2500	d3000	d3500	d4000	d4500	d5000	d5500	d6000	d6500
d2005	d2505	d3005	d3505	d4005	d4505	d5005	d5505	d6005	d6505
d2010	d2510	d3010	d3510	d4010	d4510	d5010	d5510	d6010	d6510
d2100	d2600	d3100	d3600	d4100	d4600	d5100	d5600	d6100	d6600
d2105	d2605	d3105	d3605	d4105	d4605	d5105	d5605	d6105	d6605
d2110	d2610	d3110	d3610	d4110	d4610	d5110	d5610	d6110	d6610
d2115	d2615	d3115	d3615	d4115	d4615	d5115	d5615	d6115	d6615
d2120	d2620	d3120	d3620	d4120	d4620	d5120	d5620	d6120	d6620
d2125	d2625	d3125	d3625	d4125	d4625	d5125	d5625	d6125	d6625
d2130	d2630	d3130	d3630	d4130	d4630	d5130	d5630	d6130	d6630
d2135	d2635	d3135	d3635	d4135	d4635	d5135	d5635	d6135	d6635
d2140	d2640	d3140	d3640	d4140	d4640	d5140	d5640	d6140	d6640
d2150	d2650	d3150	d3650	d4150	d4650	d5150	d5650	d6150	d6650
d2160	d2660	d3160	d3660	d4160	d4660	d5160	d5660	d6160	d6660
d2170	d2670	d3170	d3670	d4170	d4670	d5170	d5670	d6170	d6670
d2180	d2680	d3180	d3680	d4180	d4680	d5180	d5680	d6180	d6680
d2181	d2681	d3181	d3681	d4181	d4681	d5181	d5681	d6181	d6681
d2184	d2684	d3184	d3684	d4184	d4684	d5184	d5684	d6184	d6684
d2200	d2700	d3200	d3700	d4200	d4700	d5200	d5700	d6200	d6700
d2210	d2710	d3210	d3710	d4210	d4710	d5210	d5710	d6210	d6710
d2220	d2720	d3220	d3720	d4220	d4720	d5220	d5720	d6220	d6720
d2230	d2730	d3230	d3730	d4230	d4730	d5230	d5730	d6230	d6730
d2240	d2740	d3240	d3740	d4240	d4740	d5240	d5740	d6240	d6740
d2250	d2750	d3250	d3750	d4250	d4750	d5250	d5750	d6250	d6750

Char. 1	Char. 2	Char. 3	Char. 4	Char. 5	Char. 6	Char. 7	Char. 8	Char. 9	Ch.10
d2260	d2760	d3260	d3760	d4260	d4760	d5260	d5760	d6260	d6760
d2270	d2770	d3270	d3770	d4270	d4770	d5270	d5770	d6270	d6770
d2280	d2780	d3280	d3780	d4280	d4780	d5280	d5780	d6280	d6780
d2300	d2800	d3300	d3800	d4300	d4800	d5300	d5800	d6300	d6800
d2310	d2810	d3310	d3810	d4310	d4810	d5310	d5810	d6310	d6810
d2320	d2820	d3320	d3820	d4320	d4820	d5320	d5820	d6320	d6820
d2330	d2830	d3330	d3830	d4330	d4830	d5330	d5830	d6330	d6830
d2340	d2840	d3340	d3840	d4340	d4840	d5340	d5840	d6340	d6840
d2350	d2850	d3350	d3850	d4350	d4850	d5350	d5850	d6350	d6850
d2360	d2860	d3360	d3860	d4360	d4860	d5360	d5860	d6360	d6860
d2370	d2870	d3370	d3870	d4370	d4870	d5370	d5870	d6370	d6870
d2380	d2880	d3380	d3880	d4380	d4880	d5380	d5880	d6380	d6880
d2390	d2890	d3390	d3890	d4390	d4890	d5390	d5890	d6390	d6890
d2400	d2900	d3400	d3900	d4400	d4900	d5400	d5900	d6400	d6900
d2410	d2910	d3410	d3910	d4410	d4910	d5410	d5910	d6410	d6910

Suffix Codes are available for characteristics (n=1..20).

Char. n
d2000.n
d2005.n
d2010.n
d2100.n
d2105.n
d2110.n
d2115.n
d2120.n
d2125.n
d2130.n
d2135.n
d2140.n
d2150.n
d2160.n
d2170.n
d2180.n
d2181.n
d2184.n
d2200.n
d2210.n
d2220.n
d2230.n
d2240.n
d2250.n
d2260.n
d2270.n
d2280.n
d2300.n
d2310.n
d2320.n
d2330.n
d2340.n
d2350.n
d2360.n
d2370.n
d2380.n
d2390.n
d2400.n
d2410.n

2.5.3 Not characteristic related current Coil Values

d7000	Current Line Speed x.xxxx (length resolution) (m/min or ft/min, depending on length units)	Value	R
d7010	Current coil number xx (fixed resolution)	Value	R/W
d7020	Current coil date begin mm/dd/yyyy dd/mm/yyyy yyyy/mm/dd (determined by system date format)	String	R
d7030	Current coil time begin hh:mm hh:mm AM/PM hh:mm:ss hh:mm:ss AM/PM (determined by system time format)	String	R
d7040	Reserved		
d7050	Reserved		
d7060	Current coil length x.xxxx (length resolution)	Value	R
d7500	Current coil event 1 count xx (fixed resolution)	Value	R

The codes for other events are as the next table:

Event 1	Event 2	Event 3	Event 4	Event 5
d7500	d7550	d7600	d7650	d7700

Suffix Codes are available for events (n=1..6).

Event n
d7500.n

2.5.4 Not characteristic related Current Session Values

d8000	Current session code xxxxxxxxxxxxxxxx (15 characters)	String	R/W
d8010	Current session operator code xxxxxxxxxxxxxxxx (15 characters)	String	R/W
d8020	Current session date begin mm/dd/yyyy dd/mm/yyyy yyyy/mm/dd (determined by system date format)	String	R
d8030	Current session time begin hh:mm hh:mm AM/PM hh:mm:ss hh:mm:ss AM/PM (determined by system time format)	String	R
d8040	Reserved		

d8050	Reserved		
d8060	Current session length x.xxxx (length resolution)	Value	R
d8100	Current produced coils xx (fixed resolution)	Value	R
d8110	Current produced coils with alarms xx (fixed resolution)	Value	R
d8120	Current discarded coils xx (fixed resolution)	Value	R
d8500	Current session event 1 count xx (fixed resolution)	Value	R

The codes for other events are as the next table:

Event 1	Event 2	Event 3	Event 4	Event 5
d8500	d8550	d8600	d8650	d8700

Suffix Codes are available for events (n=1..6).

Event n
c8500.n

2.5.5 Not characteristic related Current SPC Values

d9020	Current SPC date begin mm/dd/yyyy dd/mm/yyyy yyyy/mm/dd (determined by system date format)	String	R
d9030	Current SPC time begin hh:mm hh:mm AM/PM hh:mm:ss hh:mm:ss AM/PM (determined by system time format)	String	R
d9040	Reserved		
d9050	Reserved		

2.6 GROUP X - FFT CURRENT MEASURES AND STATISTICS

2.6.1 Characteristic related FFT Current Values

Characteristic 1 FFT Current Values

x1000	Frame Count xxxx (fixed resolution)	Value	R
x1005	Worst Peak Frequency x.x (Hz units, Frequency resolution)	Value	R
x1015	Worst Peak Amplitude x.x (Unit and resolution in dB)	Value	R

x1025	Worst Peak Amplitude x.x (Unit and resolution in characteristic units)	Value	R
x1035	Worst Peak Wavelength x.xxx (Length units, Wavelength resolution)	Value	R
x1045	Worst Peak Pulley x.xxx (mm or Inch, Pulley resolution)	Value	R
Characteristic 1 SRL Current Values			
x1200	Frame Count xxxx (fixed resolution)	Value	R
x1205	Worst Peak Frequency x.x (MHz units, Frequency resolution)	Value	R
x1215	Worst Peak Amplitude x.x (Unit and resolution in dB)	Value	R
x1235	Worst Peak Wavelength x.xxx(Length Units, Wavelength resolution)	Value	R
x1245	Worst Peak Pulley x.xxx (mm or Inch units, Pulley resolution)	Value	R
Characteristic 1 VSWR Current Values			
x1400	Frame Count xxxx (fixed resolution)	Value	R
x1405	Worst Peak Frequency x.x (MHz units, Frequency resolution)	Value	R
x1415	Worst Peak Amplitude x.x (Unit and resolution in dB)	Value	R
x1435	Worst Peak Wavelength x.xxx (Length units, Wavelength resolution)	Value	R
x1445	Worst Peak Pulley x.xxx(mm or Inch, Pulley resolution)	Value	R
Suffix Codes are available for characteristics (n=1..20).			
Characteristic 1 FFT Current Coil Values			
x2000	Worst peak Zone 1 Frequency x.x (Hz units, Frequency resolution)	Value	R
x2010	Worst Peak Zone 1 Amplitude x.x (Unit and resolution in dB)	Value	R
x2020	Worst Peak Zone 1 Amplitude x.x (Unit and resolution in characteristic units)	Value	R
x2030	Worst Peak Zone 1 Wavelength x.xxx(Length units, Wavelength resolution)	Value	R
x2040	Worst Peak Zone 1 Pulley x.xxx (mm or Inch, Pulley resolution)	Value	R

x2050	Worst Peak Zone 1 Length x.xxx (Length Units and Resolution)	Value	R
x2060	Worst Peak Zone 1 Alarms xx (Fixed resolution)	Value	R
x2070	Worst Peak Zone 1 Prealarms xx (Fixed resolution)	Value	R

Suffix Codes are available for characteristics (n=1..20).

Zone 1 Char n	Zone 2 Char n	Zone 3 Char n	Zone 4 Char n	Zone 5 Char n	Zone 6 Char n	Zone 7 Char n	Zone 8 Char n	Zone 9 Char n	Zone 10 Char n
x2000.n	x2001.n	x2002.n	x2003.n	x2004.n	x2005.n	x2006.n	x2007.n	x2008.n	x2009.n
x2010.n	x2011.n	x2012.n	x2013.n	x2014.n	x2015.n	x2016.n	x2017.n	x2018.n	x2019.n
x2020.n	x2021.n	x2022.n	x2023.n	x2024.n	x2025.n	x2026.n	x2027.n	x2028.n	x2029.n
x2030.n	x2031.n	x2032.n	x2033.n	x2034.n	x2035.n	x2036.n	x2037.n	x2038.n	x2039.n
x2040.n	x2041.n	x2042.n	x2043.n	x2044.n	x2045.n	x2046.n	x2047.n	x2048.n	x2049.n
x2050.n	x2051.n	x2052.n	x2053.n	x2054.n	x2055.n	x2056.n	x2057.n	x2058.n	x2059.n
x2060.n	x2061.n	x2062.n	x2063.n	x2064.n	x2065.n	x2066.n	x2067.n	x2068.n	x2069.n
x2070.n	x2071.n	x2072.n	x2073.n	x2074.n	x2075.n	x2076.n	x2077.n	x2078.n	x2079.n

x2100	Total Alarms xx(Fixed Resolution)	Value	R
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x2105	Total Prelarms xx(Fixed Resolution)	Value	R
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Characteristic 1 SRL Current Coil Values

x2200	Worst peak Zone 1 Frequency x.x (Hz units, Frequency resolution)	Value	R
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x2210	Worst Peak Zone 1 Amplitude x.x (Unit and resolution in dB)	Value	R
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x2230	Worst Peak Zone 1 Wavelength x.xxx(Length units, Wavelength resoltion)	Value	R
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x2240	Worst Peak Zone 1 Pulley x.xxx (mm or Inch, Pulley resolution)	Value	R
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x2250	Worst Peak Zone 1 Length x.xxx (Length Units and Resolution)	Value	R
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x2260	Worst Peak Zone 1 Alarms xx (Fixed resolution)	Value	R
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x2270	Worst Peak Zone 1 Prealarms xx (Fixed resolution)	Value	R
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Zone 1 Char n	Zone 2 Char n	Zone 3 Char n	Zone 4 Char n	Zone 5 Char n	Zone 6 Char n	Zone 7 Char n	Zone 8 Char n	Zone 9 Char n	Zone 10 Char n
x2200.n	x2201.n	x2202.n	x2203.n	x2204.n	x2205.n	x2206.n	x2207.n	x2208.n	x2209.n
x2210.n	x2211.n	x2212.n	x2213.n	x2214.n	x2215.n	x2216.n	x2217.n	x2218.n	x2219.n
x2220.n	x2221.n	x2222.n	x2223.n	x2224.n	x2225.n	x2226.n	x2227.n	x2228.n	x2229.n
x2230.n	x2231.n	x2232.n	x2233.n	x2234.n	x2235.n	x2236.n	x2237.n	x2238.n	x2239.n
x2240.n	x2241.n	x2242.n	x2243.n	x2244.n	x2245.n	x2246.n	x2247.n	x2248.n	x2249.n
x2250.n	x2251.n	x2252.n	x2253.n	x2254.n	x2255.n	x2256.n	x2257.n	x2258.n	x2259.n
x2260.n	x2261.n	x2262.n	x2263.n	x2264.n	x2265.n	x2266.n	x2267.n	x2268.n	x2269.n

x2270.n	x2271.n	x2272.n	x2273.n	x2274.n	x2275.n	x2276.n	x2277.n	x2278.n	x2279.n
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x2300 Total Alarms
 xx(Fixed Resolution) Value R

x2305 Total Prelarms
 xx(Fixed Resolution) Value R

Suffix Codes are available for characteristics (n=1..20).

Characteristic 1 VSWR Current Coil Values

x2400 Worst peak Zone 1 Frequency
 x.x (Hz units, Frequency resolution) Value R

x2410 Worst Peak Zone 1 Amplitude
 x.x (Unit and resolution in **dB**) Value R

x2430 Worst Peak Zone 1 Wavelength
 x.xxx(Length units, Wavelength resolution) Value R

x2440 Worst Peak Zone 1 Pulley
 x.xxx (mm or Inch, Pulley resolution) Value R

x2450 Worst Peak Zone 1 Length
 x.xxx (Length Units and Resolution) Value R

x2460 Worst Peak Zone 1 Alarms
 xx (Fixed resolution) Value R

x2470 Worst Peak Zone 1 Prelarms
 xx (Fixed resolution) Value R

Suffix Codes are available for characteristics (n=1..20).

Zone 1 Char n	Zone 2 Char n	Zone 3 Char n	Zone 4 Char n	Zone 5 Char n	Zone 6 Char n	Zone 7 Char n	Zone 8 Char n	Zone 9 Char n	Zone 10 Char n
x2400.n	x2401.n	x2402.n	x2403.n	x2404.n	x2405.n	x2406.n	x2407.n	x2408.n	x2409.n
x2410.n	x2411.n	x2412.n	x2413.n	x2414.n	x2415.n	x2416.n	x2417.n	x2418.n	x2419.n
x2430.n	x2431.n	x2432.n	x2433.n	x2434.n	x2435.n	x2436.n	x2437.n	x2438.n	x2439.n
x2440.n	x2441.n	x2442.n	x2443.n	x2444.n	x2445.n	x2446.n	x2447.n	x2448.n	x2449.n
x2450.n	x2451.n	x2452.n	x2453.n	x2454.n	x2455.n	x2456.n	x2457.n	x2458.n	x2459.n
x2460.n	x2461.n	x2462.n	x2463.n	x2464.n	x2465.n	x2466.n	x2467.n	x2468.n	x2469.n
x2470.n	x2471.n	x2472.n	x2473.n	x2474.n	x2475.n	x2476.n	x2477.n	x2478.n	x2479.n

x2500 Total Alarms
 xx(Fixed Resolution) Value R

x2505 Total Prelarms
 xx(Fixed Resolution) Value R

2.7 GROUP E - ALARM MESSAGES AND DEVICE STATUS

2.7.1 Not characteristic related Alarm Messages

e1000	Upload/Download Error 0 No Error 200 The received value was invalid 201 The received characters format was invalid 202 The received parameter code was invalid 203 The requested value is not available 204 Upload and Download are not allowed	Value	R
e1010	Last order in error xxxxx Returns the order that produced the error. In the Grouped messages it returns the last single order in error, so that it permits identify which one of the messages in the group is wrong.	String	R
e1015	Grouped Upload/Download error (Same code as e1000) When last order in error is into a grouped message, it is the error produced into this order.	Value	R
e1020	Line Stopped 00 Line not stopped 01 Line stopped (below minimum line speed)	Selection	R
e1030	Last product activation error 000 Validation Ok 001 Bad Characteristic 002 Bad Event 003 Too many Characteristics 004 Too many Events 010 No Characteristics Selected 011 Bad Nominal Line Speed 012 Bad Pre-alarms Value 013 Bad Nominal Diameter 0 014 Bad Nominal Piece Length 015 Bad Piece start inhibit 016 Bad Piece end inhibit 100 Characteristic not Allowed (see char. on Index value) 101 Disabled Characteristic (see char. on Index value) 102 All Set Points are Zero (see char. on Index value) 103 Negative Value (see char. on Index value) 104 Nominal smaller than Minimum (see char. on Index value) 105 Nominal bigger than Maximum (see char. on Index value) 106 Nominal smaller than Lower Pre-alarm (see char. on Index value) 107 Nominal bigger than Upper Pre-alarm (see char. on Index value) 108 Maximum must be greater than Minimum (see char. on Index value) 109 Upper Pre-alarm smaller than Lower Pre-alarm (see ch. on Index value) 110 UCL smaller than LCL (see char. on Index value) 111 RCL is Negative or Zero (see char. on Index value) 190 Event not available (see event on Index value) 200 Bad Lay Length 201 Bad Number of Strands 202 Bad Averaging Time 203 Invalid OD Lump (see ODAC/ODEX on Index value) 204 Invalid OD Neck (see ODAC/ODEX on Index value)	Value	R

205 Invalid OD Period (see ODAC/ODEX on Index value)
210 Invalid KW Lump
211 Invalid KW Neck
220 Invalid SPARKER setup
221 Invalid SPARKER tolerances
240 Invalid Gravimetric Extruder Throughput(see Gravimetric on Index)
241 Invalid Gravimetric Extruder Weight (see Gravimetric on Index)
242 Invalid Gravimetric Extruder Density (see Gravimetric on Index)
252 Invalid ODEX Inner Diameter
253 Invalid ODEX Twist Angle
260 Invalid Accumulator Distance
300 Invalid Controlled Characteristic (see controller on Index value)
302 Invalid Controller Type (see controller on Index value)
303 Invalid Controller Cpk (see controller on Index value)
304 Invalid Controller SRD (see controller on Index value)
305 Invalid Controller Inner Diameter (see controller on Index value)
306 Inner Diameter Larger than Diameter (see controller on Index value)
307 Incorrect Upper Deadband value (see controller on Index value)
308 Incorrect Lower Deadband value (see controller on Index value)
309 Controlled Characteristic Repeated (see controller on Index value)
350 No Characteristic Selected for Conductor Layer
351 No Wall Characteristic Selected for Inner SemiConductor Layer
352 No Wall Characteristic Selected for Insulation Layer
353 No Wall Characteristic Selected for Outer Semiconductor Layer
354 Conductor Layer not in Product
355 SemiConductor Layer not in Product
356 Insulation Layer not in Product
357 Outer Semiconductor Layer not in Product
358 Bad absorption coefficient
359 Bad Shrinkage
360 Bad Material
400 Invalid FFT Alarmed Characteristic
401 Invalid Velocity of Propagation
402 Invalid SRL Adjust Factor
403 FFT Alarm 'X From' smaller than 'X To' (see FFT on Index value)
404 FFT Alarm 'Y From' smaller than 'Y To' (see FFT on Index value)
405 Invalid FFT Alarm Color (see FFT alarm on Index value)
425 UMAC #1: Invalid UPS Transducers
426 UMAC #2: Invalid UPS Transducers
427 UMAC #1: Invalid UPS Layers
428 UMAC #2: Invalid UPS Layers
429 UMAC #1: Invalid UPS Gains
430 UMAC #2: Invalid UPS Gains
431 UMAC #1: Other Invalid UPS data
432 UMAC #2: Other Invalid UPS data
433 UMAC #1: Invalid Compensations
434 UMAC #2: Invalid Compensations
435 UMAC #1: Invalid UPS Material
436 UMAC #2: Invalid UPS Material
437 Bad Wall UPS
450 Bad Foam wire type
451 Bad Foam characteristic Process Control
452 Bad Solid Characteristic
453 Bad Foam Nominal Inner Diameter
454 Bad Foam Calculated Expansion
455 Bad Foam Calculated Weight
456 Bad Foam Setup gravity
457 Bad Foam Setup Dielectric
458 Bad Foam Setup Thickness

459 Bad Foam Ctrl Upper Dead Band
 460 Bad Foam Ctrl Lower Dead Band
 475 Bad Bar Density
 476 Bad Bar Correction Factor
 500 Bad Preheater Nominal Temperature
 501 Bad Preheater Temperature Offset
 502 Bad Preheater Cross Section
 503 Bad Preheater Material Electrical Resistivity
 504 Bad Preheater Material Temperature Coefficient
 505 Bad Preheater Material Specific Heat Capacity
 506 Bad Preheater Material Density
 910 Controller is on.
 911 RAYEX is on.
 999 Undefined Product Error

e1035	Last product activation error index	Value	R
	-1 Index not used		
	xx Index related to activation error		

2.7.2 Controllers related Alarm Messages

e1100	Controller 1 error	Selection	R
	00 No controller errors		
	01 No values received		
	02 Error exceeds set limits		
	03 Inner diameter exceeds outer diameter nominal		
	04 Measured speed is below minimum limit		
	05 No Controller Input assigned		
	06 Input V too high		
	07 Requested output V too high		
	08 Requested position out of range		
	09 Requested RegDev position out of range		
	10 Method not used in current mode		
	11 Selected PCB is not supported		
	12 Required IO assignments have not been made.		
	13 Regulated device has not been assigned.		

The codes for other controllers are as the next table:

Controller 1	Controller 2	Controller 3	Controller 4
e1100	e1200	e1300	e1400

e1500	Dual Loop 1 error	Selection	R
	00 No controller errors		
	01 No values received		
	02 Error exceeds set limits		
	03 Inner diameter exceeds outer diameter nominal		
	04 Measured speed is below minimum limit		
	05 No Controller Input assigned		
	06 Input V too high		
	07 Requested output V too high		
	08 Requested position out of range		
	09 Requested RegDev position out of range		
	10 Method not used in current mode		
	11 Selected PCB is not supported		
	12 Required IO assignments have not been made.		
	13 Regulated device has not been assigned.		

The codes for other Dual Loops are as the next table:

Dual Loop 1	Dual Loop 2	Dual Loop 3	Dual Loop 4
e1500	e1600	e1700	e1800

2.7.3 Characteristic related Alarm Messages and Device Status

Characteristic 1

e2000	General Alarm 00 Valid Value 01 Current not calculated	Selection	R
e2010	Upper Tolerance 00 Limit not exceeded 01 Limit exceeded	Selection	R
e2020	Upper Pre-Alarm 00 Limit not exceeded 01 Limit exceeded	Selection	R
e2030	Lower Tolerance 00 Limit not exceeded 01 Limit exceeded	Selection	R
e2040	Lower Pre-Alarm 00 Limit not exceeded 01 Limit exceeded	Selection	R
e2050	Upper Tolerance Latch 00 Limit not exceeded since last read 01 Limit exceeded since last read	Selection	R
e2060	Upper Pre-Alarm Latch 00 Limit not exceeded since last read 01 Limit exceeded since last read	Selection	R
e2070	Lower Tolerance Latch 00 Limit not exceeded since last read 01 Limit exceeded since last read	Selection	R
e2080	Lower Pre-Alarm Latch 00 Limit not exceeded since last read 01 Limit exceeded since last read	Selection	R

The codes for the error status of other characteristics are as the next table:

Char. 1	Char. 2	Char. 3	Char. 4	Char. 5	Char. 6	Char. 7	Char. 8	Char. 9	Ch. 10
e2000	e2500	e3000	e3500	e4000	e4500	e5000	e5500	e6000	e6500
e2010	e2510	e3010	e3510	e4010	e4510	e5010	e5510	e6010	e6510
e2020	e2520	e3020	e3520	e4020	e4520	e5020	e5520	e6020	e6520
e2030	e2530	e3030	e3530	e4030	e4530	e5030	e5530	e6030	e6530
e2040	e2540	e3040	e3540	e4040	e4540	e5040	e5540	e6040	e6540
e2050	e2550	e3050	e3550	e4050	e4550	e5050	e5550	e6050	e6550
e2060	e2560	e3060	e3560	e4060	e4560	e5060	e5560	e6060	e6560
e2070	e2570	e3070	e3570	e4070	e4570	e5070	e5570	e6070	e6570
e2080	e2580	e3080	e3580	e4080	e4580	e5080	e5580	e6080	e6580

Suffix Codes are available for characteristics (n=1..20).

Char. n
e2000.n

e2010.n
e2020.n
e2030.n
e2040.n
e2050.n
e2060.n
e2070.n
e2080.n

2.7.4 Related Alarm Messages and Device Status for any Characteristic.

e7000	Characteristics out of any tolerances	Multiple	R
	0 None		
	1 Characteristic #1		
	2 Characteristic #2		
	4 Characteristic #3		
	8 Characteristic #4		
	16 Characteristic #5		
	32 Characteristic #6		
	64 Characteristic #7		
	128 Characteristic #8		
	256 Characteristic #9		
	512 Characteristic #10		
e7010	Characteristics out of High tolerances (Same codes as e7000)	Multiple	R
e7020	Characteristics out of Low tolerances (Same codes as e7000)	Multiple	R
e7030	Characteristics out of any Pre-Alarm (Same codes as e7000)	Multiple	R
e7040	Characteristics out of High Pre-Alarm (Same codes as e7000)	Multiple	R
e7050	Characteristics out of Low Pre-Alarm (Same codes as e7000)	Multiple	R
e7100	Characteristics out of any tolerances Latch (Same codes as e7000)	Multiple	R
e7110	Characteristics out of High tolerances Latch (Same codes as e7000)	Multiple	R
e7120	Characteristics out of Low tolerances Latch (Same codes as e7000)	Multiple	R
e7130	Characteristics out of any Pre-Alarm Latch (Same codes as e7000)	Multiple	R
e7140	Characteristics out of High Pre-Alarm Latch (Same codes as e7000)	Multiple	R
e7150	Characteristics out of Low Pre-Alarm Latch (Same codes as e7000)	Multiple	R

2.7.5 Device Errors and Status

e8100	Spark Tester Active 00 Off 01 On	Selection	R
e8110	Spark Tester High Voltage 00 Off 01 On	Selection	R
e8120	Spark Tester Error Detected Latch 00 Error not detected since last read 01 Error detected since last read	Selection	R
e8130	Spark Tester Status 00 Ok 01 Hardware error 02 Feedback voltage out of tolerance 03 The 2 previous errors occur together 04 Configuring 05 Pause 06 IO channel is not assigned 07 RS port is not assigned 08 Does not answer 20 Not Measuring 21 Current Voltage out of range	Selection	R

Suffix Codes are available for Spark Tester Gauges (n=1..6):

Spark Tester n
e8100.n
e8110.n
e8120.n
e8130.n

e8150	Preheater WST Heater 00 Off 01 On	Selection	R
e8152	Preheater WST Status 00 Ok 01 Not Measuring 02 Connecting 03 Configuring 04 Communication is not assigned 05 Does not answer 06 Configuration Error 07 Internal Error (use e8154)	Selection	R
e8154	Preheater WST Error 0 No alarm 1 Emergency stop activated or not connected 2 Remote heat enabling not activated 4 Door open 8 Insufficient power for desired temperature 16 Ambient temperature sensor not connected 32 Not used 64 Heatsink temperature too high 128 Inductor temperature too high	Multiple	R

	256	Power part error		
	512	Main voltage out of tolerance		
	1024	Line speed below threshold		
	2048	Wire temperature before heating too high		
	4096	Wire temperature before heating too low		
	8192	AUTAC 300 - No communication		
	16384	AUTAC 300 - Error E1: Difference between head A and B		
	32768	AUTAC 300 - Error E2: Setpoint not reached		
	65536	AUTAC 300 - Error E9: Failure		
	131072	AUTAC 300 - Fluctuating product temperature		
e8200	KW Neck Detected Latch		Selection	R
	00 Error not detected since last read			
	01 Error detected since last read			
e8210	KW Lump Detected Latch		Selection	R
	00 Error not detected since last read			
	01 Error detected since last read			
e8220	KW Lump/Neck Detected Latch		Selection	R
	00 Error not detected since last read			
	01 Error detected since last read			
e8230	KW Status		Selection	R
	00 Ok			
	01 Hardware error			
	02 Configuring			
	03 Does not answer			
	05 RS port is not assigned			
	06 IO channel is not assigned			
	10 Not Measuring			

Suffix Codes are available for KW Gauges (n=1..6):

KW n
e8200.n
e8210.n
e8220.n
e8230.n

e8350	UMAC #1 Status		Selection	R
	00 Not Measuring			
	01 Configuring			
	02 Connecting			
	03 Ok			
	11 RS port is not assigned			
	12 Does not answer			
	13 Received value is invalid			
	14 Received code is invalid			
	15 Requested download value not available			
	16 Requested upload value not available			
	17 Telegram too short			
	18 Telegram too long			
	19 Protocol error			
	20 Block check failure			
	21 Block aborted (no ETX)			
	22 Communication watchdog failure			
	23 Generic hardware error			
	24 Generic power supply error			

25 The configuration process failed

e8351	UMAC #2 Status Same as e8350	Selection	R
e8400	Measuring Head 1 error Errors for J-Interface gauges 0 No errors xxx Error code (See CP-310 manual) Errors for Analog gauges 0 No errors 1 Error on last calibration Gain. 2 Error on last calibration Offset.	Value	R
e8410	Measuring Head 1 status Status for J-Interface gauges 00 Ok 01 Not Measuring 02 Calibrating 03 Error Status for Analog gauges 00 Calibrating 01 End of Calibration 02 Ok 03 Error 04 Not Measuring 05 IO Channel not assigned	Value	R

The codes for other Measuring Heads are as the table :

M. H. 1	M. H. 2	M. H. 3	M. H. 4	M. H. 5	M. H. 6	M. H. 7	M. H. 8	M. H. 9	M.H.10
e8400	e8420	e8440	e8460	e8480	e8500	e8520	e8540	e8560	e8580
e8410	e8430	e8450	e8470	e8490	e8510	e8530	e8550	e8570	e8590

e8600	Event Gauge 1 Detected Latch 00 Event not detected since last read 01 Event detected since last read	Selection	R
e8610	Event Gauge 1 Status 00 Ok 01 IO channel is not assigned 02 Not Measuring	Selection	R

The codes for other Event Gauges are as the next table:

Event Gauge 1	Event Gauge 2
e8600	e8620
e8610	e8630

e8700	Marker is Marking 00 Off 01 On	Selection	R
e8710	Marker Detected Latch 00 No mark since last read 01 Some marks since last read	Selection	R

e8720	Marker Status 00 Marker OK 01 Interface not detected	Selection	R
e8850	RAYEX D Status 00 To Stop 01 Stop 02 To Init 03 Init 04 Ready 05 To Warm up 06 Warm up 07 To Reference Scan 08 Reference Scan 09 To Tube Scan 10 Tube Scan 11 To Measure 12 Measure 13 Measure Unstable 14 Preparing to activate 15 To Activate Product	Selection	R
e8851	RAYEX D Errors 00 None 01 Parity 02 Error Checksum 03 Error Communication 04 Command not accepted 05 Received command too long 06 Temperature too high 07 Door open 08 SF6 Low Pressure 09 Too many X-Ray Flases 10 Watchdog Timeout 11 PC Watch Timeout 12 CPU Reset 13 KV/ μ A Out of Tolerances 14 KV/ μ A Out of Limits 15 KV/ μ A not stable 16 More than expected numbers of samples 17 Motor position out of limits 18 Command not executed within time 19 Xray file parameter error 20 Setup file does not exist 21 Recipe file parameter error 22 Recipe file does not exist 23 Machine file parameter error 24 Machine file does not exist 25 Fatal number of oscalc errors 26 Fatal Error, Cable dimensions completely wrong 27 Outer diameter not found within profile 28 Unexpected high X-ray absorption in CaOdB 29 Cable movement 30 Not sufficient insulator thickness found 31 Unexpected high X-ray absorption in OsIIB 32 Outer zone: signal processor error 33 Points for Mu calculation not within profile 34 Outer zone: Auto. correction of invalid result	Selection	R

35 Fatal number of iscalc errors
 36 Core not found within profile
 37 Unexpected high X-ray absorption in IsCoB
 38 Not sufficient inner screen thickness found
 39 Unexpected high X-ray absorption in IllsB
 40 Inner zone: signal processor error
 41 Automatic correction of invalid result
 42 Edge not found in profile (coarse search scan)
 43 Edge not found in profile (fine search scan)
 44 Edge found in the first point (coarse search scan)
 45 Edge found in the first point (fine search scan)
 46 Scan limits out of range (coarse search scan)
 47 Scan limits out of range (fine search scan)
 48 Asymmetric reference current
 49 NewRefCur deviating too much from RefCurAct level
 50 RefCurSlo outside limits
 51 RefCurAct outside max min limit
 52 Position outside reference file
 53 Ref current > 16000
 54 Not all points in ref. profile are valid
 55 Multi Scan: Rope not found
 56 Fatal level of condensate
 57 Warning level of condensate
 58 Fatal level of dark current fluctuations
 59 Fatal level of dark current deviating from expected level
 60 Warning level of dark current deviating from expected level
 61 Long term dark current deviating from expected level
 62 Short term dark current deviating from expected level
 63 New scan positions not in increasing order
 64 New scan positions out of limits
 65 Scan zone too big
 66 Command cannot be executed because comma is already busy
 67 Command cannot be executed because other master is running
 68 Software error: Out of memory.
 69 Software error: Unhandled error

e8852	RAYEX D Measuring Mode 01 Conductor 02 Inner Semi-Conductor 04 Outer Semi-Conductor 08 Outer Diameter	Multiple	R
e8860	RAYEX S/D XT Status 00 Stopped 01 Powering on 02 Connecting 03 Configuring 04 Ready off 05 Warming up 06 Ready on 07 Cleaning 08 Reference Scan 09 Tube Scan 10 Measuring 11 Measure internal error 12 Empty measuring field 13 Object touching borders 14 Dirty borders 15 Dark scan is saturated low	Selection	R

16 Ref. scan is saturated high
 17 No valid Reference/Dark scan
 18 Invalid Recipe
 19 Too much cable movement
 20 Undefined Error
 21 Ref. Scan Error
 22 Error
 23 Irrecoverable Error
 24 Switched off

e8861	RAYEX S/D XT Errors	Selection	R
	00 None		
	01 Communication not assigned		
	02 Not answer		
	03 Configuration error		
	04 Product error		
	05 RAYEX S Internal error		
	06 RAYEX S High voltage not achieved on warmup		
	07 RAYEX S High voltage not achieved on cleaning		
	08 RAYEX S Switched off		
	09 RAYEX S Emergency button pressed		
	10 RAYEX S Internal Safety Loop Open		
	11 RAYEX S External Safety Loop Open		
	12 RAYEX S Hardware error.		
	13 RAYEX D XT Internal error		
	14 RAYEX D XT Temperature Error		
	15 RAYEX D XT Door open,		
	16 RAYEX D XT Too many flashed,		
	17 RAYEX D XT Communication error,		
	18 Xray on error		
	19 Clean up error		
	20 Warm up error		
	20 Reference scan error		
	21 Dark on scan error		
	22 Dark off acquisition error		
	23 White acquisition error		
	24 Stop error		
	25 White scan error		
	26 Error starting eccentricity		
	27 Error starting wallthickness		
	28 Error starting outer diameter		
	29 Error starting tube scan		
	30 Undefined error		
	31 Error application not licensed		
	32 Hardware Communication error		

2.8 GROUP F - CONTROL FUNCTIONS

f1000	Statistics Pause Selection	Selection	R/W
	00 Off		
	01 On		
f1100	Stop/Start all statistics (no message)	--	W
f1110	Stop/Start SPC (no message)	--	W
f1120	Stop/Start session (no message)	--	W

f1130	Stop/Start coil (no message)	--	W
f1140	Stop or Start all statistics 00 Stop 01 Start	Selection	W
f1150	Stop or Start SPC 00 Stop 01 Start	Selection	R/W
f1160	Stop or Start session 00 Stop 01 Start	Selection	R/W
f1170	Stop or Start coil 00 Stop 01 Start	Selection	R/W
f1200	Reset all statistics (no message)	--	W
f1210	Reset SPC (no message)	--	W
f1220	Reset session (no message)	--	W
f1230	Reset coil (no message)	--	W
f2100	Controller 1 on/off 00 Off 01 On	Selection	R/W
f2200	Reset Controller 1 (no message)	--	W
f2300	Freeze Controller 1 00 Off 01 On	Selection	R/W
f2400	Increase Controller 1 (no message)	--	W
f2500	Decrease Controller 1 (no message)	--	W

The codes for other controllers are as the next table:

Controller 1	Controller 2	Controller 3	Controller 4
f2100	f2110	f2120	f2130
f2200	f2210	f2220	f2230
f2300	f2310	f2320	f2330
f2400	f2410	f2420	f2430
f2500	f2510	f2520	f2530

Suffix Codes are available for all controllers (n=1..4):

Controller 1
f2100.n
f2200.n
f2300.n
f2400.n
f2500.n

2.8.1 Gravimetric Extruder Controllers

f2610	Gravimetric Extruder Controller 1 on/off	Selection	R/W
	00 Off		
	01 On		

The codes for other controllers are as the next table:

Gravimetric 1	Gravimetric 2	Gravimetric 3	Gravimetric 4
f2610	f2611	f2612	f2613

2.8.2 RAYEX D Functions

f2700	Warm Up on/off (no message)	--	W
f2701	Stop (no message)	--	W
f2702	Start Wallthickness 01 Conductor 02 Inner SemiConductor 04 Outer SemiConductor	Multiple	W
f2703	Start Eccentricity (no message)	--	W

2.8.3 RAYEX S/D XT Functions

f2751	Stop (no message)	--	W
f2752	Start Wallthickness (no message)	—	W
f2753	Start Eccentricity (no message)	--	W
f2754	Start Outer Diameter (no message)	--	W

2.8.4 Length Counters

f3100	Line Speed x.xxxx (length resolution, m/min or ft/min)	Value	R/W
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The codes for other length counters are as the next table:

Length Counter 1	Length Counter 2
F3100	F3110

2.8.5 Accumulator

f3200	Velocity after Accumulator x.xxxx (length resolution, m/s or ft/s)	Value	R/W
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2.8.6 Spark Tester Series A

f3300	Test Voltage On/Off 00 Off 01 On	Value	R/W
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Suffix Codes are available for Spark Tester Gauges (n=1..6):

Spark Tester n
f3300.n

2.8.7 Preheater WST

f3310	Heat On/Off 00 Off 01 On	Value	R/W
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2.9 GROUP G - MEASURING HEAD INPUT GROUP

2.9.1 Measuring Heads

Measuring Head 1

g1000	Fast Diameter (ODAC Single, BENDCHECK) Fast Diameter X (ODAC XY, ODAC TRIO, ODAC Dual XY, ODAC Dual 2XY, ODEX, MSD XY,, BENDCHECK) x.xxxx (global resolution) Fast Capacity (CAPAC J, CAPAC Analog) x.xxxx (capacity resolution) Fast Measure (Analog Gauge) Fast Diameter 1 (ODAC Multi-Wire, MSD XY/4) Fast Diameter X1 (ODAC Multi-Wire XY) x.xxxx (gauge resolution)	Value	R
g1002	Fast Diameter Y1 (ODAC Multi-Wire XY)	Value	R
g1010	Fast Diameter Y (ODAC XY, ODAC TRIO, ODAC Dual XY, ODAC Dual 2XY, ODEX, MSD XY, BENDCHECK) Fast Diameter 2 (ODAC Multi-Wire, MSD XY/4) Fast Diameter X2 (ODAC Multi-Wire XY) x.xxxx (global resolution)	Value	R
g1011	Fast Diameter Y2 (ODAC Multi-Wire XY)	Value	R
g1012	Fast Diameter Z (ODAC TRIO, ODAC Dual 2XY, BENDCHECK) Fast Diameter 3 (ODAC Multi-Wire, MSD XY/4) Fast Measure X3 (ODAC Multi-Wire XY) x.xxxx (global resolution)	Value	R
g1013	Fast Diameter Y3 (ODAC Multi-Wire XY)	Value	R
g1014	Fast Diameter T (ODAC Dual 2XY) Fast Measure 4 (ODAC Multi-Wire, MSD XY/4) Fast Measure X4 (ODAC Multi-Wire XY) x.xxxx (global resolution)	Value	R

g1015	Fast Diameter Y4 (ODAC Multi-Wire XY)	Value	R
g1020	Fast RMS (X Y) (ODAC XY, ODAC Dual XY, ODEX, MSD XY, BENDCHECK) Fast RMS (X Y Z) (ODAC TRIO, BENDCHECK) Fast RMS (X Y Z T) (ODAC Dual 2XY) Fast RMS (X1,Y1) (ODAC Multi-Wire XY) x.xxxx (global resolution)	Value	R
g1022	Fast RMS (X2,Y2) (ODAC Multi-Wire XY)	Value	R
g1024	Fast RMS (X3,Y3) (ODAC Multi-Wire XY)	Value	R
g1026	Fast RMS (X4,Y4) (ODAC Multi-Wire XY)	Value	R
g1030	Fast Smaller (X Y) (ODAC XY, ODAC Dual XY, ODEX, MSD XY, BENDCHECK) Fast Smaller (X Y Z) (ODAC TRIO, BENDCHECK) Fast Smaller (X Y Z T) (ODAC Dual 2XY) Fast Smaller (X1,Y1) (ODAC Multi-Wire XY) x.xxxx (global resolution)	Value	R
g1032	Fast Smaller (X2,Y2) (ODAC Multi-Wire XY)	Value	R
g1034	Fast Smaller (X3,Y3) (ODAC Multi-Wire XY)	Value	R
g1036	Fast Smaller (X4,Y4) (ODAC Multi-Wire XY)	Value	R
g1040	Fast Larger (X Y) (ODAC XY, ODAC Dual XY, ODEX, MSD XY, BENDCHECK) Fast Larger (X Y Z) (ODAC TRIO, BENDCHECK) Fast Larger (X Y Z T) (ODAC Dual 2XY) Fast Larger (X1,Y1) (ODAC Multi-Wire XY) x.xxxx (global resolution)	Value	R
g1042	Fast Larger (X2,Y2) (ODAC Multi-Wire XY)	Value	R
g1044	Fast Larger (X3,Y3) (ODAC Multi-Wire XY)	Value	R
g1046	Fast Larger (X4,Y4) (ODAC Multi-Wire XY)	Value	R

g1050	Fast Ovality (X Y) (ODAC XY, ODAC Dual XY, ODEX, MSD XY) Fast Ovality (X Y Z) (ODAC TRIO) Fast Ovality (X Y Z, T) (ODAC Dual 2XY) Fast Ovality (X1,Y1) (ODAC Multi-Wire XY) x.xxxx (global resolution)	Value	R
g1052	Fast Ovality (X2,Y2) (ODAC Multi-Wire XY)	Value	R
g1054	Fast Ovality (X3,Y3) (ODAC Multi-Wire XY)	Value	R
g1056	Fast Ovality (X4,Y4) (ODAC Multi-Wire XY)	Value	R
g1060	Fast Ovality 90° (X Y Z T) (ODAC Dual 2XY) x.xxxx (global resolution)	Value	R
g1100	Fast Position (ODAC Single, BENDCHECK) Fast Position X (ODAC XY, ODAC TRIO, ODAC Dual XY, ODAC Dual 2XY,ODEX, MSD XY, BENDCHECK) x.xxxx (global resolution)	Value	R
g1110	Fast Position Y (ODAC XY, ODAC TRIO, ODAC Dual XY, ODAC Dual 2XY,ODEX, MSD XY, BENDCHECK) x.xxxx (global resolution)	Value	R
g1112	Fast Position Z (ODAC TRIO, ODAC Dual 2XY, BENDCHECK) x.xxxx (global resolution)	Value	R
g1114	Fast Position T (ODAC Dual 2XY) x.xxxx (global resolution)	Value	R

The codes for other Measuring Heads are as the table :

M. H. 1	M. H. 2	M. H. 3	M. H. 4	M. H. 5	M. H. 6	M. H. 7	M. H. 8	M. H. 9	M.H.10
g1000	g1500	g2000	g2500	g3000	g3500	g4000	g4500	g5000	g5500
g1002	g1502	g2002	g2502	g3002	g3502	g4002	g4502	g5002	g5502
g1010	g1510	g2010	g2510	g3010	g3510	g4010	g4510	g5010	g5510
g1011	g1511	g2011	g2511	g3011	g3511	g4011	g4511	g5011	g5511
g1012	g1512	g2012	g2512	g3012	g3512	g4012	g4512	g5012	g5512
g1013	g1513	g2013	g2513	g3013	g3513	g4013	g4513	g5013	g5513
g1014	g1514	g2014	g2514	g3014	g3514	g4014	g4514	g5014	g5514
g1015	g1515	g2015	g2515	g3015	g3515	g4015	g4515	g5015	g5515
g1020	g1520	g2020	g2520	g3020	g3520	g4020	g4520	g5020	g5520
g1022	g1522	g2022	g2522	g3022	g3522	g4022	g4522	g5022	g5522
g1024	g1524	g2024	g2524	g3024	g3524	g4024	g4524	g5024	g5524
g1026	g1526	g2026	g2526	g3026	g3526	g4026	g4526	g5026	g5526

g1030	g1530	g2030	g2530	g3030	g3530	g4030	g4530	g5030	g5530
g1032	g1532	g2032	g2532	g3032	g3532	g4032	g4532	g5032	g5532
g1034	g1534	g2034	g2534	g3034	g3534	g4034	g4534	g5034	g5534
g1036	g1536	g2036	g2536	g3036	g3536	g4036	g4536	g5036	g5536
g1040	g1540	g2040	g2540	g3040	g3540	g4040	g4540	g5040	g5540
g1042	g1542	g2042	g2542	g3042	g3542	g4042	g4542	g5042	g5542
g1044	g1544	g2044	g2544	g3044	g3544	g4044	g4544	g5044	g5544
g1046	g1546	g2046	g2546	g3046	g3546	g4046	g4546	g5046	g5546
g1050	g1550	g2050	g2550	g3050	g3550	g4050	g4550	g5050	g5550
g1052	g1552	g2052	g2552	g3052	g3552	g4052	g4552	g5052	g5552
g1054	g1554	g2054	g2554	g3054	g3554	g4054	g4554	g5054	g5554
g1056	g1556	g2056	g2556	g3056	g3556	g4056	g4556	g5056	g5556
g1060	g1560	g2060	g2560	g3060	g3560	g4060	g4560	g5060	g5560
g1100	g1600	g2100	g2600	g3100	g3600	g4100	g4600	g5100	g5600
g1110	g1610	g2110	g2610	g3110	g3610	g4110	g4610	g5110	g5610
g1112	g1612	g2112	g2612	g3112	g3612	g4112	g4612	g5112	g5612
g1114	g1614	g2114	g2614	g3114	g3614	g4114	g4614	g5114	g5614

Suffix Codes are available for other Wires (n=1..MaxWires):

Wire n
g1000.n
g1002.n
g1020.n
g1030.n
g1040.n
g1050.n

2.9.2 KW

g7000	Lump / Neck errors counter xx (fixed resolution)	Value	R
g7010	Lump errors counter xx (fixed resolution)	Value	R
g7020	Neck errors counter xx (fixed resolution)	Value	R
g7030	Last Lump Size (KW RS) x.xxxx (global units, global resolution)	Value	R
g7040	Last Neck Size (KW RS) x.xxxx (global units, global resolution)	Value	R

Suffix Codes are available for KW Gauges (n=1..6):

KW n
g7000.n
g7010.n
g7020.n
g7030.n
g7040.n

2.9.3 Spark Tester

g7100	Errors counter xx (fixed resolution)	Value	R
g7110	Current Voltage (CI-SP/CMK) x.xxxx (kV, voltage resolution)	Value	R

Suffix Codes are available for Spark Tester Gauges (n=1..6):

Spark Tester n
g7100.n
g7110.n

2.9.4 Preheater WST

g7150	Calculated Temperature x.xxxx (global resolution)	Value	R
g7152	Heated Wire Temperature x.xxxx (global resolution)	Value	R
g7154	Unheated Wire Temperature x.xxxx (global resolution)	Value	R
g7156	Loop Voltage x.xxx (V,Fixed Resolution)		
g7158	Power x.xx (KW,Fixed Resolution)		
g7160	Required Power x.xx (KW,Fixed Resolution)		
g7162	Available Power x.xx (KW,Fixed Resolution)		
g7164	Power Rate x.xx (global resolution)		

2.9.5 Event Gauges

g7200	Events counter xx (fixed resolution)	Value	R
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The codes for other Event Gauges are as the next table:

Event Gauge 1	Event Gauge 2
g7200	g7220

2.9.6 BENDCHECK

g7300	Local Bend x.xxxx (global resolution)	Value	R
g7310	Local Bend Angle x.xx (fixed resolution)	Value	R
g7320	Begin Bar Bend x.xxxx (global resolution)	Value	R
g7330	Begin Bar Bend Angle x.xx (fixed resolution)	Value	R
g7340	End Bar Bend x.xxxx (global resolution)	Value	R

g7350	End Bar Bend Angle x.xx (fixed resolution)	Value	R
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2.9.7 ODEX

g8000	Eccentricity x.xxxx (global resolution)	Value	R
g8010	Angle xxx (0..359 degrees, fixed resolution)	Value	R
g8020	Eccentricity / Angle x.xxxx<SP>xxx (global and fixed resolution)	String	R
g8030	Eccentricity X-component x.xxxx (global resolution)	Value	R
g8040	Eccentricity Y-component x.xxxx (global resolution)	Value	R
g8050	Eccentricity X/Y-component x.xxxx<SP>x.xxxx (global resolution)	String	R
g8060	Relative Standard Deviation Eccentricity x.xxxx<SP>x.xxxx (global resolution)	Value	R
g8070	Concentricity x.xxxx<SP>x.xxxx (global resolution)	Value	R
g8080	Minimum Wallthickness x.xxxx<SP>x.xxxx (global resolution)	Value	R

Suffix Codes are available for ODEX Gauges (n=1..6):

ODEX n
g8000.n
g8010.n
g8020.n
g8030.n
g8040.n
g8050.n
g8060.n
g8070.n
g8080.n

2.9.8 Extruder & Gravimetric Extruder

g8100	Extruder Speed Extruder & Gravimetric Extruder xx (fixed resolution)	Value	R
g8101	Gravimetric Extruder Line Speed x.xxxx (length resolution, m/s or ft/s)	Value	R
g8102	Gravimetric Extruder Screw Yield x.xxxx (Screw Yield resolution)	Value	R
g8104	Gravimetric Extruder Net Weight x.xxxx (Weight resolution)	Value	R

g8105	Gravimetric Extruder Throughput x.xxxx (Throughput resolution)	Value	R
g8106	Gravimetric Extruder Meter Weight x.xxxx (Meter Weight resolution)	Value	R

Gravimetric #1	Gravimetric #2	Gravimetric #3	Gravimetric #4
g8100	g8110	g8120	g8130
g8101	g8111	g8121	g8131
g8102	g8112	g8122	g8132
g8104	g8114	g8124	g8134
g8105	g8115	g8125	g8135
g8106	g8116	g8126	g8136

2.9.9 Host Gauges

g8300	Host Gauge #1 Characteristic 1 Value x.xxxx (global units, global resolution)	Value	R/W
g8310	Host Gauge #1 Characteristic 2 Value x.xxxx (global units, global resolution)	Value	R/W
g8320	Host Gauge #1 Characteristic 3 Value x.xxxx (global units, global resolution)	Value	R/W
g8330	Host Gauge #1 Characteristic 4 Value x.xxxx (global units, global resolution)	Value	R/W

Suffix Codes are available for Host Gauges (n=1..4):

Host Gauge n
g8300.n
g8310.n
g8320.n
g8330.n

2.9.10 UMAC

LMax is the maximum number of Layers of all active transducers.

Wall Thickness Measures

g9000	Average L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	String	R
g9001	Minimum L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	String	R
g9002	Minimum WallVector L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	String	R
g9003	Maximum L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	String	R
g9004	Maximum WallVector L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	String	R

Eccentricity Measures

g9010	Shift Center L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	String	R
g9011	Maximum - Minimum L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	String	R
g9012	Maximum - Minimum Vector L1_xxx<SP>...<SP>LMax_xxx xxxx (global units, global resolution)	String	R
g9013	Shift Center X L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	String	R
g9014	Shift Center Y L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	String	R

Concentricity Measures

g9020	(Maximum - Minimum) / Average L1_xxx<SP>...<SP>LMax_xxx x.xx (%) (fixed resolution)	String	R
g9021	(Minimum / Maximum) L1_xxx<SP>...<SP>LMax_xxx x.xx (%) (fixed resolution)	String	R
g9030	Wall Layer 1 – Transducer Values T1_xxx<SP>...<SP>TMax_xxx x.xxxx (global units, global resolution)	String	R

The codes for other Measured Layers are as the next table:

Layer 1	Layer 2	Layer 3	Layer 4	Layer 5	Layer 6	Layer 7	Layer 8
g9030	g9031	g9032	g9033	g9034	g9035	g9036	g9037

g9050	Wall Transducer 1 – Layer Values L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	String	R
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The codes for other Measured Layers are as the next table:

T1	T2	T3	T4	T5	T6	T7	T8
g9050	g9051	g9052	g9053	g9054	g9055	g9056	g9057

g9100	Position X x.xxxx (global units, global resolution)	Value	R
g9101	Position Y x.xxxx (global units, global resolution)	Value	R
g9102	Distance from Transducer to Product T1_xxx<SP>...<SP>TMax_xxx x.xxxx (global units, global resolution)	String	R

Overall Wall thickness

g9105	Average Overall Transducers Value x.xxxx (global units, global resolution)	Value	R
g9106	Minimum Overall Transducers Value x.xxxx (global units, global resolution)	Value	R
g9107	Minimum WallVector Overall Transducers Value x.xxxx (global units, global resolution)	Value	R
g9108	Maximum Overall Transducers Value x.xxxx (global units, global resolution)	Value	R
g9109	Maximum WallVector Overall Transducers Value x.xxxx (global units, global resolution)	Value	R

Overall Eccentricity

g9115	Shift Center Overall L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	Value	R
g9116	Maximum - Minimum Overall L1_xxx<SP>...<SP>LMax_xxx xxxx (global units, global resolution)	Value	R
g9117	Maximum - Minimum Vector Overall L1_xxx<SP>...<SP>LMax_xxx xxxx (global units, global resolution)	Value	R
g9118	Shift Center X Overall L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	Value	R
g9119	Shift Center Y Overall L1_xxx<SP>...<SP>LMax_xxx x.xxxx (global units, global resolution)	Value	R

Temperature

g9125	Water Temperature x.xxxx (global units, global resolution)	Value	R
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Outer Diameter

g9130	Minimum Outer Diameter x.xxxx (global units, global resolution)	Value	R
g9131	Maximum Outer Diameter x.xxxx (global units, global resolution)	Value	R
g9132	Average Outer Diameter x.xxxx (global units, global resolution)	Value	R
g9133	Outer Ovality x.xxxx (global units, global resolution)	Value	R
g9134	Outer Diameter – Transducer Values T1_xxx<SP>...<SP>TMax_xxx x.xxxx (global units, global resolution)	Value	R

Inner Diameter

g9140	Minimum Inner Diameter x.xxxx (global units, global resolution)	Value	R
g9141	Maximum Inner Diameter x.xxxx (global units, global resolution)	Value	R
g9142	Average Inner Diameter x.xxxx (global units, global resolution)	Value	R
g9143	Inner Ovality x.xxxx (global units, global resolution)	Value	R
g9144	Inner Diameter – Transducer Values T1_xxx<SP>...<SP>TMax_xxx x.xxxx (global units, global resolution)	Value	R

The codes for UMACs are as the table:

UMAC 1	UMAC 2
g9000	g9200
g9001	g9201
g9002	g9202
g9003	g9203
g9004	g9204
g9010	g9210
g9011	g9211
g9012	g9212
g9013	g9213
g9014	g9214
g9020	g9220
g9021	g9221
g9030	g9230
g9050	g9250
g9100	g9300
g9101	g9301
g9105	g9305
g9106	g9306
g9107	g9307
g9108	g9308
g9109	g9309
g9115	g9315
g9116	g9316
g9117	g9317
g9118	g9318
g9119	g9319
g9125	g9325
g9130	g9330
g9131	g9331
g9132	g9332
g9133	g9333
g9134	g9334
g9140	g9340
g9141	g9341
g9142	g9342
g9143	g9343
g9144	g9344

2.9.11 RAYEX D

LMax is the maximum number of Layers + Overall.

L1 is the conductor and have no valid measure for the Wallthickness, eccentricity and concentricity.

WallThickness Measures

g9500	Average Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xxxx (global units, global resolution)	String	R
g9501	Minimum Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xxxx (global units, global resolution)	String	R
g9502	Maximum Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xxxx (global units, global resolution)	String	R
g9503	X+ Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xxxx (global units, global resolution)	String	R
g9504	X- Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xxxx (global units, global resolution)	String	R
g9505	Y+ Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xxxx (global units, global resolution)	String	R
g9506	Y- Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xxxx (global units, global resolution)	String	R

Suffix Codes are available for RAYEX Gauges (n=2..5):

Layer n
g9500.n
g9501.n
g9502.n
g9503.n
g9504.n
g9505.n
g9506.n

n=2: Inner Semiconductor

n=3: Insulation

n=4: Outer SemiConductor

n=5: Overall

Eccentricity Measures

g9510	Shift Center Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xxxx (global units, global resolution)	String	R
g9511	Maximum - Minimum Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xxxx (global units, global resolution)	String	R

g9512	Shift Center X Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xxxx (global units, global resolution)	String	R
g9513	Shift Center Y Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xxxx (global units, global resolution)	String	R

Suffix Codes are available for RAYEX Gauges (n=2..5):

Layer n
g9510.n
g9511.n
g9512.n
g9513.n

n=2: Inner Semiconductor
n=3: Insulation
n=4: Outer SemiConductor
n=5: Overall

Concentricity Measures

g9520	(Maximum - Minimum) / Average Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xx (% , fixed resolution)	String	R
g9521	(Minimum / Maximum) Inner Semi<SP>Insulation<SP> Outer Semi<SP>Overall x.xx (% , fixed resolution)	String	R

Suffix Codes are available for RAYEX Gauges (n=2..5):

Layer n
g9520.n
g9521.n

n=2: Inner Semiconductor
n=3: Insulation
n=4: Outer SemiConductor
n=5: Overall

g9550	Position X x.xxxx (global units, global resolution)	Value	R
g9551	Position Y x.xxxx (global units, global resolution)	Value	R
g9552	Ovality x.xxxx (global units, global resolution)	Value	R
Outer Diameter			
g9575	Minimum Outer Diameter x.xxxx (global units, global resolution)	Value	R
g9576	Maximum Outer Diameter x.xxxx (global units, global resolution)	Value	R

g9577	Average Outer Diameter x.xxxx (global units, global resolution)	Value	R
g9578	X Outer Diameter x.xxxx (global units, global resolution)	Value	R
g9579	Y Outer Diameter x.xxxx (global units, global resolution)	Value	R
Inner Diameter			
g9590	Minimum Inner Diameter x.xxxx (global units, global resolution)	Value	R
g9591	Maximum Inner Diameter x.xxxx (global units, global resolution)	Value	R
g9592	Average Inner Diameter x.xxxx (global units, global resolution)	Value	R
g9593	X Inner Diameter x.xxxx (global units, global resolution)	Value	R
g9594	Y Inner Diameter x.xxxx (global units, global resolution)	Value	R
g9599	Active Layers 01 Conductor 02 Inner SemiConductor 04 Insulation 08 Outer SemiConductor	Multiple	R
Temperatures			
g9710	Xray Temperature X xx(fixed resolution)	Value	R
g9711	Xray Temperature Y xx(fixed resolution)	Value	R
g9715	Control Board Temperature X x.xxxx (global units, global resolution)	Value	R
g9716	Control Board Temperature Y x.xxxx (global units, global resolution)	Value	R

2.9.12 RAYEX S/D XT

LMax is the maximum number of Layers.

LMax is the conductor and have no valid measure for the Wallthickness, eccentricity and concentricity.

WallThickness Measures

g9600	Average L1<SP>...<SP>LMax-1<SP>Overall x.xxxx (global units, global resolution)	String	R
g9601	Minimum L1<SP>...<SP>LMax-1<SP>Overall x.xxxx (global units, global resolution)	String	R

g9602	Maximum L1<SP>...<SP>LMax-1<SP>Overall x.xxxx (global units, global resolution)	String	R
g9603	X+ L1<SP>...<SP>LMax-1<SP>Overall x.xxxx (global units, global resolution)	String	R
g9604	X- L1<SP>...<SP>LMax-1<SP>Overall x.xxxx (global units, global resolution)	String	R
g9605	Y+ L1<SP>...<SP>LMax-1<SP>Overall x.xxxx (global units, global resolution)	String	R
g9606	Y- L1<SP>...<SP>LMax-1<SP>Overall x.xxxx (global units, global resolution)	String	R

Suffix Codes are available for RAYEX Layers (n=1..LMax-1) + Overall (LMax):

Layer n
g9600.n
g9601.n
g9602.n
g9603.n
g9604.n
g9605.n
g9606.n

Eccentricity Measures

g9610	Shift Center L1<SP>...<SP>LMax-1<SP>Overall x.xxxx (global units, global resolution)	String	R
g9611	Maximum - Minimum L1<SP>...<SP>LMax-1<SP>Overall x.xxxx (global units, global resolution)	String	R
g9612	Shift Center X L1<SP>...<SP>LMax-1<SP>Overall x.xxxx (global units, global resolution)	String	R
g9613	Shift Center Y L1<SP>...<SP>LMax-1<SP>Overall x.xxxx (global units, global resolution)	String	R

Suffix Codes are available for RAYEX Layers (n=1..LMax-1) + Overall (LMax):

Layer n
g9610.n
g9611.n
g9612.n
g9613.n

Concentricity Measures

g9620	(Maximum - Minimum) / Average L1<SP>...<SP>LMax-1<SP>Overall x.xx (% , fixed resolution)	String	R
g9621	(Minimum / Maximum) L1<SP>...<SP>LMax-1<SP>Overall x.xx (% , fixed resolution)	String	R

Suffix Codes are available for RAYEX Layers (n=1..LMax-1) + Overall (LMax):

Layer n
g9620.n
g9621.n

g9650	Position X x.xxxx (global units, global resolution)	Value	R
g9651	Position Y x.xxxx (global units, global resolution)	Value	R
g9652	Ovality x.xxxx (global units, global resolution)	Value	R

Outer Diameter

g9675	Minimum Outer Diameter x.xxxx (global units, global resolution)	Value	R
g9676	Maximum Outer Diameter x.xxxx (global units, global resolution)	Value	R
g9677	Average Outer Diameter x.xxxx (global units, global resolution)	Value	R
g9678	X Outer Diameter x.xxxx (global units, global resolution)	Value	R
g9679	Y Outer Diameter x.xxxx (global units, global resolution)	Value	R

Inner Diameter

g9690	Minimum Inner Diameter x.xxxx (global units, global resolution)	Value	R
g9691	Maximum Inner Diameter x.xxxx (global units, global resolution)	Value	R
g9692	Average Inner Diameter x.xxxx (global units, global resolution)	Value	R
g9693	X Inner Diameter x.xxxx (global units, global resolution)	Value	R
g9694	Y Inner Diameter x.xxxx (global units, global resolution)	Value	R

g9699	Active Layers 01 Conductor 02 Inner SemiConductor 04 Insulation 08 Outer SemiConductor	Multiple	R
Temperatures (Only for RAYEX D XT)			
g9710	Xray Temperature X xx(fixed resolution)	Value	R
g9711	Xray Temperature Y xx(fixed resolution)	Value	R
g9715	Control Board Temperature X x.xxxx (global units, global resolution)	Value	R
g9716	Control Board Temperature Y x.xxxx (global units, global resolution)	Value	R

2.10 GROUP H - STATISTIC PREVIOUS VALUES

2.10.1 Characteristic related Previous Coil Values

Characteristic 1			
h2100	Previous coil average x.xxxx (characteristic units and resolution)	Value	R
h2110	Previous coil maximum x.xxxx (characteristic units and resolution)	Value	R
h2120	Previous coil minimum x.xxxx (characteristic units and resolution)	Value	R
h2130	Previous coil standard deviation x.xxxx (characteristic units and resolution)	Value	R
h2140	Previous coil high alarms xx (fixed resolution)	Value	R
h2150	Previous coil low alarms xx (fixed resolution)	Value	R
h2160	Previous coil high pre-alarms xx (fixed resolution)	Value	R
h2170	Previous coil low pre-alarms xx (fixed resolution)	Value	R
h2179	Previous coil Cp x.xxxx (characteristic resolution)	Value	R
h2180	Previous coil Cpk x.xxxx (characteristic resolution)	Value	R

2.10.2 Characteristic related Previous Session Values

Characteristic 1

h2200	Previous session average x.xxxx (characteristic units and resolution)	Value	R
h2210	Previous session maximum x.xxxx (characteristic units and resolution)	Value	R
h2220	Previous session minimum x.xxxx (characteristic units and resolution)	Value	R
h2230	Previous session standard deviation x.xxxx (characteristic units and resolution)	Value	R
h2240	Previous session high alarms xx (fixed resolution)	Value	R
h2250	Previous session low alarms xx (fixed resolution)	Value	R
h2260	Previous session high pre-alarms xx (fixed resolution)	Value	R
h2270	Previous session low pre-alarms xx (fixed resolution)	Value	R
h2279	Previous session Cp x.xxxx (characteristic resolution)	Value	R
h2280	Previous session Cpk x.xxxx (characteristic resolution)	Value	R

2.10.3 Characteristic related Previous SPC Values

Characteristic 1

h2300	SPC - X double bar x.xxxx (characteristic units and resolution)	Value	R
h2310	SPC - R bar x.xxxx (characteristic units and resolution)	Value	R
h2320	SPC - Standard deviation x.xxxx (characteristic units and resolution)	Value	R
h2330	SPC - Cpk x.xxxx (characteristic resolution)	Value	R
h2340	SPC - Cp x.xxxx (characteristic resolution)	Value	R
h2350	SPC - # of subgroups xx (fixed resolution)	Value	R
h2360	SPC - # of samples xx (fixed resolution)	Value	R

h2370	SPC - UCLx x.xxxx (characteristic units and resolution)	Value	R
h2380	SPC - LCLx x.xxxx (characteristic units and resolution)	Value	R
h2390	SPC - UCLr x.xxxx (characteristic units and resolution)	Value	R
h2400	SPC – X bar alarms xx (fixed resolution)	Value	R
h2410	SPC – Range alarms xx (fixed resolution)	Value	R

The codes for the statistic values of other characteristics are as the next table:

Char. 1	Char. 2	Char. 3	Char. 4	Char. 5	Char. 6	Char. 7	Char. 8	Char. 9	Ch.10
h2100	h2600	h3100	h3600	h4100	h4600	h5100	h5600	h6100	h6600
h2110	h2610	h3110	h3610	h4110	h4610	h5110	h5610	h6110	h6610
h2120	h2620	h3120	h3620	h4120	h4620	h5120	h5620	h6120	h6620
h2130	h2630	h3130	h3630	h4130	h4630	h5130	h5630	h6130	h6630
h2140	h2640	h3140	h3640	h4140	h4640	h5140	h5640	h6140	h6640
h2150	h2650	h3150	h3650	h4150	h4650	h5150	h5650	h6150	h6650
h2160	h2660	h3160	h3660	h4160	h4660	h5160	h5660	h6160	h6660
h2170	h2670	h3170	h3670	h4170	h4670	h5170	h5670	h6170	h6670
h2180	h2680	h3180	h3680	h4180	h4680	h5180	h5680	h6180	h6680
h2200	h2700	h3200	h3700	h4200	h4700	h5200	h5700	h6200	h6700
h2210	h2710	h3210	h3710	h4210	h4710	h5210	h5710	h6210	h6710
h2220	h2720	h3220	h3720	h4220	h4720	h5220	h5720	h6220	h6720
h2230	h2730	h3230	h3730	h4230	h4730	h5230	h5730	h6230	h6730
h2240	h2740	h3240	h3740	h4240	h4740	h5240	h5740	h6240	h6740
h2250	h2750	h3250	h3750	h4250	h4750	h5250	h5750	h6250	h6750
h2260	h2760	h3260	h3760	h4260	h4760	h5260	h5760	h6260	h6760
h2270	h2770	h3270	h3770	h4270	h4770	h5270	h5770	h6270	h6770
h2280	h2780	h3280	h3780	h4280	h4780	h5280	h5780	h6280	h6780
h2300	h2800	h3300	h3800	h4300	h4800	h5300	h5800	h6300	h6800
h2310	h2810	h3310	h3810	h4310	h4810	h5310	h5810	h6310	h6810
h2320	h2820	h3320	h3820	h4320	h4820	h5320	h5820	h6320	h6820
h2330	h2830	h3330	h3830	h4330	h4830	h5330	h5830	h6330	h6830
h2340	h2840	h3340	h3840	h4340	h4840	h5340	h5840	h6340	h6840
h2350	h2850	h3350	h3850	h4350	h4850	h5350	h5850	h6350	h6850
h2360	h2860	h3360	h3860	h4360	h4860	h5360	h5860	h6360	h6860
h2370	h2870	h3370	h3870	h4370	h4870	h5370	h5870	h6370	h6870
h2380	h2880	h3380	h3880	h4380	h4880	h5380	h5880	h6380	h6880
h2390	h2890	h3390	h3890	h4390	h4890	h5390	h5890	h6390	h6890
h2400	h2900	h3400	h3900	h4400	h4900	h5400	h5900	h6400	h6900
h2410	h2910	h3410	h3910	h4410	h4910	h5410	h5910	h6410	h6910

Suffix Codes are available for characteristics (n=1..20).

Char. n
h2100.n
h2110.n
h2120.n
h2130.n
h2140.n
h2150.n
h2160.n
h2170.n
h2180.n

h2200.n
h2210.n
h2220.n
h2230.n
h2240.n
h2250.n
h2260.n
h2270.n
h2280.n
h2300.n
h2310.n
h2320.n
h2330.n
h2340.n
h2350.n
h2360.n
h2370.n
h2380.n
h2390.n
h2400.n
h2410.n

2.10.4 Not characteristic related Previous Coil Values

h7010	Previous coil number xx (fixed resolution)	Value	R
h7020	Previous coil date begin mm/dd/yyyy dd/mm/yyyy yyyy/mm/dd (determined by system date format)	String	R
h7030	Previous coil time begin hh:mm hh:mm AM/PM hh:mm:ss hh:mm:ss AM/PM (determined by system time format)	String	R
h7040	Previous coil date end mm/dd/yyyy dd/mm/yyyy yyyy/mm/dd (determined by system date format)	String	R
h7050	Previous coil time end hh:mm hh:mm AM/PM hh:mm:ss hh:mm:ss AM/PM (determined by system time format)	String	R
h7060	Previous coil length x.xxxx (length resolution)	Value	R
h7500	Previous coil event 1 count xx (fixed resolution)	Value	R

The codes for other events are as the next table:

Event 1	Event 2	Event 3	Event 4	Event 5
h7500	h7550	h7600	h7650	h7700

Suffix Codes are available for events (n=1..6).

Event n
h7500.n

2.10.5 Not characteristic related Previous Session Values

h8000	Previous session code xxxxxxxxxxxxxxxx (15 characters)	Wide-String	R
h8010	Previous session operator code xxxxxxxxxxxxxxxx (15 characters)	Wide-String	R
h8020	Previous session date begin mm/dd/yyyy dd/mm/yyyy yyyy/mm/dd (determined by system date format)	String	R
h8030	Previous session time begin hh:mm hh:mm AM/PM hh:mm:ss hh:mm:ss AM/PM (determined by system time format)	String	R
h8040	Previous session date end mm/dd/yyyy dd/mm/yyyy yyyy/mm/dd (determined by system date format)	String	R
h8050	Previous session time end hh:mm hh:mm AM/PM hh:mm:ss hh:mm:ss AM/PM (determined by system time format)	String	R
h8060	Previous session length x.xxxx (length resolution)	Value	R
h8100	Previous produced coils xx (fixed resolution)	Value	R
h8110	Previous produced coils with alarms xx (fixed resolution)	Value	R
h8120	Previous discarded coils xx (fixed resolution)	Value	R
h8500	Previous session event 1 count xx (fixed resolution)	Value	R

The codes for other events are as the next table:

Event 1	Event 2	Event 3	Event 4	Event 5
h8500	h8550	h8600	h8650	h8700

Suffix Codes are available for events (n=1..6).

Event n
h8500.n

2.10.6 Not characteristic related Previous SPC Values

h9020	Previous SPC date begin mm/dd/yyyy dd/mm/yyyy yyyy/mm/dd (determined by system date format)	String	R
h9030	Previous SPC time begin hh:mm hh:mm AM/PM hh:mm:ss hh:mm:ss AM/PM (determined by system time format)	String	R
h9040	Previous SPC date end mm/dd/yyyy dd/mm/yyyy yyyy/mm/dd (determined by system date format)	String	R
h9050	Previous SPC time end hh:mm hh:mm AM/PM hh:mm:ss hh:mm:ss AM/PM (determined by system time format)	String	R

2.11 GROUP L - PC-LINK CODES

I1100	PC-Link command xxxxx...xxxxx (80 characters)	Wide-String	W
I1300	PC-Link screen visibility 00 Off 01 On	Selection	R
I1400	Confirmation of PC-Link screen 00 Off 01 On	Selection	R
I1500	If confirmed, contents of INPUT 1 xxxxx...xxxxx (50 characters)	Wide-String	R

For the other INPUT's it only changes the numbers as the next Table:

INPUT 1	INPUT 2	...	INPUT 99
I1500	I1501	...	I1599

I1600	If confirmed, index selected in SELECT 1 xx (0..255, fixed resolution)	Value	R
-------	---	-------	---

For the other SELECTS's it only changes the numbers as the next Table:

SELECT 1	SELECT 2	...	SELECT 99
I1600	I1601	...	I1699

I1700	If confirmed, contents RADIO_BUTTON 1 xx (0..255, fixed resolution)	Value	R
-------	--	-------	---

For the other RADIO_BUTTON's it only changes the numbers as the next Table:

RADIO_BUTTON 1	RADIO_BUTTON 2	...	RADIO_BUTTON 99
I1700	I1701	...	I1799

I1800 If confirmed, contents of CHECK_BOX #1 Value R
 xx ($0..2^{32}-1$, fixed resolution)

For the other CHECK_BOX it only changes the numbers as the next Table:

CHECK_BOX 1	CHECK_BOX 2	...	CHECK_BOX 99
I1800	I1801	...	I1899

I1900 If confirmed, selection of SELECT 1 Wide-String R
 xxxxx...xxxxx (50 characters)

For the other SELECT's it only changes the numbers as the next Table:

SELECT 1	SELECT 2	...	SELECT 99
I1900	I1901	...	I1999

2.12 GROUP M - GROUPED MESSAGES

To make simpler the communication, it is possible to combine several simple messages in a grouped message. You can define up to 20 different combined messages.

For each one of the possible configuration messages there are two associated orders, one to configure (or read) the message format, and other to get/set the present values previously defined.

m1000	Grouped message # 0 configuration Message configuration string	String	R/W
m1005	Get/Set grouped message #0 Several values as configured	Grouped	R/W

The other grouped messages have the following codes:

Message #	Format	Values
0	m1000	M1005
1	m1010	M1015
2	m1020	M1025
3	m1030	M1035
4	m1040	M1045
5	m1050	M1055
6	m1060	M1065
7	m1070	M1075
8	m1080	M1085
9	m1090	M1095
10	m1100	M1105
11	m1110	M1115
12	m1120	M1125
13	m1130	M1135
14	m1140	M1145
15	m1150	M1155
16	m1160	M1165
17	m1170	M1175
18	m1180	M1185
19	m1190	M1195

All these messages can be placed to work in Automatic mode (see Group Z - Automatic values). In the case they were not used in automatic mode they can be used in standard form.

2.12.1 Errors process

When programming a new grouped order using the messages m1000 to m1190, the USYS will answer an <ACK> when all the single messages in the group are accepted, and with a <NAK> when there is at least one in error. For example, the USYS will answer a <NAK> to the following grouped message configuration because there is an unknown order (c2820):

```
M1000<SP>c1110<HT>c2100<HT>c2820<HT>c2140<HT>c2600<HT>c2620<CR>
```

It is not possible to combine in the same grouped message orders "Read Only" with orders "Write Only". For example the following message will be answered by a <NAK> :

```
M1000<SP>f1120<HT>d8060<CR>
```

This grouped order is not valid because we are mixing in the same grouped message a Write Only order (Stop/Start Lot F1120) and a Read Only order (Current lot length: d8060)

It is not valid either to add in a grouped message an alarm error command (e1000, e1010, e1015) or a group M command.

Once programmed and accepted by an <ACK> a new order, we can begin to use the corresponding Set/Get message using the orders m1005 to m1195. The USYS will execute the orders of a grouped message sequentially, beginning with the first one placed on the grouped message.

In the DOWNLOAD grouped messages (orders M1005 to M1195) the USYS will answer an <ACK> when all the single messages in the group are accepted, and with a <NAK> when there is at least one in error, stopping execution on this erroneous message. The Host computer can use the messages e1000 and e1010 to know the single message inside a group that produced the error.

In the UPLOAD grouped messages (m1005 to m1195) the USYS will answer with a complete grouped message placing a <ESC> in the position of the order in error (if any). Using the messages e1010 and e1015 the Host computer can know the error and the first message not executed.

2.12.2 Examples

If we want to get the current value of the characteristics, we define a grouped message with:

```
Number of active Characteristics (c1195)
Characteristic #1 Current Value (d2000)
Characteristic #2 Current Value (d2500)
Characteristic #3 Current Value (d3000)
Characteristic #4 Current Value (d3500)
Characteristic #5 Current Value (d4000)
```

```
M1000<SP>c1195<HT>d2000<HT>d2500<HT>d3000<HT>d3500<HT>d4000<CR>
or
M1000<SP>c1195<HT>d2000.1.5<CR>
```

The same order can be used to read the present configuration i.e. if the Host sends the Upload order m1000<CR> it will receive the present configuration of the grouped message #1:

```
m1000<SP>c1195<HT>d2000<HT>d2500<HT>d3000<HT>d3500<HT>d4000<CR>
or
m1000<SP>c1195<HT>d2000.1.5<CR>
```

Once the grouped message has been configured, we will get current characteristic values with upload message:

```
m1005<CR>
```

USYS will answer:

```
m1005<SP>4<HT>2.324<HT>4.354<HT>2.345<HT>1.958<CR>
```

However, if we try to use this grouped command with a product with two characteristics, USYS will answer:

```
m1005<SP>2<HT>2.324<HT>4.354<HT><ESC><HT><ESC><HT><ESC><CR>
```

Characteristic #1, #2 values has been answered, but not characteristic #3, #4 and #5. If we request the error messages, the answers will be:

e1000<SP>0<CR>	No error on m1005 order
e1010<SP>d4000<CR>	Last grouped order with error
e1015<SP>203<CR>	Error code on d4000

To get the characteristic value of all product characteristic we can also use the suffix 0, then, it doesn't matter how many characteristics are defined in the product recipe.

```
M1000<SP>c1195<HT>d2000.0<CR>
```

With a product of two characteristics,USYS will answer:

```
m1005<SP>2<HT>2.324<HT>4.354<CR>
```

With a product of four characteristics,USYS will answer:

```
m1005<SP>4<HT>2.324<HT>4.354<CR><HT>2.345<HT>1.958<CR>
```

To change the active product Set points we can define a grouped message with the values to be changed:

Product Name (c1110)
Characteristic #1 Nominal (c2100)
Characteristic #1 Upper Tolerance (c2120)
Characteristic #1 Lower Tolerance (c2140)
Characteristic #2 Nominal (c2600)
Characteristic #2 Upper Tolerance (c2620)
Characteristic #2 Lower Tolerance (c2640)

```
M1000<SP>c1110<HT>c2100<HT>c2120<HT>c2140<HT>c2600<HT>c2620<HT>c2640<CR>
```

Upload order m1000<CR> will receive:

```
m1000<SP>c1110<HT>c2100<HT>c2120<HT>c2140<HT>c2600<HT>c2620<HT>c2640<CR>
```

When the grouped message has been configured, we can use it to change or read the Active product values. For example, to set new values we must send the following orders:

Open the active Product:

```
C0100<SP>1<CR>
C1010<CR>
```

Send the new values:

```
M1005<SP><HT>5.30<HT>5.35<HT>5.25<HT>12.70<HT>12.75<HT>12.65<CR>
```

Activate the product and close working area:

```
C1000<CR>
C0100<SP>0<CR>
```

The three previous messages could be combined in a single grouped message that include the orders to Open the active product, change the Set points and Activate the product (see forward).

Conversely we can read the values in the active product:

Open the active Product:

```
C1010<CR>
```

Request the grouped message:

```
m1005<CR>
```

As answer the USYS will send:

```
m1005<SP><HT>5.30<HT>5.35<HT>5.25<HT>12.70<HT>12.75<HT>12.65<CR>
```

We can combine in a grouped message any type of orders. We can mix orders without a value associated with orders that must carry values.

The following order will program a grouped message that will reset the current statistics, load a new product from the product table, change the line speed and activate it:

```
M1010<SP>f1200<HT>c0100<HT>c1025<HT>c1140<HT>c1000<HT>c0100<CR>
```

To use the previous message Host will send the following order:

```
M1015<SP>1<SP>Product Name<HT>20.45<SP>0<CR>
```

It is important to note that the orders without values (like the f1200 or c1000 in the previous example) must not be included in the grouped order.

2.13 GROUP Z - AUTOMATIC VALUES

Using the following commands the Host can program the USYS to send automatically some messages at the end of one statistical period, at the end of a time or length period, or for a change in the working environment.

To summarize, the next list shows the entire automatic events defined in this chapter:

- Statistics automatic grouped messages
 - Piece end
 - Lot end
 - SPC end
- Periodic automatic grouped messages
 - Periodic 1
 - Periodic 2
 - Periodic 3
 - Periodic 4
- Not-grouped automatic messages
 - Change in working conditions
 - Change in statistics status
 - New SPC subgroup

2.13.1 Statistics automatic grouped messages

These automatic messages are generated at the end of one statistical period: Piece, Lot, or SPC.

They must be associated to a grouped message, which will be generated when event occurs. The grouped message could be configured with any message from m1005 to m1195. For example, if piece end event is configured with the grouped message m1005, when a piece end is detected m1005 will be generated containing the values previously selected by the means of the message m1000.

The following table shows the relation between Automatic and the pre-assigned Grouped messages:

Event	Automatic On/Off	Grouped selection
Piece end	z0110	z0115
Lot end	z0120	z0125
SPC end	z0140	z0145

When these events are generated, related statistics values should be read using group D commands (see Group D - Current Measurements and Statistics). When a not-automatic upload is done, remember that group D values will be valid until next start event (see statistic status event). So when distance between stop and start is short it is better to wait for the start event and upload previous statistics using group H commands (see Group H - Statistic Previous Values).

2.13.2 Piece automatic message

At the end of each piece the USYS will send the configured grouped message with the user programmed values.

z0110	Piece Automatic Values 0 Off 1 On	Selection	R/W
z0115	Piece end automatic grouped answer 00 m1005 01 m1015 02 m1025 03 m1035 04 m1045 05 m1055 06 m1065 07 m1075	Selection	R/W

08 m1085
 09 m1095
 10 m1105
 11 m1115
 12 m1125
 13 m1135
 14 m1145
 15 m1155
 16 m1165
 17 m1175
 18 m1185
 19 m1195

2.13.3 Lot automatic message

At the end of each lot the USYS will send the configured grouped message with the user programmed values.

z0120	Lot Automatic Values 0 Off 1 On	Selection	R/W
z0125	Lot end automatic grouped answer (see codes on z0115 command)	Selection	R/W

2.13.4 SPC End automatic message

At the end of each SPC the USYS will send the configured grouped message with the user programmed values.

z0140	SPC Automatic Values 0 Off 1 On	Selection	R/W
z0145	SPC end automatic grouped answer (see codes on z0115 command)	Selection	R/W

2.13.5 Periodic automatic grouped messages

Automatic periodic commands generate an event at the end of a configurable period. When event occurs a configurable grouped message will be sent. USYS Host provides four periodic events to allow the user to use different periods at same time.

The period size could be measured with time (seconds) or length (meters or feet) units. When period uses time units, the event occurs periodically since it is activated. When using length units, event occurs only while producing piece, resetting period at piece begin.

z0200	Periodic automatic values #1 00 Off 01 On	Selection	R/W
z0201	Periodic automatic period type #1 00 Length 01 Time	Selection	R/W
z0202	Periodic automatic period size #1 x.x Length period type: length resolution, length units Time period type: time resolution, seconds	Value	R/W

Note: The minimum valid value for periodic interval is 0.1 seconds / meters / feet for all the applications but for the ones running in a le7 Hardware where the minimum value is 0.05.

z0205 Periodic automatic grouped answer #1 Selection R/W
(see codes on z0115 command)

Periodic # 1	Periodic # 2	Periodic # 3	Periodic # 4
z0200	z0210	z0220	z0230
z0201	z0211	z0221	z0231
z0202	z0212	z0222	z0232
z0205	z0215	z0225	z0235

2.13.6 Not-grouped automatic messages

These messages are not associated with a grouped message and are intended to inform to the Host computer of the system events in Real Time.

The following table summarizes the relation between not-grouped automatic events and its configuration and automatic commands:

Automatic On/Off	Command	Event
z0300	z1000	Change in working conditions
z0310	z1050	Change in statistics status
z0320	z1100...z1109...	New SPC subgroup
z0340	z1200...z1209...	Measuring Head Errors
z0350	z1250	Lot Log Incidences
z0360	z1260	Event Log Incidences
z0370	z1300	General Status

2.13.7 Working Conditions

When there is any change in the working conditions the USYS sends an automatic message.

z0300 Automatic working conditions message Value R/W
0 Off
1 On

USYS answer:

z1000 Status Change Selection R
00 New Product (Activation)
01 New Product (Apply)
02 New Operator Code
03 New Lot Code
06 Pause ON
07 Pause OFF

2.13.8 Statistics Status

z0310 Automatic statistics conditions messages Value R/W
0 Off
1 On

USYS answer:

z1050 Statistics Status Change Selection R
00 Start Piece
01 Start Lot
03 Start SPC
04 Stop Piece

05 Stop Lot
 07 Stop SPC
 08 Reset Piece
 09 Reset Lot
 11 Reset SPC
 12 Discard Piece
 13 Discard SPC
 14 Start FFT
 15 Stop FFT

2.13.9 Characteristics SPC subgroups

At the end of a SPC subgroup of any characteristic the USYS sends an automatic message with all the SPC samples that form the subgroup.

z0320 SPC Automatic Subgroup Values Selection R/W
 0 Off
 1 On

USYS answer:

z1100 SPC subgroup samples String R
 x.xxxx<HT>x.xxxx<HT>...<HT>x.xxxx

The other automatic SPC-characteristics subgroups messages have the following codes:

Char #1	Char #2	Char #3	Char #4	Char #XX
z1100	z1101	z1102	z1103	...

2.13.10 Measuring Heads State

Each new Measuring Head Error is sent by USYS through an automatic message

z0340 Measuring Head State Selection R/W
 0 Off
 1 On

USYS answer:

z1200 Measuring Head State Value R
 Status for J-Interface gauges
 00 Ok
 01 Not Measuring
 02 Calibrating
 03 Error
 xxx Error code (See CP-310 manual)
 800 Communication not assigned
 900 Gauge not answer.
 Status for Analog gauges
 00 Calibrating
 01 End of Calibration
 02 Ok
 03 Error
 04 Not Measuring
 05 IO Channel not assigned

2.13.11 Lot Log Incidences

Each new Lot Log Incidence is sent by USYS through an automatic message with the same format at the screen.

z0350	Lot Log Incidences 0 Off 1 On	Selection	R/W
-------	-------------------------------------	-----------	-----

z0355	Lot Log Incidences Selection 01 Alarms 02 SPC Alarms 04 Periodic Data 08 Line Status 16 Gauge Pause 32 FFT Alarms 64 Prealarms 128 Events	Multiple	R/W
-------	---	----------	-----

USYS answer:

z1250	Lot Log Incidence String x.xxxx<HT>x.xxxx<HT>...<HT>x.xxxx	String	R
-------	---	--------	---

2.13.12 Event Log Incidences

Each new Event Log Incidence is sent by USYS through an automatic message with the same format at the screen.

z0360	Event Log Incidence 0 Off 1 On	Selection	R/W
-------	--------------------------------------	-----------	-----

USYS answer:

z1260	Event Log Incidence String x.xxxx<HT>x.xxxx<HT>...<HT>x.xxxx	String	R
-------	---	--------	---

2.13.13 General Status

Automatic transmission when there is any change in the error status of the system.

z0370	Automatic general status message 0 Off 1 On	Selection	R/W
-------	---	-----------	-----

USYS answer:

z1300	General Status 00 No Errors 01 Any Characteristic Alarm 02 Any Characteristic Prealarm 04 Any Event 08 Low Line Speed 16 Gauge Error 32 Printer Error 64 Host Error 128 Remote Display Error 256 Analog Peripheral Error 512 Controllers Limits 1024 DLP Limits	Multiple	R
-------	---	----------	---

2.13.14 Weight Cutter

Each new cutter length calculated by the Weight Cutter gauge, is sent by USYS through an automatic message.

Host



USYS

z1450<SP>2.2345<CR>
z1450<SP>2.3355<CR>
z1450<SP>2.3225<CR>
Etc.

2.13.15 Examples

Program a grouped message that the USYS will send at the end of each piece.

```
M1060<SP>d7010<HT>d7060<HT>d2100<HT>d2110<HT>d2120<HT>...  
...d2600<HT>d2610<HT>d2620<HT>d3100<HT>d3110<HT>d3120<CR>
```

With the previous command Host programs the grouped message #6 with the following fields:

- Piece Number
- Piece Length
- Characteristic #1 Average
- Characteristic #1 Maximum
- Characteristic #1 Minimum
- Characteristic #2 Average
- Characteristic #2 Maximum
- Characteristic #2 Minimum
- Characteristic #3 Average
- Characteristic #3 Maximum
- Characteristic #3 Minimum

```
Z0115<SP>6<CR> //Assign grouped message #6 to piece end event  
Z0110<SP>1<CR> //Activate grouped message #6  
C1040<SP>54<CR> //Activate product from table #54
```

At this point the USYS will send the programmed piece summary at the end of each piece:

```
m1065<SP>345<HT>1024.5<HT>43.8<HT>44.3<HT>43.2<HT>...  
...<HT>24.5<HT>24.9<HT>24.2<HT>750<HT>775<HT>745<CR>
```

Program a grouped message that the USYS will send periodically.

```
M1100<SP>d2000<HT>d2500<HT>d3000<HT>d3500<HT>d4000<CR>
```

With the previous order, the Host programs the Grouped message #10 with the following fields:

- Current Characteristic #1 Value
- Current Characteristic #2 Value
- Current Characteristic #3 Value
- Current Characteristic #4 Value
- Current Characteristic #5 Value

Select automatic period type, period size, period value and grouped message associated:

```
Z0201<SP>1<CR> //Period type: by time  
Z0202<SP>1.5<CR> //Period size: 1.5 seconds  
Z0205<SP>10<CR> //Assign grouped message #10 to the first  
//automatic periodic command  
Z0200<SP>1<CR> //Activate the automatic periodic command
```

At this point the USYS will send the programmed message at each programmed period, in this case each 1.5 seconds.

Activate the automatic working conditions message.

z0300<SP>1<CR>

As answer, every time there is a change in the working conditions, USYS will send:

z1000<SP>0<CR> //New product activation

z1000<SP>6<CR> //Pause On

z1000<SP>7<CR> //Pause Off

Host computer can also read directly the status of this message:

z1000<CR>

As answer, Usys will send the last event registered:

z1000<SP>7<CR> //Pause Off

Activate the automatic Statistics Status message.

z0310<SP>1>CR>

As answer, every time there is a statistical change, USYS will send:

z0150<SP>00<CR> //USYS notifies a Piece Start

z0150<SP>04<CR> //USYS notifies a Piece Stop

z0150<SP>08<CR> //USYS notifies a Piece Reset

...

Host computer can also read directly the status of this message:

z0150<CR>

As answer, Usys will send the last change registered:

z0150<SP>08<CR> //USYS notifies a Piece Reset

2.14 APPENDIX A: LIST OF CHARACTERISTICS AND ITS CODES

2.14.1 Single Characteristics

0 No characteristic

Single Plane ODAC

1000 Diameter 1

1001 Position 1

1010 Diameter 2

1011 Position 2

1020 Diameter 3

1021 Position 3

1030 Diameter 4

1031 Position 4

1040 Diameter 5

1041 Position 5

1050 Diameter 6

1051 Position 6

1060 Diameter 7

1061 Position 7

1070 Diameter 8

1071 Position 8

19000 Perimeter 1

19001 Area 1

19002 Meter Weight 1

19010 Perimeter 2

19011 Area 2

19012 Meter Weight 2
 19020 Perimeter 3
 19021 Area 3
 19022 Meter Weight 3
 19030 Perimeter 4
 19031 Area 4
 19032 Meter Weight 4
 19040 Perimeter 5
 19041 Area 5
 19042 Meter Weight 5
 19050 Perimeter 6
 19051 Area 6
 19052 Meter Weight 6
 19060 Perimeter 7
 19061 Area 7
 19062 Meter Weight 7
 19070 Perimeter 8
 19071 Area 8
 19072 Meter Weight 8

ODAC XY

1100 Diameter X-Y 1
 1101 Ovality X-Y 1
 1102 Smaller X-Y 1
 1103 Larger X-Y 1
 1104 Diameter X 1
 1105 Diameter Y 1
 1106 Position X 1
 1107 Position Y 1
 1110 Diameter X-Y 2
 1111 Ovality X-Y 2
 1112 Smaller X-Y 2
 1113 Larger X-Y 2
 1114 Diameter X 2
 1115 Diameter Y 2
 1116 Position X 2
 1117 Position Y 2
 1120 Diameter X-Y 3
 1121 Ovality X-Y 3
 1122 Smaller X-Y 3
 1123 Larger X-Y 3
 1124 Diameter X 3
 1125 Diameter Y 3
 1126 Position X 3
 1127 Position Y 3
 1130 Diameter X-Y 4
 1131 Ovality X-Y 4
 1132 Smaller X-Y 4
 1133 Larger X-Y 4
 1134 Diameter X 4
 1135 Diameter Y 4
 1136 Position X 4
 1137 Position Y 4
 1140 Diameter X-Y 5
 1141 Ovality X-Y 5
 1142 Smaller X-Y 5
 1143 Larger X-Y 5
 1144 Diameter X 5
 1145 Diameter Y 5
 1146 Position X 5
 1147 Position Y 5
 1150 Diameter X-Y 6
 1151 Ovality X-Y 6
 1152 Smaller X-Y 6

1153	Larger X-Y 6
1154	Diameter X 6
1155	Diameter Y 6
1156	Position X 6
1157	Position Y 6
1160	Diameter X-Y 7
1161	Ovality X-Y 7
1162	Smaller X-Y 7
1163	Larger X-Y 7
1164	Diameter X 7
1165	Diameter Y 7
1166	Position X 7
1167	Position Y 7
1170	Diameter X-Y 8
1171	Ovality X-Y 8
1172	Smaller X-Y 8
1173	Larger X-Y 8
1174	Diameter X 8
1175	Diameter Y 8
1176	Position X 8
1177	Position Y 8
19100	Perimeter X-Y 1
19101	Area X-Y 1
19102	Meter Weight X-Y 1
19110	Perimeter X-Y 2
19111	Area X-Y 2
19112	Meter Weight X-Y 2
19120	Perimeter X-Y 3
19121	Area X-Y 3
19122	Meter Weight X-Y 3
19130	Perimeter X-Y 4
19131	Area X-Y 4
19132	Meter Weight X-Y 4
19140	Perimeter X-Y 5
19141	Area X-Y 5
19142	Meter Weight X-Y 5
19150	Perimeter X-Y 6
19151	Area X-Y 6
19152	Meter Weight X-Y 6
19160	Perimeter X-Y 7
19161	Area X-Y 7
19162	Meter Weight X-Y 7
19170	Perimeter X-Y 8
19171	Area X-Y 8
19172	Meter Weight X-Y 8

ODAC 2XY

1200	Diameter X-Y-S-T 1
1201	Ovality X-Y-S-T 1
1202	Ovality-90 X-Y-S-T 1
1203	Smaller X-Y-S-T 1
1204	Larger X-Y-S-T 1
1205	Diameter X 1
1206	Diameter Y 1
1207	Diameter S 1
1208	Diameter T 1
1209	Position X 1
1210	Position Y 1
1211	Position S 1
1212	Position T 1
1220	Diameter X-Y-S-T 2
1221	Ovality X-Y-S-T 2
1222	Ovality-90 X-Y-S-T 2
1223	Smaller X-Y-S-T 2

1224 Larger X-Y-S-T 2
 1225 Diameter X 2
 1226 Diameter Y 2
 1227 Diameter S 2
 1228 Diameter T 2
 1229 Position X 2
 1230 Position Y 2
 1231 Position S 2
 1232 Position T 2
 1240 Diameter X-Y-S-T 3
 1241 Ovality X-Y-S-T 3
 1242 Ovality-90 X-Y-S-T 3
 1243 Smaller X-Y-S-T 3
 1244 Larger X-Y-S-T 3
 1245 Diameter X 3
 1246 Diameter Y 3
 1247 Diameter S 3
 1248 Diameter T 3
 1249 Position X 3
 1250 Position Y 3
 1251 Position S 3
 1252 Position T 3
 1260 Diameter X-Y-S-T 4
 1261 Ovality X-Y-S-T 4
 1262 Ovality-90 X-Y-S-T 4
 1263 Smaller X-Y-S-T 4
 1264 Larger X-Y-S-T 4
 1265 Diameter X 4
 1266 Diameter Y 4
 1267 Diameter S 4
 1268 Diameter T 4
 1269 Position X 4
 1270 Position Y 4
 1271 Position S 4
 1272 Position T 4
 1280 Diameter X-Y-S-T 5
 1281 Ovality X-Y-S-T 5
 1282 Ovality-90 X-Y-S-T 5
 1283 Smaller X-Y-S-T 5
 1284 Larger X-Y-S-T 5
 1285 Diameter X 5
 1286 Diameter Y 5
 1287 Diameter S 5
 1288 Diameter T 5
 1289 Position X 5
 1290 Position Y 5
 1291 Position S 5
 1292 Position T 5
 1300 Diameter X-Y-S-T 6
 1301 Ovality X-Y-S-T 6
 1302 Ovality-90 X-Y-S-T 6
 1303 Smaller X-Y-S-T 6
 1304 Larger X-Y-S-T 6
 1305 Diameter X 6
 1306 Diameter Y 6
 1307 Diameter S 6
 1308 Diameter T 6
 1309 Position X 6
 1310 Position Y 6
 1311 Position S 6
 1312 Position T 6
 1320 Diameter X-Y-S-T 5
 1321 Ovality X-Y-S-T 5
 1322 Ovality-90 X-Y-S-T 5
 1323 Smaller X-Y-S-T 5

1324 Larger X-Y-S-T 5
 1325 Diameter X 5
 1326 Diameter Y 5
 1327 Diameter S 5
 1328 Diameter T 5
 1329 Position X 5
 1330 Position Y 5
 1331 Position S 5
 1332 Position T 5

 19200 Perimeter X-Y-S-T 1
 19201 Area X-Y-S-T 1
 19202 Meter Weight X-Y-S-T 1
 19210 Perimeter X-Y-S-T 2
 19211 Area X-Y-S-T 2
 19212 Meter Weight X-Y-S-T 2
 19220 Perimeter X-Y-S-T 3
 19221 Area X-Y-S-T 3
 19222 Meter Weight X-Y-S-T 3
 19230 Perimeter X-Y-S-T 4
 19231 Area X-Y-S-T 4
 19232 Meter Weight X-Y-S-T 4
 19240 Perimeter X-Y-S-T 5
 19241 Area X-Y-S-T 5
 19242 Meter Weight X-Y-S-T 5
 19250 Perimeter X-Y-S-T 6
 19251 Area X-Y-S-T 6
 19252 Meter Weight X-Y-S-T 6
 19260 Perimeter X-Y-S-T 7
 19261 Area X-Y-S-T 7
 19262 Meter Weight X-Y-S-T 7
 19270 Perimeter X-Y-S-T 8
 19271 Area X-Y-S-T 8
 19272 Meter Weight X-Y-S-T 8

ODAC with Statistics

1700 Averaged Statistical of ODAC 1
 1701 Ovality Statistical (Max-Min) of ODAC 1
 1702 Minimum Statistical of ODAC 1
 1703 Maximum Statistical of ODAC 1
 1710 Averaged Statistical of ODAC 2
 1711 Ovality Statistical (Max-Min) of ODAC 2
 1712 Minimum Statistical of ODAC 2
 1713 Maximum Statistical of ODAC 2
 1720 Averaged Statistical of ODAC 3
 1721 Ovality Statistical (Max-Min) of ODAC 3
 1722 Minimum Statistical of ODAC 3
 1723 Maximum Statistical of ODAC 3
 1730 Averaged Statistical of ODAC 4
 1731 Ovality Statistical (Max-Min) of ODAC 4
 1732 Minimum Statistical of ODAC 4
 1733 Maximum Statistical of ODAC 4
 1740 Averaged Statistical of ODAC 5
 1741 Ovality Statistical (Max-Min) of ODAC 5
 1742 Minimum Statistical of ODAC 5
 1743 Maximum Statistical of ODAC 5
 1750 Averaged Statistical of ODAC 6
 1751 Ovality Statistical (Max-Min) of ODAC 6
 1752 Minimum Statistical of ODAC 6
 1753 Maximum Statistical of ODAC 6
 19500 Averaged Statistical of ODAC 7
 19501 Ovality Statistical (Max-Min) of ODAC 7
 19502 Minimum Statistical of ODAC 7
 19503 Maximum Statistical of ODAC 7
 19510 Averaged Statistical of ODAC 8

19511 Ovality Statistical (Max-Min) of ODAC 8
 19512 Minimum Statistical of ODAC 8
 19513 Maximum Statistical of ODAC 8

 1760 Averaged Statistical of ODAC X 1
 1761 Ovality Statistical (Max-Min) of ODAC X 1
 1762 Minimum Statistical of ODAC X 1
 1763 Maximum Statistical of ODAC X 1
 1770 Averaged Statistical of ODAC X 2
 1771 Ovality Statistical (Max-Min) of ODAC X 2
 1772 Minimum Statistical of ODAC X 2
 1773 Maximum Statistical of ODAC X 2
 1780 Averaged Statistical of ODAC X 3
 1781 Ovality Statistical (Max-Min) of ODAC X 3
 1782 Minimum Statistical of ODAC X 3
 1783 Maximum Statistical of ODAC X 3
 1790 Averaged Statistical of ODAC X 4
 1791 Ovality Statistical (Max-Min) of ODAC X 4
 1792 Minimum Statistical of ODAC X 4
 1793 Maximum Statistical of ODAC X 4
 1800 Averaged Statistical of ODAC X 5
 1801 Ovality Statistical (Max-Min) of ODAC X 5
 1802 Minimum Statistical of ODAC X 5
 1803 Maximum Statistical of ODAC X 5
 1810 Averaged Statistical of ODAC X 6
 1811 Ovality Statistical (Max-Min) of ODAC X 6
 1812 Minimum Statistical of ODAC X 6
 1813 Maximum Statistical of ODAC X 6
 19520 Averaged Statistical of ODAC X 7
 19521 Ovality Statistical (Max-Min) of ODAC X 7
 19522 Minimum Statistical of ODAC X 7
 19523 Maximum Statistical of ODAC X 7
 19530 Averaged Statistical of ODAC X 8
 19531 Ovality Statistical (Max-Min) of ODAC X 8
 19532 Minimum Statistical of ODAC X 8
 19533 Maximum Statistical of ODAC X 8

 1820 Averaged Statistical of ODAC Y 1
 1821 Ovality Statistical (Max-Min) of ODAC Y 1
 1822 Minimum Statistical of ODAC Y 1
 1823 Maximum Statistical of ODAC Y 1
 1830 Averaged Statistical of ODAC Y 2
 1831 Ovality Statistical (Max-Min) of ODAC Y 2
 1832 Minimum Statistical of ODAC Y 2
 1833 Maximum Statistical of ODAC Y 2
 1840 Averaged Statistical of ODAC Y 3
 1841 Ovality Statistical (Max-Min) of ODAC Y 3
 1842 Minimum Statistical of ODAC Y 3
 1843 Maximum Statistical of ODAC Y 3
 1850 Averaged Statistical of ODAC Y 4
 1851 Ovality Statistical (Max-Min) of ODAC Y 4
 1852 Minimum Statistical of ODAC Y 4
 1853 Maximum Statistical of ODAC Y 4
 1860 Averaged Statistical of ODAC Y 5
 1861 Ovality Statistical (Max-Min) of ODAC Y 5
 1862 Minimum Statistical of ODAC Y 5
 1863 Maximum Statistical of ODAC Y 5
 1870 Averaged Statistical of ODAC Y 6
 1871 Ovality Statistical (Max-Min) of ODAC Y 6
 1872 Minimum Statistical of ODAC Y 6
 1873 Maximum Statistical of ODAC Y 6
 19540 Averaged Statistical of ODAC Y 7
 19541 Ovality Statistical (Max-Min) of ODAC Y 7
 19542 Minimum Statistical of ODAC Y 7
 19543 Maximum Statistical of ODAC Y 7

19550 Averaged Statistical of ODAC Y 8
 19551 Ovality Statistical (Max-Min) of ODAC Y 8
 19552 Minimum Statistical of ODAC Y 8
 19553 Maximum Statistical of ODAC Y 8

 1880 Averaged Statistical of ODAC Z 1
 1881 Ovality Statistical (Max-Min) of ODAC Z 1
 1882 Minimum Statistical of ODAC Z 1
 1883 Maximum Statistical of ODAC Z 1
 1890 Averaged Statistical of ODAC Z 2
 1891 Ovality Statistical (Max-Min) of ODAC Z 2
 1892 Minimum Statistical of ODAC Z 2
 1893 Maximum Statistical of ODAC Z 2
 1900 Averaged Statistical of ODAC Z 3
 1901 Ovality Statistical (Max-Min) of ODAC Z 3
 1902 Minimum Statistical of ODAC Z 3
 1903 Maximum Statistical of ODAC Z 3
 1910 Averaged Statistical of ODAC Z 4
 1911 Ovality Statistical (Max-Min) of ODAC Z 4
 1912 Minimum Statistical of ODAC Z 4
 1913 Maximum Statistical of ODAC Z 4
 1920 Averaged Statistical of ODAC Z 5
 1921 Ovality Statistical (Max-Min) of ODAC Z 5
 1922 Minimum Statistical of ODAC Z 5
 1923 Maximum Statistical of ODAC Z 5
 1930 Averaged Statistical of ODAC Z 6
 1931 Ovality Statistical (Max-Min) of ODAC Z 6
 1932 Minimum Statistical of ODAC Z 6
 1933 Maximum Statistical of ODAC Z 6
 19560 Averaged Statistical of ODAC Z 7
 19561 Ovality Statistical (Max-Min) of ODAC Z 7
 19562 Minimum Statistical of ODAC Z 7
 19563 Maximum Statistical of ODAC Z 7
 19570 Averaged Statistical of ODAC Z 8
 19571 Ovality Statistical (Max-Min) of ODAC Z 8
 19572 Minimum Statistical of ODAC Z 8
 19573 Maximum Statistical of ODAC Z 8

 1940 Averaged Statistical of ODAC T 1
 1941 Ovality Statistical (Max-Min) of ODAC T 1
 1942 Minimum Statistical of ODAC T 1
 1943 Maximum Statistical of ODAC T 1
 1950 Averaged Statistical of ODAC T 2
 1951 Ovality Statistical (Max-Min) of ODAC T 2
 1952 Minimum Statistical of ODAC T 2
 1953 Maximum Statistical of ODAC T 2
 1960 Averaged Statistical of ODAC T 3
 1961 Ovality Statistical (Max-Min) of ODAC T 3
 1962 Minimum Statistical of ODAC T 3
 1963 Maximum Statistical of ODAC T 3
 1970 Averaged Statistical of ODAC T 4
 1971 Ovality Statistical (Max-Min) of ODAC T 4
 1972 Minimum Statistical of ODAC T 4
 1973 Maximum Statistical of ODAC T 4
 1980 Averaged Statistical of ODAC T 5
 1981 Ovality Statistical (Max-Min) of ODAC T 5
 1982 Minimum Statistical of ODAC T 5
 1983 Maximum Statistical of ODAC T 5
 1990 Averaged Statistical of ODAC T 6
 1991 Ovality Statistical (Max-Min) of ODAC T 6
 1992 Minimum Statistical of ODAC T 6
 1993 Maximum Statistical of ODAC T 6
 19580 Averaged Statistical of ODAC T 7
 19581 Ovality Statistical (Max-Min) of ODAC T 7
 19582 Minimum Statistical of ODAC T 7

19583 Maximum Statistical of ODAC T 7
 19590 Averaged Statistical of ODAC T 8
 19591 Ovality Statistical (Max-Min) of ODAC T 8
 19592 Minimum Statistical of ODAC T 8
 19593 Maximum Statistical of ODAC T 8

CAPAC

2000 Capacitance 1
 2010 Capacitance 2

ODEX

2100 Eccentricity 1
 2101 Eccentricity X 1
 2102 Eccentricity Y 1
 2110 Eccentricity 2
 2111 Eccentricity X 2
 2112 Eccentricity Y 2
 2120 Eccentricity 3
 2121 Eccentricity X 3
 2122 Eccentricity Y 3
 2130 Eccentricity 4
 2131 Eccentricity X 4
 2132 Eccentricity Y 4
 2140 Eccentricity 5
 2141 Eccentricity X 5
 2142 Eccentricity Y 5
 2150 Eccentricity 6
 2151 Eccentricity X 6
 2152 Eccentricity Y 6

ODEX with Statistics

2200 Eccentricity Statistical 1
 2201 Eccentricity Statistical 2
 2202 Eccentricity Statistical 3
 2203 Eccentricity Statistical 4
 2204 Eccentricity Statistical 5
 2205 Eccentricity Statistical 6
 2206 Eccentricity Statistical X 1
 2207 Eccentricity Statistical X 2
 2208 Eccentricity Statistical X 3
 2209 Eccentricity Statistical X 4
 2210 Eccentricity Statistical X 5
 2211 Eccentricity Statistical X 6
 2212 Eccentricity Statistical Y 1
 2213 Eccentricity Statistical Y 2
 2214 Eccentricity Statistical Y 3
 2215 Eccentricity Statistical Y 4
 2216 Eccentricity Statistical Y 5
 2217 Eccentricity Statistical Y 6

SPARKER

3500 High Voltage SPARKER 1
 3510 High Voltage SPARKER 2
 3520 High Voltage SPARKER 3
 3530 High Voltage SPARKER 4
 3540 High Voltage SPARKER 5
 3550 High Voltage SPARKER 6

EXTRUDER

3600 RPM EXTRUDER 1
 3601 Meter Weight EXTRUDER 1
 3602 Throughput EXTRUDER 1
 3620 RPM EXTRUDER 2
 3621 Meter Weight EXTRUDER 2

3632 Throughput EXTRUDER 2
 3640 RPM EXTRUDER 3
 3641 Meter Weight EXTRUDER 3
 3642 Throughput EXTRUDER 3
 3660 RPM EXTRUDER 4
 3661 Meter Weight EXTRUDER 4
 3662 Throughput EXTRUDER 4

VELOCITY

4500 Line velocity before the accumulator
 4510 Line velocity at the line end

Analog Gauge

4000 Analog input 1
 4010 Analog input 2
 4020 Analog input 3
 4030 Analog input 4

UMAC

UMAC 1

2298 Smaller Wall Layer 1
 2299 Larger Wall Layer 1
 2300 Smaller Wall vector calculation Layer 1
 2301 Larger Wall vector calculation Layer 1
 2302 Average Wall Layer 1
 2303 Eccentricity :shift of center using vector calculation Layer 1
 2304 Eccentricity :max transducer minus min transducer Layer 1
 2305 Eccentricity :max transducer minus min using vector calculation Layer 1
 2306 Concentricity :max transducer minus min divided by average Layer 1
 2307 Concentricity : min divided by max Layer 1
 2308 Eccentricity :X component of shift of center using vector calculation Layer 1
 2309 Eccentricity :Y component of shift of center using vector calculation Layer 1
 2310 Wall Transducer 1 Layer 1
 2311 Wall Transducer 2 Layer 1
 2312 Wall Transducer 3 Layer 1
 2313 Wall Transducer 4 Layer 1
 2314 Wall Transducer 5 Layer 1
 2315 Wall Transducer 6 Layer 1
 2316 Wall Transducer 7 Layer 1
 2317 Wall Transducer 8 Layer 1

2348 Smaller Wall Layer 2
 2349 Larger Wall Layer 2
 2350 Smaller Wall vector calculation Layer 2
 2351 Larger Wall vector calculation Layer 2
 2352 Average Wall Layer 2
 2353 Eccentricity :shift of center using vector calculation Layer 2
 2354 Eccentricity :max transducer minus min transducer Layer 2
 2355 Eccentricity :max transducer minus min using vector calculation Layer 2
 2356 Concentricity :max transducer minus min divided by average Layer 2
 2357 Concentricity : min divided by max Layer 2
 2358 Eccentricity :X component of shift of center using vector calculation Layer 2
 2359 Eccentricity :Y component of shift of center using vector calculation Layer 2
 2360 Wall Transducer 1 Layer 2
 2361 Wall Transducer 2 Layer 2
 2362 Wall Transducer 3 Layer 2
 2363 Wall Transducer 4 Layer 2
 2364 Wall Transducer 5 Layer 2
 2365 Wall Transducer 6 Layer 2
 2366 Wall Transducer 7 Layer 2
 2367 Wall Transducer 8 Layer 2

2398 Smaller Wall Layer 3

2399 Larger Wall Layer 3
 2400 Smaller Wall using vector calculation Layer 3
 2401 Larger Wall using vector calculation Layer 3
 2402 Average Wall Layer 3
 2403 Eccentricity :shift of center using vector calculation Layer 3
 2404 Eccentricity :max transducer minus min transducer Layer 3
 2405 Eccentricity :max transducer minus min using vector calculation Layer 3
 2406 Concentricity :max transducer minus min divided by average Layer 3
 2407 Concentricity : min divided by max Layer 3
 2408 Eccentricity :X component of shift of center using vector calculation Layer 3
 2409 Eccentricity :Y component of shift of center using vector calculation Layer 3
 2410 Wall Transducer 1 Layer 3
 2411 Wall Transducer 2 Layer 3
 2412 Wall Transducer 3 Layer 3
 2413 Wall Transducer 4 Layer 3
 2414 Wall Transducer 5 Layer 3
 2415 Wall Transducer 6 Layer 3
 2416 Wall Transducer 7 Layer 3
 2417 Wall Transducer 8 Layer 3

 2448 Smaller Wall Layer 4
 2449 Larger Wall Layer 4
 2450 Smaller Wall using vector calculation Layer 4
 2451 Larger Wall using vector calculation Layer 4
 2452 Average Wall Layer 4
 2453 Eccentricity :shift of center using vector calculation Layer 4
 2454 Eccentricity :max transducer minus min transducer Layer 4
 2455 Eccentricity :max transducer minus min using vector calculation Layer 4
 2456 Concentricity :max transducer minus min divided by average Layer 4
 2457 Concentricity : min divided by max Layer 4
 2458 Eccentricity :X component of shift of center using vector calculation Layer 4
 2459 Eccentricity :Y component of shift of center using vector calculation Layer 4
 2460 Wall Transducer 1 Layer 4
 2461 Wall Transducer 2 Layer 4
 2462 Wall Transducer 3 Layer 4
 2463 Wall Transducer 4 Layer 4
 2464 Wall Transducer 5 Layer 4
 2465 Wall Transducer 6 Layer 4
 2466 Wall Transducer 7 Layer 4
 2467 Wall Transducer 8 Layer 4

 2498 Smaller Wall Layer 5
 2499 Larger Wall Layer 5
 2500 Smaller Wall using vector calculation Layer 5
 2501 Larger Wall using vector calculation Layer 5
 2502 Average Wall Layer 5
 2503 Eccentricity :shift of center using vector calculation Layer 5
 2504 Eccentricity :max transducer minus min transducer Layer 5
 2505 Eccentricity :max transducer minus min using vector calculation Layer 5
 2506 Concentricity :max transducer minus min divided by average Layer 5
 2507 Concentricity : min divided by max Layer 5
 2508 Eccentricity :X component of shift of center using vector calculation Layer 5
 2509 Eccentricity :Y component of shift of center using vector calculation Layer 5
 2510 Wall Transducer 1 Layer 5
 2511 Wall Transducer 2 Layer 5
 2512 Wall Transducer 3 Layer 5
 2513 Wall Transducer 4 Layer 5
 2514 Wall Transducer 5 Layer 5
 2515 Wall Transducer 6 Layer 5
 2516 Wall Transducer 7 Layer 5
 2517 Wall Transducer 8 Layer 5

 2548 Smaller Wall Layer 6
 2549 Larger Wall Layer 6
 2550 Smaller Wall using vector calculation Layer 6

2551 Larger Wall using vector calculation Layer 6
 2552 Average Wall Layer 6
 2553 Eccentricity :shift of center using vector calculation Layer 6
 2554 Eccentricity :max transducer minus min transducer Layer 6
 2555 Eccentricity :max transducer minus min using vector calculation Layer 6
 2556 Concentricity :max transducer minus min divided by average Layer 6
 2557 Concentricity : min divided by max Layer 6
 2558 Eccentricity :X component of shift of center using vector calculation Layer 6
 2559 Eccentricity :Y component of shift of center using vector calculation Layer 6
 2560 Wall Transducer 1 Layer 6
 2561 Wall Transducer 2 Layer 6
 2562 Wall Transducer 3 Layer 6
 2563 Wall Transducer 4 Layer 6
 2564 Wall Transducer 5 Layer 6
 2565 Wall Transducer 6 Layer 6
 2566 Wall Transducer 7 Layer 6
 2567 Wall Transducer 8 Layer 6

 2598 Smaller Wall Layer 7
 2599 Larger Wall Layer 7
 2600 Smaller Wall using vector calculation Layer 7
 2601 Larger Wall using vector calculation Layer 7
 2602 Average Wall Layer 7
 2603 Eccentricity :shift of center using vector calculation Layer 7
 2604 Eccentricity :max transducer minus min transducer Layer 7
 2605 Eccentricity :max transducer minus min using vector calculation Layer 7
 2606 Concentricity :max transducer minus min divided by average Layer 7
 2607 Concentricity : min divided by max Layer 7
 2608 Eccentricity :X component of shift of center using vector calculation Layer 7
 2609 Eccentricity :Y component of shift of center using vector calculation Layer 7
 2610 Wall Transducer 1 Layer 7
 2611 Wall Transducer 2 Layer 7
 2612 Wall Transducer 3 Layer 7
 2613 Wall Transducer 4 Layer 7
 2614 Wall Transducer 5 Layer 7
 2615 Wall Transducer 6 Layer 7
 2616 Wall Transducer 7 Layer 7
 2617 Wall Transducer 8 Layer 7

 2648 Smaller Wall Layer 8
 2649 Larger Wall Layer 8
 2650 Smaller Wall using vector calculation Layer 8
 2651 Larger Wall using vector calculation Layer 8
 2652 Average Wall Layer 8
 2653 Eccentricity :shift of center using vector calculation Layer 8
 2654 Eccentricity :max transducer minus min transducer Layer 8
 2655 Eccentricity :max transducer minus min using vector calculation Layer 8
 2656 Concentricity :max transducer minus min divided by average Layer 8
 2657 Concentricity : min divided by max Layer 8
 2658 Eccentricity :X component of shift of center using vector calculation Layer 8
 2659 Eccentricity :Y component of shift of center using vector calculation Layer 8
 2660 Wall Transducer 1 Layer 8
 2661 Wall Transducer 2 Layer 8
 2662 Wall Transducer 3 Layer 8
 2663 Wall Transducer 4 Layer 8
 2664 Wall Transducer 5 Layer 8
 2665 Wall Transducer 6 Layer 8
 2666 Wall Transducer 7 Layer 8
 2667 Wall Transducer 8 Layer 8

 2748 UMAC Temperature
 2749 UMAC Position X
 2750 UMAC Position Y

 2798 Smaller Overall Wall

2799 Larger Overall Wall
 2800 Smaller Overall Wall vector calculation
 2801 Larger Overall Wall vector calculation
 2802 Average Overall Wall
 2803 Overall Eccentricity :shift of center using vector calculation
 2804 Overall Eccentricity :max transducer minus min transducer
 2805 Overall Eccentricity :max transducer minus min using vector calculation
 2806 Overall Concentricity :max transducer minus min divided by average
 2807 Overall Concentricity : min divided by max
 2808 Overall Eccentricity :X component of shift of center using vector calculation
 2809 Overall Eccentricity :Y component of shift of center using vector calculation
 2810 Overall Wall Transducer 1
 2811 Overall Wall Transducer 2
 2812 Overall Wall Transducer 3
 2813 Overall Wall Transducer 4
 2814 Overall Wall Transducer 5
 2815 Overall Wall Transducer 6
 2816 Overall Wall Transducer 7
 2817 Overall Wall Transducer 8

2848 Smaller Outer Diameter
 2849 Larger Outer Diameter
 2850 Average Outer Diameter
 2851 Outer Ovality
 2852 Outer Diameter Transducer 1
 2853 Outer Diameter Transducer 2
 2854 Outer Diameter Transducer 3
 2855 Outer Diameter Transducer 4
 2856 Outer Diameter Transducer 5
 2857 Outer Diameter Transducer 6
 2858 Outer Diameter Transducer 7
 2859 Outer Diameter Transducer 8

2898 Smaller Inner Diameter
 2899 Larger Inner Diameter
 2900 Average Inner Diameter
 2901 Inner Ovality
 2902 Inner Diameter Transducer 1
 2903 Inner Diameter Transducer 2
 2904 Inner Diameter Transducer 3
 2905 Inner Diameter Transducer 4
 2906 Inner Diameter Transducer 5
 2907 Inner Diameter Transducer 6
 2908 Inner Diameter Transducer 7
 2909 Inner Diameter Transducer 8

UMAC 2

15000 Smaller Wall Layer 1
 15001 Larger Wall Layer 1
 15002 Smaller Wall vector calculation Layer 1
 15003 Larger Wall vector calculation Layer 1
 15004 Average Wall Layer 1
 15005 Eccentricity :shift of center using vector calculation Layer 1
 15006 Eccentricity :max transducer minus min transducer Layer 1
 15007 Eccentricity :max transducer minus min using vector calculation Layer 1
 15008 Concentricity :max transducer minus min divided by average Layer 1
 15009 Concentricity : min divided by max Layer 1
 15010 Eccentricity :X component of shift of center using vector calculation Layer 1
 15011 Eccentricity :Y component of shift of center using vector calculation Layer 1
 15012 Wall Transducer 1 Layer 1
 15013 Wall Transducer 2 Layer 1
 15014 Wall Transducer 3 Layer 1
 15015 Wall Transducer 4 Layer 1
 15016 Wall Transducer 5 Layer 1
 15017 Wall Transducer 6 Layer 1

15018	Wall Transducer 7	Layer 1
15019	Wall Transducer 8	Layer 1
15050	Smaller Wall Layer 2	
15051	Larger Wall Layer 2	
15052	Smaller Wall vector calculation Layer 2	
15053	Larger Wall vector calculation Layer 2	
15054	Average Wall Layer 2	
15055	Eccentricity :shift of center using vector calculation Layer 2	
15056	Eccentricity :max transducer minus min transducer Layer 2	
15057	Eccentricity :max transducer minus min using vector calculation Layer 2	
15058	Concentricity :max transducer minus min divided by average Layer 2	
15059	Concentricity : min divided by max Layer 2	
15050	Eccentricity :X component of shift of center using vector calculation Layer 2	
15051	Eccentricity :Y component of shift of center using vector calculation Layer 2	
15052	Wall Transducer 1	Layer 2
15053	Wall Transducer 2	Layer 2
15054	Wall Transducer 3	Layer 2
15055	Wall Transducer 4	Layer 2
15056	Wall Transducer 5	Layer 2
15057	Wall Transducer 6	Layer 2
15058	Wall Transducer 7	Layer 2
15059	Wall Transducer 8	Layer 2
15100	Smaller Wall Layer 3	
15101	Larger Wall Layer 3	
15102	Smaller Wall vector calculation Layer 3	
15103	Larger Wall vector calculation Layer 3	
15104	Average Wall Layer 3	
15105	Eccentricity :shift of center using vector calculation Layer 3	
15106	Eccentricity :max transducer minus min transducer Layer 3	
15107	Eccentricity :max transducer minus min using vector calculation Layer 3	
15108	Concentricity :max transducer minus min divided by average Layer 3	
15109	Concentricity : min divided by max Layer 3	
15110	Eccentricity :X component of shift of center using vector calculation Layer 3	
15111	Eccentricity :Y component of shift of center using vector calculation Layer 3	
15112	Wall Transducer 1	Layer 3
15113	Wall Transducer 2	Layer 3
15114	Wall Transducer 3	Layer 3
15115	Wall Transducer 4	Layer 3
15116	Wall Transducer 5	Layer 3
15117	Wall Transducer 6	Layer 3
15118	Wall Transducer 7	Layer 3
15119	Wall Transducer 8	Layer 3
15150	Smaller Wall Layer 4	
15151	Larger Wall Layer 4	
15152	Smaller Wall vector calculation Layer 4	
15153	Larger Wall vector calculation Layer 4	
15154	Average Wall Layer 4	
15155	Eccentricity :shift of center using vector calculation Layer 4	
15156	Eccentricity :max transducer minus min transducer Layer 4	
15157	Eccentricity :max transducer minus min using vector calculation Layer 4	
15158	Concentricity :max transducer minus min divided by average Layer 4	
15159	Concentricity : min divided by max Layer 4	
15160	Eccentricity :X component of shift of center using vector calculation Layer 4	
15161	Eccentricity :Y component of shift of center using vector calculation Layer 4	
15162	Wall Transducer 1	Layer 4
15163	Wall Transducer 2	Layer 4
15164	Wall Transducer 3	Layer 4
15165	Wall Transducer 4	Layer 4
15166	Wall Transducer 5	Layer 4
15167	Wall Transducer 6	Layer 4
15168	Wall Transducer 7	Layer 4

15169 Wall Transducer 8 Layer 4

15200 Smaller Wall Layer 5
15201 Larger Wall Layer 5
15202 Smaller Wall vector calculation Layer 5
15203 Larger Wall vector calculation Layer 5
15204 Average Wall Layer 5
15205 Eccentricity :shift of center using vector calculation Layer 5
15206 Eccentricity :max transducer minus min transducer Layer 5
15207 Eccentricity :max transducer minus min using vector calculation Layer 5
15208 Concentricity :max transducer minus min divided by average Layer 5
15209 Concentricity : min divided by max Layer 5
15210 Eccentricity :X component of shift of center using vector calculation Layer 5
15211 Eccentricity :Y component of shift of center using vector calculation Layer 5
15212 Wall Transducer 1 Layer 5
15213 Wall Transducer 2 Layer 5
15214 Wall Transducer 3 Layer 5
15215 Wall Transducer 4 Layer 5
15216 Wall Transducer 5 Layer 5
15217 Wall Transducer 6 Layer 5
15218 Wall Transducer 7 Layer 5
15219 Wall Transducer 8 Layer 5

15250 Smaller Wall Layer 6
15251 Larger Wall Layer 6
15252 Smaller Wall vector calculation Layer 6
15253 Larger Wall vector calculation Layer 6
15254 Average Wall Layer 6
15255 Eccentricity :shift of center using vector calculation Layer 6
15256 Eccentricity :max transducer minus min transducer Layer 6
15257 Eccentricity :max transducer minus min using vector calculation Layer 6
15258 Concentricity :max transducer minus min divided by average Layer 6
15259 Concentricity : min divided by max Layer 6
15260 Eccentricity :X component of shift of center using vector calculation Layer 6
15261 Eccentricity :Y component of shift of center using vector calculation Layer 6
15262 Wall Transducer 1 Layer 6
15263 Wall Transducer 2 Layer 6
15264 Wall Transducer 3 Layer 6
15265 Wall Transducer 4 Layer 6
15266 Wall Transducer 5 Layer 6
15267 Wall Transducer 6 Layer 6
15268 Wall Transducer 7 Layer 6
15269 Wall Transducer 8 Layer 6

15300 Smaller Wall Layer 7
15301 Larger Wall Layer 7
15302 Smaller Wall vector calculation Layer 7
15303 Larger Wall vector calculation Layer 7
15304 Average Wall Layer 7
15305 Eccentricity :shift of center using vector calculation Layer 7
15306 Eccentricity :max transducer minus min transducer Layer 7
15307 Eccentricity :max transducer minus min using vector calculation Layer 7
15308 Concentricity :max transducer minus min divided by average Layer 7
15309 Concentricity : min divided by max Layer 7
15310 Eccentricity :X component of shift of center using vector calculation Layer 7
15311 Eccentricity :Y component of shift of center using vector calculation Layer 7
15312 Wall Transducer 1 Layer 7
15313 Wall Transducer 2 Layer 7
15314 Wall Transducer 3 Layer 7
15315 Wall Transducer 4 Layer 7
15316 Wall Transducer 5 Layer 7
15317 Wall Transducer 6 Layer 7
15318 Wall Transducer 7 Layer 7
15319 Wall Transducer 8 Layer 7

15350 Smaller Wall Layer 8
 15351 Larger Wall Layer 8
 15352 Smaller Wall vector calculation Layer 8
 15353 Larger Wall vector calculation Layer 8
 15354 Average Wall Layer 8
 15355 Eccentricity :shift of center using vector calculation Layer 8
 15356 Eccentricity :max transducer minus min transducer Layer 8
 15357 Eccentricity :max transducer minus min using vector calculation Layer 8
 15358 Concentricity :max transducer minus min divided by average Layer 8
 15359 Concentricity : min divided by max Layer 8
 15360 Eccentricity :X component of shift of center using vector calculation Layer 8
 15361 Eccentricity :Y component of shift of center using vector calculation Layer 8
 15362 Wall Transducer 1 Layer 8
 15363 Wall Transducer 2 Layer 8
 15364 Wall Transducer 3 Layer 8
 15365 Wall Transducer 4 Layer 8
 15366 Wall Transducer 5 Layer 8
 15367 Wall Transducer 6 Layer 8
 15368 Wall Transducer 7 Layer 8
 15369 Wall Transducer 8 Layer 8

15400 UMAC Temperature
 15401 UMAC Position X
 15402 UMAC Position Y

15450 Smaller Overall Wall
 15451 Larger Overall Wall
 15452 Smaller Overall Wall vector calculation
 15453 Larger Overall Wall vector calculation
 15454 Average Overall Wall
 15455 Overall Eccentricity :shift of center using vector calculation
 15456 Overall Eccentricity :max transducer minus min transducer
 15457 Overall Eccentricity :max transducer minus min using vector calculation
 15458 Overall Concentricity :max transducer minus min divided by average
 15459 Overall Concentricity : min divided by max
 15460 Overall Eccentricity :X component of shift of center using vector calculation
 15461 Overall Eccentricity :Y component of shift of center using vector calculation
 15462 Overall Wall Transducer 1
 15463 Overall Wall Transducer 2
 15464 Overall Wall Transducer 3
 15465 Overall Wall Transducer 4
 15466 Overall Wall Transducer 5
 15467 Overall Wall Transducer 6
 15468 Overall Wall Transducer 7
 15469 Overall Wall Transducer 8

15500 Smaller Outer Diameter
 15501 Larger Outer Diameter
 15502 Average Outer Diameter
 15503 Outer Ovality
 15504 Outer Diameter Transducer 1
 15505 Outer Diameter Transducer 2
 15506 Outer Diameter Transducer 3
 15507 Outer Diameter Transducer 4
 15508 Outer Diameter Transducer 5
 15509 Outer Diameter Transducer 6
 15510 Outer Diameter Transducer 7
 15511 Outer Diameter Transducer 8

15550 Smaller Inner Diameter
 15551 Larger Inner Diameter
 15552 Average Inner Diameter
 15553 Inner Ovality
 15554 Inner Diameter Transducer 1
 15555 Inner Diameter Transducer 2

15556 Inner Diameter Transducer 3
 15557 Inner Diameter Transducer 4
 15558 Inner Diameter Transducer 5
 15559 Inner Diameter Transducer 6
 15560 Inner Diameter Transducer 7
 15562 Inner Diameter Transducer 8

ODAC TRIO

17500 RMS X-Y-Z 1
 17501 Ovality X-Y-Z 1
 17502 Smaller X-Y-Z 1
 17503 Larger X-Y-Z 1
 17504 Diameter X 1
 17505 Diameter Y 1
 17506 Diameter Z 1
 17507 Position X 1
 17508 Position Y 1
 17509 Position Z 1
 17510 AVG X-Y-Z 2
 17520 RMS X-Y-Z 2
 17521 Ovality X-Y-Z 2
 17522 Smaller X-Y-Z 2
 17523 Larger X-Y-Z 2
 17524 Diameter X 2
 17525 Diameter Y 2
 17526 Diameter Z 2
 17527 Position X 2
 17528 Position Y 2
 17529 Position Z 2
 17530 AVG X-Y-Z 2
 17540 RMS X-Y-Z 3
 17541 Ovality X-Y-Z 3
 17542 Smaller X-Y-Z 3
 17543 Larger X-Y-Z 3
 17544 Diameter X 3
 17545 Diameter Y 3
 17546 Diameter Z 3
 17547 Position X 3
 17548 Position Y 3
 17549 Position Z 3
 17550 AVG X-Y-Z 3
 17560 RMS X-Y-Z 4
 17561 Ovality X-Y-Z 4
 17562 Smaller X-Y-Z 4
 17563 Larger X-Y-Z 4
 17564 Diameter X 4
 17565 Diameter Y 4
 17566 Diameter Z 4
 17567 Position X 4
 17568 Position Y 4
 17569 Position Z 4
 17570 AVG X-Y-Z 4
 17580 RMS X-Y-Z 5
 17581 Ovality X-Y-Z 5
 17582 Smaller X-Y-Z 5
 17583 Larger X-Y-Z 5
 17584 Diameter X 5
 17585 Diameter Y 5
 17586 Diameter Z 5
 17587 Position X 5
 17588 Position Y 5
 17589 Position Z 5
 17590 AVG X-Y-Z 5
 17600 RMS X-Y-Z 6
 17601 Ovality X-Y-Z 6

17602 Smaller X-Y-Z 6
 17603 Larger X-Y-Z 6
 17604 Diameter X 6
 17605 Diameter Y 6
 17606 Diameter Z 6
 17607 Position X 6
 17608 Position Y 6
 17609 Position Z 6
 17610 AVG X-Y-Z 6
 17620 RMS X-Y-Z 7
 17621 Ovality X-Y-Z 7
 17622 Smaller X-Y-Z 7
 17623 Larger X-Y-Z 7
 17624 Diameter X 7
 17625 Diameter Y 7
 17626 Diameter Z 7
 17627 Position X 7
 17628 Position Y 7
 17629 Position Z 7
 17640 AVG X-Y-Z 8
 17640 RMS X-Y-Z 8
 17641 Ovality X-Y-Z 8
 17642 Smaller X-Y-Z 8
 17643 Larger X-Y-Z 8
 17644 Diameter X 8
 17645 Diameter Y 8
 17646 Diameter Z 8
 17647 Position X 8
 17648 Position Y 8
 17649 Position Z 8
 17640 AVG X-Y-Z 8

 19300 Perimeter X-Y-Z 1
 19301 Area X-Y-Z 1
 19302 Meter Weight X-Y-Z 1
 19310 Perimeter X-Y-Z 2
 19311 Area X-Y-Z 2
 19312 Meter Weight X-Y-Z 2
 19320 Perimeter X-Y-Z 3
 19321 Area X-Y-Z 3
 19322 Meter Weight X-Y-Z 3
 19330 Perimeter X-Y-Z 4
 19331 Area X-Y-Z 4
 19332 Meter Weight X-Y-Z 4
 19340 Perimeter X-Y-Z 5
 19341 Area X-Y-Z 5
 19342 Meter Weight X-Y-Z 5
 19350 Perimeter X-Y-Z 6
 19351 Area X-Y-Z 6
 19352 Meter Weight X-Y-Z 6
 19360 Perimeter X-Y-Z 7
 19361 Area X-Y-Z 7
 19362 Meter Weight X-Y-Z 7
 19370 Perimeter X-Y-Z 8
 19371 Area X-Y-Z 8
 19372 Meter Weight X-Y-Z 8

RAYEX D

18000 OD Diameter Average Overall
 18001 OD Diameter Maximum Overall
 18002 OD Diameter Minimum Overall
 18003 OD Diameter X Overall
 18004 OD Diameter Y Overall
 18005 Conductor Average
 18006 Conductor Maximum

18007 Conductor Minimum
 18008 Conductor X
 18009 Conductor Y
 18010 Ovality
 18011 WallThickness Average
 18012 WallThickness Maximum
 18013 WallThickness Minimum
 18014 WallThickness X+
 18015 WallThickness X-
 18016 WallThickness Y+
 18017 WallThickness Y-
 18018 Eccentricity Shift of Center Overall
 18019 Eccentricity Max-Min Overall
 18020 Concentricity (Max-Min) / Avg Overall
 18021 Concentricity Min / Max Overall
 18022 Eccentricity Shift of Center X Overall
 18023 Eccentricity Shift of Center Y Overall
 18024 Position X Overall
 18025 Position Y Overall

18050 WallThickness Average ISL
 18051 WallThickness Maximum ISL
 18052 WallThickness Minimum ISL
 18053 WallThickness X+ ISL
 18054 WallThickness X- ISL
 18055 WallThickness Y+ ISL
 18056 WallThickness Y- ISL
 18057 Eccentricity Shift of Center ISL
 18058 Eccentricity Max-Min ISL
 18059 Concentricity (Max-Min) / Avg ISL
 18060 Concentricity Min / Max ISL
 18061 Eccentricity Shift of Center X ISL
 18062 Eccentricity Shift of Center Y ISL

18100 WallThickness Average ILL
 18101 WallThickness Maximum ILL
 18102 WallThickness Minimum ILL
 18103 WallThickness X+ ILL
 18104 WallThickness X- ILL
 18105 WallThickness Y+ ILL
 18106 WallThickness Y- ILL
 18107 Eccentricity Shift of Center ILL
 18108 Eccentricity Max-Min ILL
 18109 Concentricity (Max-Min) / Avg ILL
 18110 Concentricity Min / Max ILL
 18111 Eccentricity Shift of Center X ILL
 18112 Eccentricity Shift of Center Y ILL

18150 WallThickness Average OSL
 18151 WallThickness Maximum OSL
 18152 WallThickness Minimum OSL
 18153 WallThickness X+ OSL
 18154 WallThickness X- OSL
 18155 WallThickness Y+ OSL
 18156 WallThickness Y- OSL
 18157 Eccentricity Shift of Center OSL
 18158 Eccentricity Max-Min OSL
 18159 Concentricity (Max-Min) / Avg OSL
 18160 Concentricity Min / Max OSL
 18161 Eccentricity Shift of Center X OSL
 18162 Eccentricity Shift of Center Y OSL

RAYEX S/D XT

18400 OD Diameter Average Overall
 18401 OD Diameter Maximum Overall

18402 OD Diameter Minimum Overall
 18403 OD Diameter X Overall
 18404 OD Diameter Y Overall
 18405 Conductor Average
 18406 Conductor Maximum
 18407 Conductor Minimum
 18408 Conductor X
 18409 Conductor Y
 18410 Ovality
 18411 WallThickness Average
 18412 WallThickness Maximum
 18413 WallThickness Minimum
 18414 WallThickness X+
 18415 WallThickness X-
 18416 WallThickness Y+
 18417 WallThickness Y-
 18418 Eccentricity Shift of Center Overall
 18419 Eccentricity Max-Min Overall
 18420 Concentricity (Max-Min) / Avg Overall
 18421 Concentricity Min / Max Overall
 18422 Eccentricity Shift of Center X Overall
 18423 Eccentricity Shift of Center Y Overall
 18424 Position X Overall
 18425 Position Y Overall

18450 WallThickness Average L1
 18451 WallThickness Maximum L1
 18452 WallThickness Minimum L1
 18453 WallThickness X+ L1
 18454 WallThickness X- L1
 18455 WallThickness Y+ L1
 18456 WallThickness Y- L1
 18457 Eccentricity Shift of Center L1
 18458 Eccentricity Max-Min L1
 18459 Concentricity (Max-Min) / Avg L1
 18460 Concentricity Min / Max L1
 18461 Eccentricity Shift of Center X L1
 18462 Eccentricity Shift of Center Y L1

18500 WallThickness Average L2
 18501 WallThickness Maximum L2
 18502 WallThickness Minimum L2
 18503 WallThickness X+ L2
 18504 WallThickness X- L2
 18505 WallThickness Y+ L2
 18506 WallThickness Y- L2
 18507 Eccentricity Shift of Center L2
 18508 Eccentricity Max-Min L2
 18509 Concentricity (Max-Min) / Avg L2
 18510 Concentricity Min / Max L2
 18511 Eccentricity Shift of Center X L2
 18512 Eccentricity Shift of Center Y L2

18550 WallThickness Average L3
 18551 WallThickness Maximum L3
 18552 WallThickness Minimum L3
 18553 WallThickness X+ L3
 18554 WallThickness X- L3
 18555 WallThickness Y+ L3
 18556 WallThickness Y- L3
 18557 Eccentricity Shift of Center L3
 18558 Eccentricity Max-Min L3
 18559 Concentricity (Max-Min) / Avg L3
 18560 Concentricity Min / Max L3
 18561 Eccentricity Shift of Center X L3

18562 Eccentricity Shift of Center Y L3

18600 WallThickness Average L4
18601 WallThickness Maximum L4
18602 WallThickness Minimum L4
18603 WallThickness X+ L4
18604 WallThickness X- L4
18605 WallThickness Y+ L4
18606 WallThickness Y- L4
18607 Eccentricity Shift of Center L4
18608 Eccentricity Max-Min L4
18609 Concentricity (Max-Min) / Avg L4
18610 Concentricity Min / Max L4
18611 Eccentricity Shift of Center X L4
18612 Eccentricity Shift of Center Y L4

18650 WallThickness Average L5
18651 WallThickness Maximum L5
18652 WallThickness Minimum L5
18653 WallThickness X+ L5
18654 WallThickness X- L5
18655 WallThickness Y+ L5
18656 WallThickness Y- L5

RAYEX USYS

20000 OD Diameter Average Overall
20001 OD Diameter Maximum Overall
20002 OD Diameter Minimum Overall
20003 OD Diameter X Overall
20004 OD Diameter Y Overall

BENDCHECK

20100 Diameter 1
20101 Position 1
20102 Diameter 2
20103 Position 2
20104 Diameter 3
20105 Position 3
20106 Diameter 4
20107 Position 4
20108 Diameter 5
20109 Position 5
20110 Diameter 6
20111 Position 6
20112 Diameter 7
20113 Position 7
20114 Diameter 8
20115 Position 8
20115 Local Bend
20116 Local Bend Angle

BENDCHECK Ends

20200 Diameter 1
20201 Position 1
20202 Diameter 2
20203 Position 2
20204 Diameter 3
20205 Position 3
20206 Diameter 4
20207 Position 4
20208 Diameter 5
20209 Position 5
20210 Diameter 6
20211 Position 6

20212 Diameter 7
 20213 Position 7
 20214 Diameter 8
 20215 Position 8
 20215 Local Bend
 20216 Local Bend Angle
 20217 Begin Bar Bend
 20218 Begin Bar Bend Angle
 20219 End Bar Bend
 20220 End Bar Bend Angle

BENDCHECK XY

20300 Diameter X-Y 1
 20301 Ovality X-Y 1
 20302 Smaller X-Y 1
 20303 Larger X-Y 1
 20304 Diameter X 1
 20305 Diameter Y 1
 20306 Position X 1
 20307 Position Y 1
 20308 Average X-Y 1
 20309 Diameter X-Y 2
 20310 Ovality X-Y 2
 20311 Smaller X-Y 2
 20312 Larger X-Y 2
 20313 Diameter X 2
 20314 Diameter Y 2
 20315 Position X 2
 20316 Position Y 2
 20317 Average X-Y 2
 20318 Diameter X-Y 3
 20319 Ovality X-Y 3
 20320 Smaller X-Y 3
 20321 Larger X-Y 3
 20322 Diameter X 3
 20323 Diameter Y 3
 20324 Position X 3
 20325 Position Y 3
 20326 Average X-Y 3
 20327 Diameter X-Y 4
 20328 Ovality X-Y 4
 20329 Smaller X-Y 4
 20330 Larger X-Y 4
 20331 Diameter X 4
 20332 Diameter Y 4
 20333 Position X 4
 20334 Position Y 4
 20335 Average X-Y 4
 20336 Diameter X-Y 5
 20337 Ovality X-Y 5
 20338 Smaller X-Y 5
 20339 Larger X-Y 5
 20340 Diameter X 5
 20341 Diameter Y 5
 20342 Position X 5
 20343 Position Y 5
 20344 Average X-Y 5
 20345 Diameter X-Y 6
 20346 Ovality X-Y 6
 20347 Smaller X-Y 6
 20348 Larger X-Y 6
 20349 Diameter X 6
 20350 Diameter Y 6
 20351 Position X 6
 20352 Position Y 6

20353 Average X-Y 6
 20354 Diameter X-Y 7
 20355 Ovality X-Y 7
 20356 Smaller X-Y 7
 20357 Larger X-Y 7
 20358 Diameter X 7
 20359 Diameter Y 7
 20360 Position X 7
 20361 Position Y 7
 20362 Average X-Y 7
 20363 Diameter X-Y 8
 20364 Ovality X-Y 8
 20365 Smaller X-Y 8
 20366 Larger X-Y 8
 20367 Diameter X 8
 20368 Diameter Y 8
 20369 Position X 8
 20370 Position Y 8
 20371 Average X-Y 8

 20372 Local Bend
 20373 Local Bend Angle

BENDCHECK XY Ends

20400 Diameter X-Y 1
 20401 Ovality X-Y 1
 20402 Smaller X-Y 1
 20403 Larger X-Y 1
 20404 Diameter X 1
 20405 Diameter Y 1
 20406 Position X 1
 20407 Position Y 1
 20408 Average X-Y 1
 20409 Diameter X-Y 2
 20410 Ovality X-Y 2
 20411 Smaller X-Y 2
 20412 Larger X-Y 2
 20413 Diameter X 2
 20414 Diameter Y 2
 20415 Position X 2
 20416 Position Y 2
 20417 Average X-Y 2
 20418 Diameter X-Y 3
 20419 Ovality X-Y 3
 20420 Smaller X-Y 3
 20421 Larger X-Y 3
 20422 Diameter X 3
 20423 Diameter Y 3
 20424 Position X 3
 20425 Position Y 3
 20426 Average X-Y 3
 20427 Diameter X-Y 4
 20428 Ovality X-Y 4
 20429 Smaller X-Y 4
 20430 Larger X-Y 4
 20431 Diameter X 4
 20432 Diameter Y 4
 20433 Position X 4
 20434 Position Y 4
 20435 Average X-Y 4
 20436 Diameter X-Y 5
 20437 Ovality X-Y 5
 20438 Smaller X-Y 5
 20439 Larger X-Y 5
 20440 Diameter X 5

20441 Diameter Y 5
 20442 Position X 5
 20443 Position Y 5
 20444 Average X-Y 5
 20445 Diameter X-Y 6
 20446 Ovality X-Y 6
 20447 Smaller X-Y 6
 20448 Larger X-Y 6
 20449 Diameter X 6
 20450 Diameter Y 6
 20451 Position X 6
 20452 Position Y 6
 20453 Average X-Y 6
 20454 Diameter X-Y 7
 20455 Ovality X-Y 7
 20456 Smaller X-Y 7
 20457 Larger X-Y 7
 20458 Diameter X 7
 20459 Diameter Y 7
 20460 Position X 7
 20461 Position Y 7
 20462 Average X-Y 7
 20463 Diameter X-Y 8
 20464 Ovality X-Y 8
 20465 Smaller X-Y 8
 20466 Larger X-Y 8
 20467 Diameter X 8
 20468 Diameter Y 8
 20469 Position X 8
 20470 Position Y 8
 20471 Average X-Y 8

 20472 Local Bend
 20473 Local Bend Angle
 20474 Begin Bar Bend
 20475 Begin Bar Bend Angle
 20476 End Bar Bend
 20477 End Bar Bend Angle

BENDCHECK TRIO

20500 Diameter X-Y-Z 1
 20501 Ovality X-Y-Z 1
 20502 Smaller X-Y-Z 1
 20503 Larger X-Y-Z 1
 20504 Diameter X 1
 20505 Diameter Y 1
 20506 Diameter Z 1
 20507 Position X 1
 20508 Position Y 1
 20509 Position Z 1
 20510 Average X-Y-Z 1
 20511 Diameter X-Y-Z 2
 20512 Ovality X-Y-Z 2
 20513 Smaller X-Y-Z 2
 20514 Larger X-Y-Z 2
 20515 Diameter X 2
 20516 Diameter Y 2
 20517 Diameter Z 2
 20518 Position X 2
 20519 Position Y 2
 20520 Position Z 2
 20521 Average X-Y-Z 2
 20522 Diameter X-Y-Z 3
 20523 Ovality X-Y-Z 3
 20524 Smaller X-Y-Z 3

20525 Larger X-Y-Z 3
 20526 Diameter X 3
 20527 Diameter Y 3
 20528 Diameter Z 3
 20529 Position X 3
 20530 Position Y 3
 20531 Position Z 3
 20532 Average X-Y-Z 3
 20533 Diameter X-Y-Z 4
 20534 Ovality X-Y-Z 4
 20535 Smaller X-Y-Z 4
 20536 Larger X-Y-Z 4
 20537 Diameter X 4
 20538 Diameter Y 4
 20539 Diameter Z 4
 20540 Position X 4
 20541 Position Y 4
 20542 Position Z 4
 20543 Average X-Y-Z 4
 20544 Diameter X-Y-Z 5
 20545 Ovality X-Y-Z 5
 20546 Smaller X-Y-Z 5
 20547 Larger X-Y-Z 5
 20548 Diameter X 5
 20549 Diameter Y 5
 20550 Diameter Z 5
 20551 Position X 5
 20552 Position Y 5
 20553 Position Z 5
 20554 Average X-Y-Z 5
 20555 Diameter X-Y-Z 6
 20556 Ovality X-Y-Z 6
 20557 Smaller X-Y-Z
 20558 Larger X-Y-Z 6
 20559 Diameter X 6
 20560 Diameter Y 6
 20561 Diameter Z 6
 20562 Position X 6
 20563 Position Y 6
 20564 Position Z 6
 20565 Average X-Y-Z 6
 20566 Diameter X-Y-Z 7
 20567 Ovality X-Y-Z 7
 20568 Smaller X-Y-Z 7
 20569 Larger X-Y-Z 7
 20570 Diameter X 7
 20571 Diameter Y 7
 20572 Diameter Z 7
 20573 Position X 7
 20574 Position Y 7
 20575 Position Z 7
 20576 Average X-Y-Z 7
 20577 Diameter X-Y-Z 8
 20578 Ovality X-Y-Z 8
 20579 Smaller X-Y-Z 8
 20580 Larger X-Y-Z 8
 20581 Diameter X 8
 20582 Diameter Y 8
 20583 Diameter Z 8
 20584 Position X 8
 20585 Position Y 8
 20586 Position Z 8
 20587 Average X-Y-Z 8

20588 Local Bend

BENDCHECK TRIO Ends

20600 Diameter X-Y-Z 1
 20601 Ovality X-Y-Z 1
 20602 Smaller X-Y-Z 1
 20603 Larger X-Y-Z 1
 20604 Diameter X 1
 20605 Diameter Y 1
 20606 Diameter Z 1
 20607 Position X 1
 20608 Position Y 1
 20609 Position Z 1
 20610 Average X-Y-Z 1
 20611 Diameter X-Y-Z 2
 20612 Ovality X-Y-Z 2
 20613 Smaller X-Y-Z 2
 20614 Larger X-Y-Z 2
 20615 Diameter X 2
 20616 Diameter Y 2
 20617 Diameter Z 2
 20618 Position X 2
 20619 Position Y 2
 20620 Position Z 2
 20621 Average X-Y-Z 2
 20622 Diameter X-Y-Z 3
 20623 Ovality X-Y-Z 3
 20624 Smaller X-Y-Z 3
 20625 Larger X-Y-Z 3
 20626 Diameter X 3
 20627 Diameter Y 3
 20628 Diameter Z 3
 20629 Position X 3
 20630 Position Y 3
 20631 Position Z 3
 20632 Average X-Y-Z 3
 20633 Diameter X-Y-Z 4
 20634 Ovality X-Y-Z 4
 20635 Smaller X-Y-Z 4
 20636 Larger X-Y-Z 4
 20637 Diameter X 4
 20638 Diameter Y 4
 20639 Diameter Z 4
 20640 Position X 4
 20641 Position Y 4
 20642 Position Z 4
 20643 Average X-Y-Z 4
 20644 Diameter X-Y-Z 5
 20645 Ovality X-Y-Z 5
 20646 Smaller X-Y-Z 5
 20647 Larger X-Y-Z 5
 20648 Diameter X 5
 20649 Diameter Y 5
 20650 Diameter Z 5
 20651 Position X 5
 20652 Position Y 5
 20653 Position Z 5
 20654 Average X-Y-Z 5
 20655 Diameter X-Y-Z 6
 20656 Ovality X-Y-Z 6
 20657 Smaller X-Y-Z 6
 20658 Larger X-Y-Z 6
 20659 Diameter X 6
 20660 Diameter Y 6
 20661 Diameter Z 6

20662 Position X 6
 20663 Position Y 6
 20664 Position Z 6
 20665 Average X-Y-Z 6
 20666 Diameter X-Y-Z 7
 20667 Ovality X-Y-Z 7
 20668 Smaller X-Y-Z 7
 20669 Larger X-Y-Z 7
 20670 Diameter X 7
 20671 Diameter Y 7
 20672 Diameter Z 7
 20673 Position X 7
 20674 Position Y 7
 20675 Position Z 7
 20676 Average X-Y-Z 7
 20677 Diameter X-Y-Z 8
 20678 Ovality X-Y-Z 8
 20679 Smaller X-Y-Z 8
 20680 Larger X-Y-Z 8
 20681 Diameter X 8
 20682 Diameter Y 8
 20683 Diameter Z 8
 20684 Position X 8
 20685 Position Y 8
 20686 Position Z 8
 20687 Average X-Y-Z 8

 20688 Local Bend
 20689 Local Bend Angle
 20690 Begin Bar Bend
 20691 Begin Bar Bend Angle
 20692 End Bar Bend
 20693 End Bar Bend Angle

2.14.2 Combined Characteristics

Combined characteristics are those calculated using values of two characteristics. Each of these two characteristics could be single or combined. A combined characteristic code (CCC) is always bigger than 32768 and it is calculated dynamically, according how characteristic has been defined on the corresponding configuration screen. So for using those characteristics from HOST user should know their CCC.

There are two ways for knowing the CCC for a given characteristic. The first one is read its value from any product modified from screen and using this characteristic. The second one is the same formula used on USYS for calculate CCC:

$$CCC = CCIndex + (CCTypeCode * 256) + 32768$$

Where

CCC: Combined Characteristic Code.
 CCIndex: Index of the combined characteristic (position into the configuration screen). The first index is 0.
 CCTypeCode: Code of the combined characteristic type, following this table:

- 01 Diameter
- 02 Position
- 03 Ovality
- 04 Capacitance
- 05 Eccentricity
- 06 Wall Thickness
- 07 Concentricity
- 08 Concentricity working as Eccentricity

- 09 Generic Analog Gauge
- 10 Voltage
- 11 DLP Diameter
- 12 DLP Capacitance
- 13 DLP Wall Thickness
- 14 Inner Diameter
- 15 Diameter Difference
- 16 ODEX

For instance, if the first combined characteristic in the configuration is a wall thickness, then its code will be:

$$\begin{aligned}
 \text{CCC} &= \text{CCIndex} + \text{Type (WallThickness)} * 256 + 32768 \\
 &= 0 + (6 * 256) + 32768 \\
 &= 1536 + 32768 \\
 &= 34304
 \end{aligned}$$

If the third combined characteristic in the configuration is an inner diameter, then its code will be:

$$\begin{aligned}
 \text{CCC} &= \text{CCIndex} + \text{Type (InnerDiameter)} * 256 + 32768 \\
 &= 2 + (14 * 256) + 32768 \\
 &= 2 + 3584 + 32768 \\
 &= 36354
 \end{aligned}$$

2.15 APPENDIX B: LIST OF EVENTS AND ITS CODES

0 No event

SPARKER

100	SPARKER 1	Spark
101	SPARKER 2	Spark
102	SPARKER 1	Bare Patch
103	SPARKER 2	Bare Patch
104	SPARKER 1	Total Defects
105	SPARKER 2	Total Defects
106	SPARKER 3	Spark
107	SPARKER 4	Spark
108	SPARKER 5	Spark
109	SPARKER 6	Spark
110	SPARKER 3	Bare Patch
111	SPARKER 4	Bare Patch
112	SPARKER 5	Bare Patch
113	SPARKER 6	Bare Patch
114	SPARKER 3	Total Defects
115	SPARKER 4	Total Defects
116	SPARKER 5	Total Defects
117	SPARKER 6	Total Defects

KW

200	KW1	KW plus
201	KW1	KW minus
202	KW1	KW plus & minus
203	KW2	KW plus
204	KW2	KW minus
205	KW2	KW plus & minus
206	KW3	KW plus
207	KW3	KW minus
208	KW3	KW plus & minus
209	KW4	KW plus
210	KW4	KW minus
211	KW4	KW plus & minus
212	KW5	KW plus
213	KW5	KW minus
214	KW5	KW plus & minus
215	KW6	KW plus
216	KW6	KW minus
217	KW6	KW plus & minus

ODAC

300	ODAC 1	KW plus
301	ODAC 1	KW minus
302	ODAC 1	KW plus & minus
303	ODAC 2	KW plus
304	ODAC 2	KW minus
305	ODAC 2	KW plus & minus
306	ODAC 3	KW plus
307	ODAC 3	KW minus
308	ODAC 3	KW plus & minus
309	ODAC 4	KW plus
310	ODAC 4	KW minus
311	ODAC 4	KW plus & minus
312	ODAC 5	KW plus

313	ODAC 5	KW minus
314	ODAC 5	KW plus & minus
315	ODAC 6	KW plus
316	ODAC 6	KW minus
317	ODAC 6	KW plus & minus
318	ODAC 1	Abs KW plus
319	ODAC 1	Abs KW minus
320	ODAC 1	Abs KW plus & minus
321	ODAC 2	Abs KW plus
322	ODAC 2	Abs KW minus
323	ODAC 2	Abs KW plus & minus
324	ODAC 3	Abs KW plus
325	ODAC 3	Abs KW minus
326	ODAC 3	Abs KW plus & minus
327	ODAC 4	Abs KW plus
328	ODAC 4	Abs KW minus
329	ODAC 4	Abs KW plus & minus
330	ODAC 5	Abs KW plus
331	ODAC 5	Abs KW minus
332	ODAC 5	AbsKW plus & minus
333	ODAC 6	Abs KW plus
334	ODAC 6	Abs KW minus
335	ODAC 6	Abs KW plus & minus

2100	ODAC 7	KW plus
2101	ODAC 7	KW minus
2102	ODAC 7	KW plus & minus
2103	ODAC 8	KW plus
2104	ODAC 8	KW minus
2105	ODAC 8	KW plus & minus
2150	ODAC 7	Abs KW plus
2151	ODAC 7	Abs KW minus
2152	ODAC 7	Abs KW plus & minus
2153	ODAC 8	Abs KW plus
2154	ODAC 8	Abs KW minus
2155	ODAC 8	Abs KW plus & minus

CAPAC

400	CAPAC Pinhole 1
401	CAPAC Pinhole 2

Event Gauge

500	Digital Event 1
501	

510	Digital Event 2
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2.16 APPENDIX C: TIME ZONE CODES

0	(GMT-12:00) International Date Line West
5	(GMT-11:00) Midway Island, Samoa
10	(GMT-10:00) Hawaii
15	(GMT-09:00) Alaska
20	(GMT-08:00) Pacific Time (US and Canada); Tijuana
25	(GMT-07:00) Mountain Time (US and Canada)
30	(GMT-07:00) Chihuahua, La Paz, Mazatlan
35	(GMT-06:00) Central Time (US and Canada)
40	(GMT-06:00) Guadalajara, Mexico City, Monterrey

45 (GMT-05:00) Eastern Time (US and Canada)
 50 (GMT-05:00) Bogota, Lima, Quito
 55 (GMT-04:00) Atlantic Time (Canada)
 60 (GMT-04:00) Caracas, La Paz
 65 (GMT-04:00) Santiago
 70 (GMT-03:30) Newfoundland and Labrador
 75 (GMT-03:00) Brasilia
 80 (GMT-03:00) Buenos Aires, Georgetown
 85 (GMT-03:00) Greenland
 90 (GMT-02:00) Mid-Atlantic
 95 (GMT-01:00) Azores
 100 (GMT-01:00) Cape Verde Islands
 105 (GMT) Dublin, Edinburgh, Lisbon, London
 110 (GMT) Casablanca, Monrovia
 115 (GMT+01:00) Belgrade, Bratislava, Budapest, Ljubljana, Prague
 120 (GMT+01:00) Sarajevo, Skopje, Warsaw, Zagreb
 125 (GMT+01:00) Brussels, Copenhagen, Madrid, Paris
 130 (GMT+01:00) Amsterdam, Berlin, Bern, Rome, Stockholm, Vienna
 135 (GMT+01:00) West Central Africa
 140 (GMT+02:00) Bucharest
 145 (GMT+02:00) Cairo
 150 (GMT+02:00) Helsinki, Kiev, Riga, Sofia, Tallinn, Vilnius
 155 (GMT+02:00) Athens, Istanbul, Minsk
 160 (GMT+02:00) Jerusalem
 165 (GMT+02:00) Harare, Pretoria
 170 (GMT+03:00) Moscow, St. Petersburg, Volgograd
 175 (GMT+03:00) Kuwait, Riyadh
 180 (GMT+03:00) Nairobi
 185 (GMT+03:00) Baghdad
 190 (GMT+03:30) Tehran
 195 (GMT+04:00) Abu Dhabi, Muscat
 200 (GMT+04:00) Baku, Tbilisi, Yerevan
 205 (GMT+04:30) Kabul
 210 (GMT+05:00) Ekaterinburg
 215 (GMT+05:00) Islamabad, Karachi, Tashkent
 220 (GMT+05:30) Chennai, Kolkata, Mumbai, New Delhi
 225 (GMT+05:30) Sri Jayawardenepura
 230 (GMT+05:45) Kathmandu
 235 (GMT+06:00) Astana, Dhaka
 240 (GMT+06:00) Almaty, Novosibirsk
 245 (GMT+06:30) Yangon Rangoon
 250 (GMT+07:00) Bangkok, Hanoi, Jakarta
 255 (GMT+07:00) Krasnoyarsk
 260 (GMT+08:00) Beijing, Chongqing, Hong Kong SAR, Urumqi
 265 (GMT+08:00) Kuala Lumpur, Singapore
 270 (GMT+08:00) Taipei
 275 (GMT+08:00) Perth
 280 (GMT+08:00) Irkutsk, Ulaanbaatar
 285 (GMT+09:00) Seoul
 290 (GMT+09:00) Osaka, Sapporo, Tokyo
 295 (GMT+09:00) Yakutsk
 300 (GMT+09:30) Darwin
 305 (GMT+09:30) Adelaide
 310 (GMT+10:00) Canberra, Melbourne, Sydney
 315 (GMT+10:00) Brisbane
 320 (GMT+10:00) Hobart
 325 (GMT+10:00) Vladivostok
 330 (GMT+10:00) Guam, Port Moresby
 335 (GMT+11:00) Magadan, Solomon Islands, New Caledonia

340 (GMT+12:00) Fiji Islands, Kamchatka, Marshall Islands
345 (GMT+12:00) Auckland, Wellington
350 (GMT+13:00) Nuku'alofa